



## Master's Thesis

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**Course:** TTK4900 Master's Thesis  
**Title (English):** Compiler-Aware Model Optimization for Edge AI  
**Title (Norwegian):** Kompileravhengig optimalisering for Edge-AI

### Thesis description:

Efficient deployment of deep learning models on embedded and edge platforms depends on both model-level and system-level factors. In addition to model design, inference performance is influenced by compiler support, runtime behavior, hardware characteristics, numerical precision, and optimization techniques such as quantization and pruning. This thesis studies compiler-aware model optimization for edge AI, with focus on real-time object detection. The aim is to evaluate how different deployment configurations affect accuracy, latency, and resource usage, and to identify the main bottlenecks that limit efficient execution on target hardware.

### The tasks will be:

1. Survey existing work on deep learning model optimization and compiler-based deployment for edge hardware.
2. Develop a framework for evaluating deep learning models across deployment configurations in terms of accuracy, latency, and resource usage.
3. Explore model optimization techniques to improve computational efficiency while maintaining acceptable performance.
4. Analyze optimized models on target hardware, identify bottlenecks, and investigate ways to improve execution efficiency.

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# Abstract

Efficient deployment of deep learning models on edge hardware is not solely a question of model size. Inference performance is shaped by the joint interaction of model architecture, numerical precision, and the degree to which the inference framework can exploit the target hardware. This thesis presents a systematic, measurement-driven study of compiler-aware optimization for edge AI, using YOLOv8n object detection on two representative edge platforms - the NVIDIA Jetson Orin Nano and the Raspberry Pi 5 as a concrete case study.

A wide range of optimization strategies is evaluated through five inference frameworks (TensorRT, Apache TVM, ONNX Runtime, Google LiteRT (Tensorflow Lite), and torch.compile) compared to the Pytorch Eager Mode, two reduced-precision paths (FP16 and INT8), structured channel pruning with retraining, and activation function substitution. The experimental design includes isolated subprocess-based repetitions, warm-up runs, confidence intervals, and power measurements for each benchmark, enabling direct comparison across frameworks and hardware targets. Roofline analysis classifies per-layer execution regimes, and GPU kernel-level profiling with Nsight Systems decomposes the TensorRT FP32 engine into concrete overhead categories. Profiling reveals that only 59 % of engine time is spent on useful convolution work. The remaining 41 % has three identifiable sources: unfused SiLU activations (19 %), explicit memory copy nodes at the C2f channel-split boundaries (6.7 %), and internal NCHW→kCHW32 format reformats within the C2f subgraph (17 % of backbone convolution time). The copy and reformat overhead arises because TensorRT’s `Split` operator fixes the layout to NCHW, while the surrounding convolutions prefer the NHWC-vectorised kCHW32 format. These findings motivated a custom TensorRT plugin that replaces the entire C2f subgraph with a single opaque node, eliminating the format boundary that standard export cannot resolve. However, the structural overhead savings were offset by the loss of TensorRT’s Tensor Core kernels inside the plugin, and end-to-end latency did not improve over the default engine.

The results reveal several non-obvious findings. On the Raspberry Pi 5, LiteRT INT8 achieves 31 ms inference, a  $9.4\times$  reduction compared to the 290 ms PyTorch eager baseline, establishing that correct deployment choices matter far more than architectural modifications for this class of hardware. TVM MetaSchedule was the only framework capable of genuine FP16 acceleration on ARM CPU, reducing latency to 122 ms, while every other framework regressed or failed entirely due to missing half-precision convolution kernels. On the Orin Nano, TVM in FP16 achieved 12.5 ms through WMMA tensorisation,

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outperforming ONNX Runtime by 26% and narrowing the gap to TensorRT (6.3 ms). Profiling further revealed that SiLU activations are not fused into convolutions by TensorRT whereas ReLU is, a finding with direct implications for architecture design on resource-constrained targets where the choice of activation function affects not only accuracy but also the quality of kernel fusion. Structured channel pruning proved ineffective as a standalone strategy: a systematic sweep of 298 per-layer configurations found no single pruned model exceeding a  $1.04\times$  speedup, while the best multi-layer result with fine-tuning achieved only a 2.3% latency reduction at the cost of a 4.0% drop in mAP50-95, confirming that channel removal does not translate to runtime gains when cuDNN’s kernel tiling granularity is coarser than the applied reductions, and deeper architectural changes are required for meaningful compression.

Taken together, the results demonstrate that deployment framework selection, precision strategy, and low-level hardware alignment, not model size alone, are the dominant factors governing inference efficiency on edge hardware.

# Sammendrag

Effektiviteten til deep learning modeller implementert på kantmaskinvare er ikke utelukkende et spørsmål om modellstørrelse. Inferensytelse bestemmes av samspillet mellom modellarkitektur, numerisk presisjon og i hvilken grad inferensrammeverket kan utnytte den anvendte maskinvaren. Denne avhandlingen presenterer en systematisk, testbasert studie av kompilatorbasert optimalisering for kant-AI, med bruk av YOLOv8n objektdeleksjon på to representative kantplatt-former, NVIDIA Jetson Orin Nano og Raspberry Pi 5, som konkret kasusstudie.

Et bredt spekter av optimaliseringsstrategier evalueres gjennom fem inferensrammeverk (TensorRT, Apache TVM med MetaSchedule-autotuning, ONNX Runtime, Google LiteRT og torch.compile) sammenlignet med PyTorch Eager Mode, to presisjonsnivåer (FP16 og INT8), strukturell kanalbeskjæring (pruning) med etterfølgende trening, og substitusjon av aktiveringsfunksjon. Det eksperimentelle opplegget inkluderer isolerte kjøring i egne delprosesser, oppstartkjøring, konfidensintervaller og måling av forbrukt energi for hvert referansemåling, noe som muliggjør direkte sammenligning på tvers av rammeverk og maskinvareplattformer. Roofline-analyse brukes til å klassifisere kjøreregimer per lag, og GPU-kjerneprofilering med Nsight Systems brukes til å bryte ned TensorRT FP32-bruk på nyttige konvolusjonsberegninger. De resterende 41% fordeles seg på tre identifiserbare kilder: usammenslåtte SiLU-aktiveringer (19%), eksplisitte minnekopieringsnoder ved C2f-kanalsplitgrensene (6,7%), og interne NCHW→kCHW32-formatkonverteringer i C2f-undergrafen (17% av backbone konvolusjonstid). Disse funnene motiverte utviklingen av en tilpasset TensorRT-plugin som erstatter hele C2f-undergrafen med en enkelt “svart” node og eliminerer format-grensen som standard eksport ikke kan løse automatisk. Pluginen fjerner den strukturelle overheaden, men gevinsten ble oppveid av tapet av TensorRTs Tensor Core-kjerner, slik at den totale inferenstiden ikke ble forbedret sammenlignet med standardmotoren.

Resultatene avdekker flere overraskende funn. På Raspberry Pi 5 oppnår LiteRT INT8 31 ms inferenstid, en  $9,4\times$  reduksjon sammenlignet med 290 ms for PyTorch eager-basislinjen, noe som viser at riktige utrullingsvalg betyr langt mer enn arkitektoniske endringer for denne klassen maskinvare. TVM MetaSchedule var det eneste rammeverket som oppnådde ekte FP16-akselerasjon på ARM-CPU og reduserte ventetiden til 122 ms, mens alle andre rammeverk enten ble tregere eller feilet på grunn av manglende halv-presisjons konvolusjons-kjerner. På Orin Nano oppnådde TVM i FP16 12,5 ms gjennom WMMA-tensorisering, noe som er 26% raskere enn ONNX Runtime og nærmer seg TensorRT (6,3 ms). Undersøkelsen avslørte videre at SiLU-aktiveringer ikke sammenslås med

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konvolusjoner av TensorRT, i motsetning til ReLU, et funn med direkte implikasjoner for arkitekturvalg på ressursbegrensede plattformer. Strukturell kanalbeskjæring viste seg å være ineffektivt som frittstående strategi: et systematisk søk gjennom 298 lagkonfigurasjoner fant ingen enkeltbeskåret modell med mer enn  $1,04\times$  hastighetsøkning, og det beste flerlagsresultatet med ny trening ga kun 2,3% redusert ventetid til en kostnad av 4,0% lavere mAP50-95. Dette bekrefter at kanalreduksjoner ikke gir proporsjonal tidsgevinst når cuDNNs kjernegranularitet er grovere enn de anvendte reduksjonene, og mere inngående arkitektonisk endring er nødvendig foren bedre komprimering.

Samlet sett viser resultatene at valg av inferensrammeverk, presisjonsstrategi og maskinvarertilpasning på lavt nivå, ikke modellstørrelsen alene, er de dominerende faktorene for inferenseffektivitet på kantmaskinvare.

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# Abbreviations

| Abbreviation | Description                                                               |
|--------------|---------------------------------------------------------------------------|
| AArch64      | ARM 64-bit instruction set architecture                                   |
| AI           | Artificial Intelligence                                                   |
| AOT          | Ahead-of-Time compilation                                                 |
| AP           | Average Precision                                                         |
| API          | Application Programming Interface                                         |
| ARM          | Advanced RISC Machine; a family of processor architectures                |
| AWQ          | Activation-aware Weight Quantization                                      |
| BN           | Batch Normalization                                                       |
| C2f          | Cross-stage Partial with 2 convolutions and feature fusion (YOLOv8 block) |
| CNN          | Convolutional Neural Network                                              |
| COCO         | Common Objects in Context (benchmark dataset)                             |
| CPU          | Central Processing Unit                                                   |
| CSP          | Cross-Stage Partial network design                                        |
| CUDA         | Compute Unified Device Architecture (NVIDIA parallel computing platform)  |
| cuDNN        | CUDA Deep Neural Network library                                          |
| CUTLASS      | CUDA Templates for Linear Algebra Subroutines and Solvers                 |
| DAG          | Directed Acyclic Graph                                                    |
| DETR         | Detection Transformer                                                     |
| DFL          | Distribution Focal Loss                                                   |
| DLA          | Deep Learning Accelerator                                                 |
| DRAM         | Dynamic Random-Access Memory                                              |
| DVFS         | Dynamic Voltage and Frequency Scaling                                     |
| FC           | Fully Connected (layer)                                                   |
| FLOP         | Floating-Point Operation                                                  |
| FP8          | 8-bit floating-point numerical format                                     |
| FP16         | 16-bit (half-precision) floating-point numerical format                   |

| Abbreviation | Description                                                   |
|--------------|---------------------------------------------------------------|
| FP32         | 32-bit (single-precision) floating-point numerical format     |
| FPGA         | Field-Programmable Gate Array                                 |
| FPN          | Feature Pyramid Network                                       |
| FPS          | Frames Per Second                                             |
| GEMM         | General Matrix Multiplication                                 |
| GELU         | Gaussian Error Linear Unit (activation function)              |
| GFLOP        | Gigaflop; $10^9$ floating-point operations                    |
| GFLOPS       | Gigaflops per second; arithmetic throughput                   |
| GPU          | Graphics Processing Unit                                      |
| HLO          | High-Level Optimizer (XLA intermediate representation)        |
| IGEMM        | Integer General Matrix Multiplication                         |
| INT8         | 8-bit integer numerical format                                |
| IoU          | Intersection over Union                                       |
| IPC          | Instructions Per Cycle                                        |
| IR           | Intermediate Representation                                   |
| JIT          | Just-In-Time compilation                                      |
| KL           | Kullback–Leibler divergence                                   |
| LiteRT       | Google Lite Runtime (formerly TensorFlow Lite)                |
| LLC          | Last-Level Cache                                              |
| LLVM         | Low-Level Virtual Machine (compiler infrastructure)           |
| LSQ          | Learned Step-size Quantization                                |
| mAP          | Mean Average Precision                                        |
| mAP50        | Mean Average Precision at 50 % IoU threshold                  |
| mAP50-95     | Mean Average Precision averaged over IoU thresholds 50–95 %   |
| MLIR         | Multi-Level Intermediate Representation                       |
| MLAS         | Microsoft Linear Algebra Subroutines                          |
| NCHW         | Channels-first tensor layout (batch, channels, height, width) |
| NEON         | ARM Advanced SIMD instruction set extension                   |
| NHWC         | Channels-last tensor layout (batch, height, width, channels)  |
| NMS          | Non-Maximum Suppression                                       |

| Abbreviation | Description                                                          |
|--------------|----------------------------------------------------------------------|
| NPU          | Neural Processing Unit                                               |
| NVTX         | NVIDIA Tools Extension (profiling annotation API)                    |
| OBB          | Oriented Bounding Box                                                |
| ONNX         | Open Neural Network Exchange (model interchange format)              |
| PAN          | Path Aggregation Network                                             |
| PTQ          | Post-Training Quantization                                           |
| PWN          | Pointwise node (TensorRT activation kernel category)                 |
| QAT          | Quantization-Aware Training                                          |
| QDQ          | QuantizeLinear–DequantizeLinear (ONNX quantization representation)   |
| ReLU         | Rectified Linear Unit (activation function)                          |
| RT-DETR      | Real-Time Detection Transformer                                      |
| SIMD         | Single Instruction, Multiple Data                                    |
| SiLU         | Sigmoid Linear Unit (activation function; also called Swish)         |
| SM           | Streaming Multiprocessor (GPU compute unit)                          |
| SPPF         | Spatial Pyramid Pooling – Fast                                       |
| STE          | Straight-Through Estimator                                           |
| TF32         | TensorFloat-32 (19-bit precision format used by NVIDIA Tensor Cores) |
| TFLite       | TensorFlow Lite                                                      |
| TensorRT     | NVIDIA TensorRT inference optimisation library                       |
| TRT          | TensorRT (abbreviated)                                               |
| TVM          | Apache TVM deep learning compiler stack                              |
| UINT8        | 8-bit unsigned integer                                               |
| WMMA         | Warp Matrix Multiply-Accumulate (NVIDIA Tensor Core intrinsic)       |
| XLA          | Accelerated Linear Algebra (Google compiler for ML workloads)        |
| XNNPACK      | Accelerated neural network operations library for ARM and x86        |
| YOLO         | You Only Look Once (real-time object detection model family)         |

# 1 Introduction

## 1.1 Motivation

Deep learning has become the dominant approach to visual perception tasks such as object detection, segmentation, and classification. Models that were once confined to data centres now need to run on embedded hardware: surveillance cameras, autonomous vehicles, industrial inspection systems, and consumer mobile devices all demand on-device inference that is fast, power-efficient, and accurate.

The challenge is that deploying a deep learning model on edge hardware is not simply a matter of copying trained weights to a device and running them. Inference performance is determined by the joint interaction of model architecture, numerical precision, and the inference framework that compiles and executes the model. Two devices with nominally similar specifications can deliver very different throughput depending on which framework is used and how it maps the computational graph to the underlying hardware. At the same time, optimization techniques that are well understood in a data-centre context, quantization, pruning, operator fusion, often behave differently on embedded hardware, where compute budgets, memory bandwidth, and driver maturity all impose additional constraints.

There is a gap between what the theoretical performance of edge hardware suggests and what users observe in practice. Roofline analysis and kernel-level profiling of TensorRT-compiled YOLOv8n on the NVIDIA Jetson Orin Nano, for example, reveals that only 59% of engine time is spent on useful convolution work; the remaining 41% is structural overhead from unfused activations, format copies, and graph-level inefficiencies that standard compilation does not resolve. Understanding where this overhead comes from, and which optimization strategies can reduce it, requires systematic measurement rather than reliance on published benchmark tables.

This thesis addresses this gap by conducting a controlled, measurement-driven study of compiler-aware optimization for edge AI. Using YOLOv8n object detection as a concrete and deployable case study, the work evaluates how inference frameworks, precision formats, model compression, and low-level kernel customization interact to determine end-to-end inference efficiency. The goal is to provide a rigorous characterization of the optimization space available to a practitioner deploying an existing model on representative edge hardware.

## 1.2 Limitations

The scope of this study is to allow controlled and reproducible experimentation. The following limitations apply throughout.

- **Two hardware platforms.** All experiments are conducted on the NVIDIA Jetson Orin Nano (Ampere GPU, SM87) and the Raspberry Pi 5 (Cortex-A76 CPU, AArch64). These two devices represent a GPU-equipped embedded platform and a CPU-only single-board computer respectively, covering the most common edge deployment profiles. Other hardware targets, dedicated NPUs (Hailo, Qualcomm AI Engine), FPGAs, mobile SoCs, or the Raspberry Pi 5 VideoCore GPU, are not evaluated experimentally.
- **One model and one task.** The study uses YOLOv8n as the sole model and object detection as the sole task. YOLOv8n is selected because it is compact, widely deployed, architecturally representative of modern one-stage detectors, and exposes realistic deployment bottlenecks. Findings may not generalise directly to larger models, segmentation or classification architectures, or transformer-based detectors.
- **Limited set of inference frameworks.** Five frameworks are evaluated: TensorRT, Apache TVM with MetaSchedule, ONNX Runtime, Google LiteRT (TensorFlow Lite), and PyTorch (eager mode and `torch.compile`). Other relevant runtimes, OpenVINO, RKNN, CoreML, Hailo Dataflow Compiler, are outside the scope of the comparison.
- **Underexplored pruning.** Structured channel pruning is evaluated using PyTorch FP32 inference as the measurement backend. The study does not evaluate pruning combined with TensorRT compilation, unstructured weight pruning, or more recent sensitivity-based and gradient-guided pruning methods. As a result, the conclusions about pruning effectiveness are specific to the PyTorch FP32 execution regime.
- **Underexplored quantization.** Post-training quantization is evaluated using the calibration tools available within each framework. Quantization-aware training (QAT), mixed-precision quantization with per-layer bit-width assignment, and joint pruning-quantization methods are not investigated. The accuracy results reported for INT8 therefore represent a conservative lower bound on what is achievable with more sophisticated quantization workflows.
- **Single-image batch size.** All latency and throughput experiments use batch size 1, reflecting a real-time, single-stream deployment scenario. Throughput-optimised configurations with larger batch sizes are not evaluated.

- **Custom plugin scope.** The custom TensorRT plugin is developed and evaluated for the C2f block at “model.2” layer (32-channel,  $160 \times 160$  spatial resolution). The plugin is not generalised across all C2f instances in the network, and the implementation is fixed to FP32.

## 1.3 Contributions

Within these limitations, the main contributions of this thesis are:

- **Systematic multi-framework, multi-precision benchmark.** A controlled end-to-end benchmark of YOLOv8n inference across five frameworks, two hardware platforms, and three precision formats (FP32, FP16, INT8) is presented. Each configuration is evaluated with isolated subprocess repetitions, warm-up runs, 95 % confidence intervals, and simultaneous power measurements, enabling direct comparison of both latency and energy efficiency.
- **Hardware-aware bottleneck analysis.** Roofline analysis is applied to classify per-layer execution regimes on both platforms. GPU kernel-level profiling with Nsight Systems and Nsight Compute is used to decompose the TensorRT FP32 engine into concrete overhead categories, identifying unfused SiLU activations (19.3 %), format copy nodes (15.1 %), and channels-first (NCHW) format reformat overhead (17.2 % of backbone time) as the dominant sources of non-compute cost.
- **Custom TensorRT C2f plugin.** A TensorRT plugin is developed that replaces the C2f subgraph (four convolutions, SiLU activations, a channel split, and a residual addition) with a single opaque node. The plugin includes a hand-written fused CUDA (Compute Unified Device Architecture) kernel for the 32-channel case and a cuDNN (CUDA Deep Neural Network library) fallback path. The experiment quantifies the gap between custom CUDA code and Tensor Core-accelerated vendor kernels, establishing that structural fusion alone is not sufficient to outperform TensorRT’s default engine without Tensor Core alignment.
- **Activation function substitution study.** The effect of replacing all SiLU activations in YOLOv8n with ReLU is evaluated as a structural latency optimization. The substitution is performed without retraining and achieves a  $1.22\times$  end-to-end speedup (11.45 ms  $\rightarrow$  9.37 ms) and a 6.2 % reduction in average power draw on the Orin Nano, demonstrating that activation function choice has a direct and measurable impact on TensorRT kernel fusion and inference cost.
- **Quantization and pruning characterization.** INT8 quantization is evaluated across multiple deployment paths on both platforms, revealing a large performance

gap between implementations: TFLite INT8 achieves a  $9.4\times$  speedup on Raspberry Pi 5 CPU while ONNX Runtime INT8 provides no practical benefit over FP32. Structured channel pruning is evaluated across 418 per-layer configurations and a full-model experiment, showing that cuDNN kernel granularity prevents proportional speedup at the channel reductions explored, and that the strategy is ineffective as a standalone optimization for this model.

## 1.4 Thesis Structure

The remainder of this report is organised as follows.

- **Chapter 2 - Literature Study** introduces object detection with deep learning models, covering CNN fundamentals, the YOLO model family, the challenges of edge deployment, and an overview of inference frameworks, compilers, quantization, and pruning techniques that form the context for the experimental work.
- **Chapter 3 - Theory** describes the analytical tools used throughout the thesis: object detection evaluation metrics (Mean Average Precision (mAP), precision, recall), the roofline model for hardware-aware performance characterization, and the activation functions ReLU and SiLU.
- **Chapter 4 - Methodology** defines the experimental workflow, model selection rationale, baseline establishment procedure, and the evaluation criteria applied consistently across all experiments.
- **Chapter 5 - Experimental Evaluation** presents the experimental results across all optimization strategies: framework and precision benchmarks on both platforms, FP16 and INT8 inference results, accuracy comparison, structured pruning evaluation, GPU kernel-level profiling, the C2f TensorRT plugin, and the SiLU-to-ReLU activation substitution.
- **Chapter 6 - Discussion** interprets the results in context, compares the effectiveness of different strategies, examines causes of unexpected or negative findings, and identifies deployment configurations not covered by the experiments.
- **Chapter 7 - Conclusion** summarises the main findings and directly answers the research questions posed by the study.
- **Chapter 8 - Future Work** outlines concrete directions for extending this work, including CUTLASS-based Tensor Core plugins, quantization-aware training, activation function retraining, TensorRT-aware pruning, and broader hardware targets.

- **Chapter 9 - Description of Use of Artificial Intelligence** declares the use of AI tools in the preparation of this thesis.
- **Appendix A - Demonstration** provides supplementary experimental data and profiling artefacts.

## 2 Literature Study

### 2.1 Scope of the chapter

This chapter introduces a literature research of the existing optimizations strategies and methods as well as the technical background necessary to understand the methodology and results of this thesis. After an introduction to object detection and the YOLO family, a dedicated section on the YOLOv8 architecture establishes the model used throughout all experiments, followed by a section on the challenges of edge deployment that motivates why optimization is necessary in the first place. The remaining sections address the specific tools and techniques applied in this work. The section on ML compilers and inference runtimes covers Apache TVM and TensorRT as subsections, representing two complementary approaches to controlling inference behaviour at the compiler level: TVM as an open-source stack that exposes the full compilation pipeline and enables hardware-aware kernel scheduling and auto-tuning through MetaSchedule, and TensorRT as the state-of-the-art NVIDIA GPU inference framework that, beyond built-in graph optimization, supports writing custom CUDA kernels through its plugin API. Quantization and mixed precision, and pruning and structural simplification are then covered as the primary model-level optimization directions; together with compiler-level and scheduling optimizations, these span most of the principal axes available for inference acceleration. Strategies not pursued here, such as knowledge distillation, neural architecture search, and dynamic sparsity, exist and are noted where relevant but fall outside the scope of the present work. Finally, a section on performance bottlenecks in ML inference introduces the analytical concepts needed to interpret the profiling results.

### 2.2 Introduction to Object Detection with Deep Learning Models

Artificial Intelligence (AI) covers a broad set of algorithms that enable computers to perform tasks in language understanding, object recognition, control, and many other domains. This thesis is concerned specifically with deep learning models for visual object detection: models that take an image as input and produce a list of bounding boxes with associated class labels as output. Before examining specific architectures, it is useful to

establish what such a model is made of and how it differs from simpler machine-learning tasks.

## From pixels to features: convolutional neural networks

Modern object detectors are built on convolutional neural networks (CNNs) [22], which process images through a sequence of learned filters. A single convolutional layer applies a small kernel (typically  $3 \times 3$  or  $1 \times 1$ ) to every spatial position of the input feature map. Each kernel learns to detect a local pattern: an edge, a texture, a colour gradient, and produces one channel of the output feature map. Stacking many such layers builds up increasingly abstract representations: shallow layers detect low-level edges and textures, while deeper layers respond to object parts and semantic categories.

Formally, a single convolution at spatial position  $(i, j)$  with kernel  $\mathbf{W}$  of size  $k \times k$  over  $C_{\text{in}}$  input channels is:

$$y_{c,i,j} = \sum_{c'=0}^{C_{\text{in}}-1} \sum_{p=0}^{k-1} \sum_{q=0}^{k-1} \mathbf{W}_{c,c',p,q} x_{c',i+s+p,j+s+q} + b_c,$$

where  $s$  is the stride,  $b_c$  is a learned bias, and  $c$  indexes the output channel. After each convolution, a non-linear activation function is applied element-wise to introduce non-linearity into the model. Common choices are ReLU ( $\max(0, x)$ ) and SiLU ( $x \cdot \sigma(x)$ ), the latter being the default in YOLOv8. Between convolutional stages, pooling or strided convolutions reduce the spatial resolution while increasing the receptive field, the area of the input image that influences each output value.

Batch Normalization (BN), applied after each convolution and before activation, normalises the distribution of activations across the batch, stabilising training and enabling the use of higher learning rates. In deployed models, BN parameters are typically folded into the preceding convolution's weight and bias, eliminating a separate operation at inference time.

Fully connected (FC) layers operate on a flat vector and learn global relationships between all features. In image classification networks, an FC layer at the end maps the final feature vector to class logits. In modern detection heads, dedicated convolutional or FC branches produce box coordinates and class scores, as described further in Section 2.3.

At a high level, the entire network is a parameterized function

$$f_{\theta} : \mathbb{R}^{H \times W \times C} \rightarrow \mathcal{Y},$$

where the input is an image of height  $H$ , width  $W$ , and  $C$  channels, and  $\theta$  contains all

trainable weights. The output space  $\mathcal{Y}$  depends on the task.

## How object detection differs from classification

In image classification,  $\mathcal{Y}$  is a vector of class probabilities: the model answers “what is in this image?” with a single label. Object detection requires answering “where are the objects, and what are they?” simultaneously for a variable number of instances. The output  $\mathcal{Y}$  is therefore a set of tuples  $\{(b_i, c_i)\}$ , where each  $b_i$  is a bounding box and  $c_i$  is a class probability vector.

This difference has important architectural consequences. A detection model must produce spatially localised outputs rather than a single global label, which requires the network to preserve spatial information through most of its depth. It also must handle multiple simultaneous detections of different sizes, which requires multiscale feature fusion: the combination of high-resolution feature maps (sensitive to small objects) with semantically rich low-resolution maps (needed for large-object classification). Modern detectors achieve this through feature pyramid networks (FPN) or PANet-style neck architectures, which are described in more detail in the context of YOLOv8 in Section 2.3.

A further distinction is post-processing. Classification outputs a single softmax distribution. Detection outputs a large number of candidate boxes, many of which overlap. A non-maximum suppression (NMS) step is therefore applied after the forward pass to remove duplicate detections, selecting the highest-confidence box among overlapping candidates. Several recent architectures (YOLOv10, RT-DETR) have sought to eliminate or simplify NMS by design.

## Two-stage versus one-stage detectors

Early deep learning detectors used a two-stage pipeline. The first stage generates a set of region proposals (candidate bounding boxes); the second stage classifies each proposal and refines its coordinates. Representative models in this family, R-CNN, Fast R-CNN, Faster R-CNN [62], achieved high accuracy but were too slow for real-time use (typically 7–18 frames per second (FPS) on contemporary hardware).

One-stage detectors eliminate the proposal stage. A single convolutional pass over the image directly predicts bounding boxes and class scores at multiple spatial positions and scales. YOLO [60] (You Only Look Once) was the first widely adopted model in this family: on PASCAL VOC 2007, the original YOLO ran at 45 FPS while achieving competitive mAP, and the faster variant reached 155 FPS. This speed advantage made one-stage detectors the dominant choice for real-time and edge deployments.

## The YOLO family and transformer-based alternatives

Since the original YOLO in 2015, eleven major versions have been released [21]. The most practically significant milestones are:

- **YOLOv5** - introduced a PyTorch implementation, multiple size variants (n/s/m/l/x), and straightforward ONNX export, making YOLO accessible for deployment across a wide range of hardware.
- **YOLOv8** - adopted an anchor-free detection head, C2f backbone blocks, and a distribution-based box regression head (DFL), improving both accuracy and training stability while retaining the same deployment ecosystem.
- **YOLOv10** - introduced NMS-free inference through dual-label assignment during training, reducing post-processing latency, and added large-kernel convolutions and partial self-attention for a wider receptive field.
- **YOLOv11** - refined the training pipeline and architecture for better accuracy per parameter, achieving higher mAP than YOLOv8 with fewer parameters.

A competing approach is the transformer-based detector. RT-DETR [32] replaces the convolutional neck and NMS post-processing with a transformer decoder that performs set-based prediction, entirely removing the duplicate-suppression step. On GPU, RT-DETR benefits from the parallelism of self-attention and achieves strong accuracy. However, transformer architectures carry a higher parameter count and memory footprint than compact YOLO variants, making them less attractive for CPU and low-power edge targets. NanoDet [57] represents the opposite extreme: a sub-megaparameter detector targeting mobile CPU inference, trading accuracy ( $\sim 34\%$  mAP) for extreme compactness.

For the purposes of this thesis, YOLOv8n is selected as the representative model. It offers a well-understood architecture with realistic deployment challenges, strong community support, a complete export ecosystem, and a parameter count (3.2 M, 8.7 GFLOPs) that makes it practical to experiment with on edge hardware. A detailed architectural description follows in Section 2.3.

## 2.3 YOLOv8 Architecture

YOLOv8 is a one-stage convolutional object detector introduced by Ultralytics in 2023 [20] as part of the YOLO family of real-time vision models. Its design follows the

standard decomposition of modern detectors into a backbone, a neck, and a prediction head, while introducing several architectural refinements compared with earlier YOLO generations. In particular, YOLOv8 employs C2f feature extraction blocks, an anchor-free split detection head, and an efficient multiscale feature aggregation pipeline. These design choices aim to improve the trade-off between detection accuracy and inference speed while maintaining practical deployability across heterogeneous hardware platforms. According to the official Ultralytics implementation, the standard detection family includes five model scales, ranging from YOLOv8n to YOLOv8x, where the nano variant contains approximately 3.2 million parameters and requires about 8.7–8.9 GFLOPs at an input resolution of  $640 \times 640$ .

**Overall structure of YOLOv8** The overall YOLOv8 architecture is illustrated in Figure 2.1. The figure also highlights the scaling variables used to derive the different model variants, which determine the network depth, width, and overall computational complexity.

The standard YOLOv8 detection network consists of three main stages. The backbone extracts hierarchical visual features from the input image. The neck fuses semantically rich low-resolution features with spatially precise higher-resolution features. Finally, the prediction head converts the fused multiscale features into bounding box coordinates and class predictions. In the reference YOLOv8 configuration, the backbone progressively downsamples the input image into feature maps at multiple pyramid levels, while the head generates predictions at three output scales commonly associated with small, medium, and large objects. In the default P5 configuration, predictions are produced from feature levels P3/8, P4/16, and P5/32.

The backbone begins with strided convolutional layers that reduce spatial resolution while increasing channel depth. These stages are interleaved with C2f blocks, which are described in the Ultralytics reference implementation as a “faster implementation of Cross-Stage Partial (CSP) bottleneck with 2 convolutions.” The main purpose of these blocks is to improve gradient flow and feature reuse while maintaining computational efficiency. At the deepest stage of the network, YOLOv8 employs an SPPF (Spatial Pyramid Pooling – Fast) module, which increases the effective receptive field and improves multiscale context aggregation with relatively low computational overhead.

Overall, the main backbone components of YOLOv8 are:

- convolutional blocks with Batch Normalization and SiLU activation,
- C2f feature extraction blocks,
- the SPPF module.

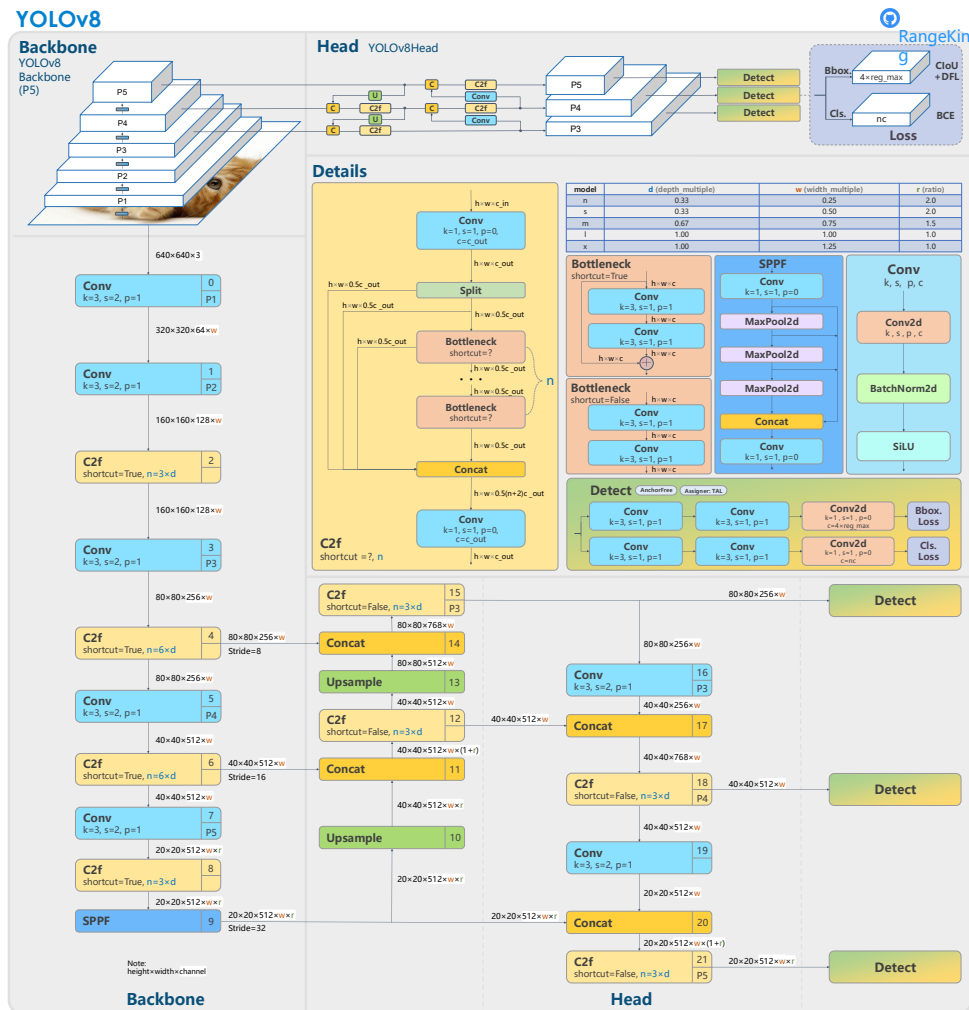


Figure 2.1: Overall YOLOv8 architecture and model scaling variables [56]

The C2f block itself is composed of several bottleneck units. Conceptually, such a block first reduces the number of channels, then performs spatial feature transformation, and finally combines intermediate representations to preserve useful information. This design improves feature propagation and reduces redundant computation while retaining informative representations. Compared with earlier CSP-style blocks, C2f enables richer feature reuse and more efficient gradient flow.

The neck follows a Path Aggregation Network (PAN)–FPN style design. In practice, this means that feature maps are upsampled, concatenated with earlier backbone features, and then processed again using C2f blocks. This combination of top-down and bottom-up feature fusion is central to YOLOv8’s ability to detect objects across multiple scales. The reference YAML architecture repeatedly uses Upsample, Concat, and C2f blocks before the final Detect module. Such multiscale aggregation is particularly important for real-world datasets, where object size can vary significantly across scenes.

The prediction head in YOLOv8 is both decoupled and anchor-free. Ultralytics describes YOLOv8 as using an anchor-free split detection head, meaning that predictions

are not tied to a predefined set of anchor box shapes. Instead, the detector predicts bounding boxes and class scores directly from the feature maps. Compared with older anchor-based YOLO variants, this simplifies the detection pipeline and removes the need for dataset-specific anchor tuning.

In an anchor-free detector, each spatial location on the feature map typically predicts:

- whether an object is present,
- the class scores,
- the box coordinates or distances to the box boundaries.

The split head further separates the localization and classification tasks into distinct branches. This decoupling allows the model to optimize these tasks more effectively, since object localization and semantic classification impose different requirements on the learned feature representations.

Another important component of the YOLOv8 detection formulation is its use of distribution-based box regression. The Ultralytics codebase includes a DFL module and the corresponding loss implementation, reflecting the use of Distribution Focal Loss for bounding box parameterization. This improves localization precision by modeling bounding box offsets as discrete probability distributions rather than relying solely on coarse direct regression.

**Relevance of YOLOv8 for this thesis** For a thesis focused on compiler-aware optimization of deep learning models for edge devices, YOLOv8, and especially YOLOv8n, represents a highly suitable case study.

First, YOLOv8 offers a strong balance between detection accuracy and inference latency, which is one reason why later detectors continue to use it as a baseline. Second, YOLOv8 has a mature deployment ecosystem. Ultralytics supports export to a wide range of formats relevant for practical deployment, including ONNX, TensorRT, CoreML, OpenVINO, and TFLite. This makes the model especially attractive for evaluating compiler and runtime behavior across different hardware platforms.

In addition, YOLOv8 is not limited to object detection alone. The model family also supports segmentation, classification, pose estimation, and oriented bounding box detection within the same software ecosystem. This is particularly relevant in applied edge AI research, where systems often evolve from simple object detection toward richer scene understanding tasks. As a result, YOLOv8 provides both architectural consistency and practical flexibility.

For this thesis, YOLOv8n is of particular interest because it is small enough to run on many embedded platforms, while still being architecturally complex enough to expose realistic deployment bottlenecks. The model includes multiscale feature fusion, repeated tensor concatenations, SiLU activations, distribution-based box regression, and post-processing stages. These properties make it a meaningful test case for studying compiler optimizations, runtime scheduling, kernel efficiency, quantization behavior, and memory-system effects.

Despite being the smallest variant, YOLOv8n is not automatically optimal for every edge target. Several practical challenges arise during deployment.

First, a low parameter count does not necessarily translate into low latency on all hardware platforms. YOLOv8n still contains nontrivial tensor reshaping, concatenation, upsampling, and multiscale fusion operations. On desktop GPUs, such operations may be efficiently hidden by optimized kernels and memory systems, whereas on edge CPUs and lightweight accelerators the cost of memory movement can dominate the arithmetic workload. In such cases, the effective execution becomes memory-bound rather than compute-bound.

Second, operator support and kernel efficiency vary significantly across deployment backends. Although YOLOv8 can be exported to ONNX, TensorRT, TFLite, and other runtimes, the achieved performance depends strongly on how well the target backend supports the specific operator set and graph structure of the model. In practice, operations such as SiLU, concatenation-heavy neck structures, and certain post-processing steps may be implemented with different levels of efficiency across frameworks. As a result, there is often a noticeable gap between the theoretical Floating-Point Operation (FLOP) count of the model and the measured end-to-end latency.

Third, post-processing introduces an additional deployment challenge. Standard YOLOv8 detection still relies on confidence filtering and non-maximum suppression after the forward pass. On GPU-based systems, this may require transferring relatively large tensors from the device to the host, followed by post-processing on the CPU. Such steps can add non-negligible latency and complicate the analysis of true end-to-end performance.

Overall, YOLOv8 is an interesting subject for deployment-oriented research because it combines architectural complexity with competitive accuracy and speed.

## 2.4 Challenges of edge deployment

With the growing interest in on-device deployment, a clear gap has emerged between the training and inference stages of machine learning models. While large models are typically trained in data centers using powerful GPUs, inference is performed across a wide variety of environments subject to strict constraints on memory, latency, and power consumption. These environments span heterogeneous hardware platforms - CPUs, GPUs, TPUs, NPUs, and FPGAs - each differing in instruction sets, memory hierarchies, and supported data types. As a result, model performance depends heavily on how well the deployment is adapted to the target hardware.

Deployment efficiency is characterized along four principal dimensions:

- **Latency** measures the time required to process a single inference request and is the dominant constraint in real-time applications such as object detection, where a response must be produced within a fixed time budget.
- **Throughput** captures the number of inference requests completed per unit of time and becomes the primary metric when processing is batched or pipelined.
- **Power consumption** reflects the energy cost of inference, which is particularly critical on battery-powered or thermally constrained embedded devices.
- **Memory requirements** encompass both the storage needed for model weights and the working memory consumed by intermediate activations during a forward pass, both of which can be hard limits on resource-constrained hardware.

Machine learning models are typically developed in high-level Python frameworks such as TensorFlow or PyTorch, which use dynamic execution: operations are dispatched immediately, control flow is resolved at runtime, and tensors are created and destroyed on the fly. This flexibility comes at a cost. Every operation passes through the Python interpreter; the Global Interpreter Lock (GIL) [51] limits parallel execution; tensors and operators are heap-allocated Python objects; and each kernel dispatch requires the framework to resolve device placement and memory management at runtime. Individually these overheads are small, but across a deep network with thousands of operations they accumulate into a non-trivial fraction of end-to-end latency.

In deployment, models are therefore not run inside Python at all. Instead, they are executed by C++-based runtimes, hardware-specific inference engines, or precompiled binaries targeting accelerators such as GPUs, NPUs, and FPGAs [6, 44, 40]. These runtimes execute pre-compiled computation graphs, call highly optimized kernels directly, and avoid Python entirely. However, simply exporting a model out of Python does not

automatically guarantee speedups. To fully benefit, the model must first be compiled into a static or semi-static graph, then optimized through operator fusion, memory planning, quantization, and layout transformations, and finally adapted to the instruction set and memory hierarchy of the target hardware. Without this compilation step, a deployed runtime may still execute inefficient kernels or suboptimal data flows.

A further challenge is that highly optimized kernel libraries such as cuDNN and oneDNN are not automatically optimal for all deployment regimes. Although these libraries support both training and inference, many of their fastest implementations rely on sufficient arithmetic intensity, data reuse, vectorization, and parallel work to amortize scheduling and dispatch overheads. These conditions are often easier to satisfy in training or high-throughput inference, where larger batches allow weights and activation tiles to be reused across multiple samples.

In real-time edge inference, however, batch size is often fixed to one in order to minimize response latency. Under this regime, the computation may expose less parallelism and less reuse per invocation. On GPUs, this can reduce occupancy and make CUDA kernel-launch overhead, memory traffic, and operator fragmentation more visible in the end-to-end latency. On CPUs, the analogous issue is not GPU occupancy but reduced Single Instruction, Multiple Data (SIMD) utilization, less effective cache blocking, thread-management overhead, and layout-conversion overheads. As a result, batch-size-one inference can be limited not only by peak compute throughput, but also by memory movement, runtime overhead, and the ability of the compiler or runtime to fuse operators and select kernels specialized for small-batch execution.

Graph-level compilers address this by operating on the entire computation graph rather than dispatching one operator at a time. With global visibility over operators, data dependencies, and tensor lifetimes, a compiler can fuse multiple operations into a single kernel, fold constants, eliminate dead branches, and reorder associative operations to improve locality. It can also select a single data layout (channels-first NCHW, channels-last NHWC, or a blocked format) that suits the target hardware and insert only the minimal number of layout conversions, avoiding the unnecessary transposes that arise when operators from different libraries are composed naively.

## 2.5 ML compilers and inference runtimes

Deep learning compilers specialize traditional compiler ideas for tensor programs and neural-network graphs [28, 73, 67, 74]. They address the gap between high-level model definitions and heterogeneous deployment targets by lowering neural-network graphs through one or more intermediate representations, applying graph-level and operator-level opti-

mizations, and generating target-specific code or calls to optimized backend libraries. So the code, expressed in a framework such as PyTorch, TensorFlow, JAX, or as an ONNX graph is transformed into an executable form suitable for a particular target, such as a CPU, GPU, NPU, FPGA, or dedicated accelerator.

Most ML compiler stacks follow a layered structure, illustrated in Figure 2.4. The frontend first captures or imports the model from a source framework such as PyTorch [49], TensorFlow [1], or JAX [9]. In practical deployment pipelines, this import step is often mediated by an exchange format such as ONNX [33]. Other representations are more closely tied to particular compiler ecosystems: TorchScript was historically used for PyTorch graph capture and deployment, FX graphs are used in the PyTorch 2.x compilation pipeline [61], High-Level Optimizer (HLO) is the central Intermediate Representation (IR) in Accelerated Linear Algebra (XLA) [45], and StableHLO provides a more portable HLO-based representation in the OpenXLA/Multi-Level Intermediate Representation (MLIR) ecosystem [46].

The imported program is then converted into the compiler’s internal intermediate representation. The graph optimization stage then applies framework-agnostic transformations: operator fusion and simplification, constant folding, dead-branch elimination, and memory planning. The resulting high-level IR is subsequently lowered to a loop-level representation where tiling, vectorization, and memory-access scheduling are applied for the specific target. Finally, the code generation backend performs instruction selection, register allocation, and emits target-specific machine code.

Once compiled, the model is managed by an ML runtime, which handles execution orchestration (loading the compiled artifact and scheduling its execution), memory allocation for intermediate tensors, hardware interfacing, and integration with vendor libraries.

Modern systems differ mainly in where they place the boundary between compilation, runtime dispatch, and vendor-specific libraries. For example, TensorFlow XLA lowers TensorFlow or JAX-style computations through HLO-based representations and uses XLA backends to generate efficient target-specific execution [47, 48]. PyTorch 2.x follows a different route: TorchDynamo captures Python-level execution into FX graphs, AOTAutograd handles ahead-of-time graph extraction for training-related graphs, and TorchInductor lowers the resulting graphs to optimized CPU or GPU code [53, 54, 52]. Apache TVM exposes a more explicit compiler stack for heterogeneous deployment, combining graph-level optimization with tensor-level scheduling and auto-tuning for CPUs, GPUs, and edge-oriented targets [2, 6]. MLIR-based systems, including IREE and TensorFlow MLIR, use multiple IR dialects to represent computation at different abstraction levels and to support flexible lowering pipelines for heterogeneous systems [24, 30, 66, 18]. ONNX Runtime, in contrast, is primarily an inference runtime built around the ONNX format; it performs graph optimizations and delegates execution to hardware-

specific execution providers such as CPU, CUDA, TensorRT, OpenVINO, CoreML, or other backends [43, 44].

These systems therefore solve overlapping but not identical problems. TVM and XLA are closer to full compiler stacks with explicit IR lowering and code generation, TensorRT is a vendor-specific inference compiler and runtime for NVIDIA GPUs, PyTorch 2.x integrates compilation into an eager framework workflow, MLIR-based systems provide reusable infrastructure for building compiler pipelines, and ONNX Runtime emphasizes portable model execution through execution providers. For edge-AI deployment, this distinction is important because performance depends not only on the model architecture, but also on the chosen representation, compiler lowering path, operator coverage, precision support, runtime overhead, and the quality of the target-specific backend.

Overall, the architecture of a deep-learning compiler can be viewed as a pipeline composed of a frontend, one or more intermediate representations, and a backend. The IR connects these stages and often appears at multiple levels of abstraction, ranging from high-level graph IR to low-level operator or tensor IR.

**The high-level IR (Graph IR)** represents the computation graph of a deep-learning model at a level above hardware-specific code generation. It captures operators, tensors, data dependencies, and in some systems also control flow, while remaining mostly hardware-independent. Its purpose is to provide a representation on which the compiler frontend can perform graph-level optimizations before lowering the model to a lower-level IR for scheduling, code generation, and hardware-specific optimization [28].

A common representation is a directed acyclic graph (DAG)(see listing 1), where nodes correspond to neural-network operators such as convolution, pooling, reshape, or activation functions, and edges correspond to tensors flowing between operators. This representation allows the compiler to analyze dependencies between operators and apply optimizations. However, pure DAG-based representations can be limited when the computation scope or control flow must be represented explicitly. For this reason, some compiler IRs also use let-binding (see listing 2), which represents each intermediate tensor as a named binding, or functional representations (see listing 3), which views computation as a collection of functions, where each function contains operations/instructions and has explicit inputs and outputs. For example, TVM Relay combines DAG-style graph representation with let-binding ideas, while XLA HLO uses a function-based representation organized around modules, computations, and instructions [28].

The high-level IR must also encode tensor-specific information that is essential for deep-learning optimization. This includes tensor shapes, possibly dynamic dimensions, data layouts such as NCHW or NHWC, broadcasting rules, reduction operations, and

support for custom operators. This information is needed to infer valid transformations, select efficient layouts, and preserve correctness when the graph is later lowered to backend-specific code. Thus, the high-level IR acts as the bridge between framework-level model definitions and lower-level compiler stages: it is expressive enough to describe deep-learning models, but abstract enough to enable hardware-independent graph optimization.

DAG-based graph IR:

```
x ----\
      conv2d ----> z ----> relu ----> a ----\
w -----/                                     add ----> y
bias -----/
```

Nodes:

conv2d, relu, add

Edges:

x, w, z, a, bias, y

Listing 1: Example of a DAG-based graph IR representation.

Let-binding-based IR:

```
let z = conv2d(x, w) in
let a = relu(z) in
let y = add(a, bias) in
return y
```

Listing 2: Example of a let-binding-based IR representation.

Function-based IR:

```
function forward(x, w, bias):
  z = conv2d(x, w)
  a = relu(z)
  y = add(a, bias)
  return y
```

Listing 3: Example of a function-based IR representation.

**The low-level IR** The low-level IR represents the computation of a deep-learning model in a more fine-grained form than the high-level graph IR. While the high-level IR is mainly used to describe operators, tensors, and graph-level dependencies, the low-level IR is closer to the actual implementation of each operator or fused subgraph. It exposes

loop structure, memory accesses, tensor indexing, scheduling decisions, and target-specific information, making it the main interface for hardware-dependent optimizations and code generation [28].

A common design in low-level IRs is to separate the description of what is computed from how it is computed. Halide-inspired IRs, used for example in TVM, follow this idea by separating the tensor computation from its schedule. The computation defines the mathematical operation, while the schedule controls implementation choices such as tiling, loop reordering, vectorization, parallelization, memory placement, and unrolling. This separation is useful because the same operator can be mapped differently depending on the target hardware, such as an Advanced RISC Machine (ARM) CPU, an NVIDIA GPU, or a specialized accelerator.

Another important family is polyhedral-based IRs. These represent loop nests and memory accesses using mathematical sets, affine transformations, and dependence relations. Such representations are well suited for optimizing regular tensor computations, because they make it possible to reason about loop transformations such as fusion, tiling, skewing, mapping, and parallelization. However, polyhedral approaches are most effective when loop bounds and memory accesses can be described statically and regularly, which can limit their applicability for highly dynamic or irregular model structures.

Some compiler systems use their own low-level IRs instead of directly adopting Halide- or polyhedral-style representations. For example, Glow [64] uses an instruction-based low-level IR over explicit memory regions, while MLIR [30] provides a multi-level compiler infrastructure based on dialects, allowing tensor, linear algebra, affine, Low-Level Virtual Machine (LLVM) [23], and hardware-specific abstractions to coexist in the same framework. XLA HLO can also serve both high-level and lower-level roles, since it is fine-grained enough to express many target-specific optimization decisions while still preserving tensor-level semantics.

The low-level IR is eventually lowered to executable code or runtime calls, often through mature compiler infrastructures such as LLVM, CUDA, OpenCL, or vendor-specific code generators. This stage may be performed just-in-time (JIT), where code is generated at runtime using available shape and hardware information, or ahead-of-time (AOT), where executable binaries are generated before deployment. In edge-AI deployment, the low-level IR is especially important because performance depends strongly on hardware-specific choices such as memory hierarchy usage, vector width, cache behavior, tensor layout, and available accelerator instructions.

**Frontend Optimizations** It is easier to identify optimization opportunities when the machine learning model is represented as a complete computation graph rather than as

isolated operators. Frontend optimizations are therefore applied at the graph or high-level IR level, before the model is lowered to hardware-specific backend code. These transformations are mostly hardware-independent: they simplify the graph, remove redundant operations, expose fusion opportunities, and reduce memory traffic, but they do not yet define how each operation is implemented on a particular CPU, GPU, or accelerator.

Li et al. [28] classify frontend optimizations in deep-learning compilers into three main categories: node-level, block-level, and dataflow-level optimizations.

- **Node-level optimizations** operate on individual graph nodes. They include node elimination and node replacement, where unnecessary or expensive nodes are removed or replaced by cheaper equivalent operations. Examples include removing identity operations, eliminating padding nodes with zero padding width, or replacing a redundant reshape with direct tensor forwarding.
- **Block-level optimizations** operate on small local subgraphs. They include algebraic simplification, strength reduction, constant folding, operator fusion, and operator sinking. For example, a sequence such as convolution, bias addition, and activation can be fused into a single higher-level operation, reducing intermediate tensors and kernel-launch overhead.
- **Dataflow-level optimizations** operate globally across the computation graph. They include common sub-expression elimination, dead code elimination, static memory planning, and layout transformation. These optimizations exploit the global graph structure to avoid recomputation, remove unused branches, reuse memory buffers, and select tensor layouts that are more suitable for later backend execution.

Figure 2.2 illustrates representative examples of these frontend graph transformations.

**Backend Optimizations** Backend optimizations form the target-specific part of a deep-learning compiler. After the model has been represented in a graph or tensor-level IR, the backend lowers this representation to executable code, optimized kernels, or calls into vendor libraries. In practice, high performance is often achieved by combining library-based execution, such as cuDNN, MIOpen, oneDNN, or BLAS kernels, with compiler-generated transformations that specialize the computation for the target architecture. These transformations include mapping low-level IR patterns to hardware intrinsics or optimized micro-kernels, assigning intermediate tensors to appropriate memory scopes, improving data locality through cooperative fetching and explicit memory allocation, hiding memory latency through scheduling and hardware thread switching, and applying loop

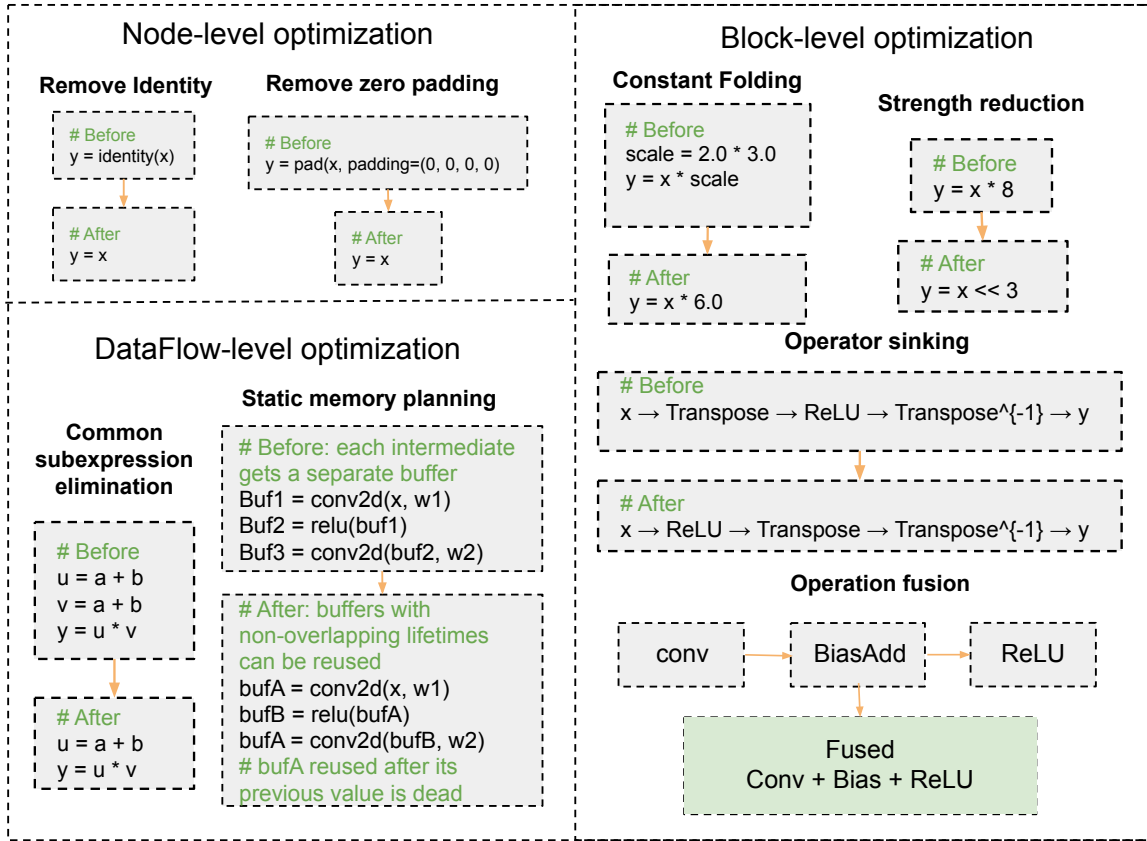


Figure 2.2: Representative examples of frontend graph optimizations in deep-learning compilers. Node-level optimizations remove or replace individual redundant nodes, block-level optimizations simplify or fuse local subgraphs, and dataflow-level optimizations exploit the global computation graph to remove redundant computation or reuse memory buffers.

transformations such as fusion, tiling, reordering, unrolling, and sliding-window optimizations. The backend also exposes parallelism at multiple granularities, including SIMD vectorization, CPU multi-threading, GPU thread/block parallelism, and accelerator-specific execution units. Figure 2.3 summarizes several representative backend-level optimization techniques used in deep-learning compilers.

A central difficulty is that these optimizations create a large, hardware-dependent search space. Even for a single convolution or matrix multiplication, the compiler may need to choose tile sizes, loop orders, memory layouts, vector widths, thread/block mappings, unrolling factors, and intrinsic implementations. The best configuration depends on the target architecture, cache hierarchy, memory bandwidth, register pressure, available parallelism, and supported instruction set. As a result, many deep-learning compilers incorporate auto-tuning mechanisms. Auto-tuning defines a space of valid schedules and searches for high-performing implementations using empirical measurements, analytical or learned cost models, and heuristic search methods such as genetic algorithm (GA) [11],

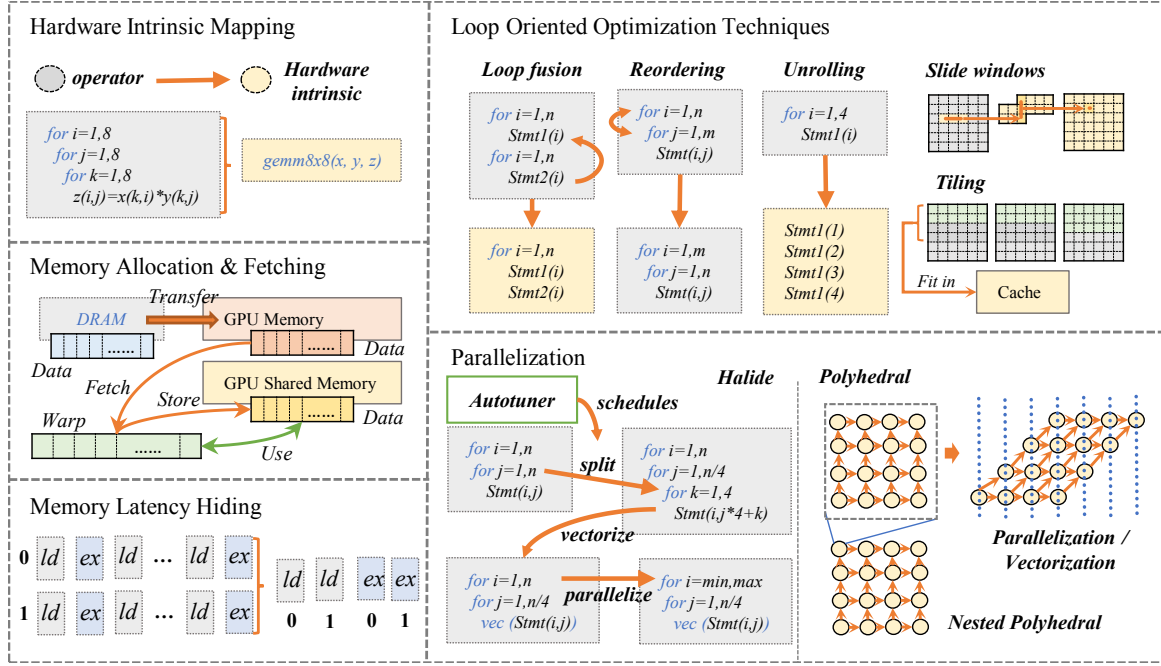


Figure 2.3: Overview of hardware-specific optimizations applied in deep-learning compilers. Reproduced from Li et al. [28].

reinforcement learning (RL) [35], simulated annealing (SA) [4], or evolutionary search [3].

Although many standard deep-learning kernels are highly optimized, model development often progresses faster than backend kernel support. New architectures can introduce operators or computation patterns that are not yet available as optimized library kernels or compiler-generated implementations. In such cases, the computation may fall back to unfused operator sequences or generic implementations, causing additional kernel launches, intermediate memory traffic, reduced locality, and significant runtime overhead.

### 2.5.1 Apache TVM

Apache TVM [6] is an open-source machine learning compiler stack that treats model deployment as a compiler problem. Rather than dispatching operators to a fixed set of vendor-provided kernels, TVM imports a model, applies graph-level transformations (constant folding, operator fusion, layout rewriting), lowers individual operators into explicit loop-based tensor programs, and then searches for hardware-efficient implementations through automated schedule tuning. This pipeline follows the general compiler flow described in the previous section, but exposes each stage to the user, making TVM useful both as a deployment tool and as a research platform for studying how compilation choices affect inference performance.

The central design principle is the separation between computation and schedule. The

computation defines what is computed; the schedule defines how it maps to hardware. For the same convolution, the compiler can explore different tile sizes, loop orders, vector widths, memory staging strategies, and thread/block mappings for the target. In practice, TVM automates this via MetaSchedule: candidate schedules are generated, benchmarked on the device, and ranked by measured latency, producing a hardware-specific binary without manual kernel engineering.

TVM’s main strength is hardware portability and the ability to unlock optimisation paths not available in fixed-kernel runtimes. On ARM CPUs, schedule search can target NEON vector intrinsics explicitly; on NVIDIA GPUs, FP16 MetaSchedule tuning enables WMMA tensorisation for convolutional operators. Its limitations are a complex compilation workflow and performance that depends heavily on the quality and duration of the tuning campaign: an untuned TVM build may not outperform generic runtimes. In this thesis, TVM is evaluated with MetaSchedule on both the Jetson Orin Nano GPU and the Raspberry Pi 5 CPU, serving as a reference for what an open, tunable compiler achieves relative to TensorRT and ONNX Runtime.

**Tensor expression and scheduling: a concrete example.** The computation–schedule separation is most clearly illustrated with matrix multiplication. The mathematical operation is:

$$C[i, j] = \sum_k A[i, k] \cdot B[k, j] \quad (2.1)$$

In TVM’s tensor expression API this is written as a pure computation definition, with no implementation detail:

```
from tvm import te
M, N, K = 1024, 1024, 1024
A = te.placeholder((M, K), name="A")
B = te.placeholder((K, N), name="B")
k = te.reduce_axis((0, K), name="k")
C = te.compute(
    (M, N),
    lambda i, j: te.sum(A[i, k] * B[k, j], axis=k),
    name="C"
)
```

This definition describes the mathematical computation only. It does not yet specify loop ordering, blocking, vectorisation, or parallel execution.

**Scheduling and kernel mapping.** To obtain an efficient implementation, TVM applies scheduling decisions that determine how the tensor computation maps to loops,

threads, and memory. Typical transformations include tiling, loop reordering, vectorisation, unrolling, and thread binding. For a CPU target these translate to cache-friendly blocking, parallelisation across cores, and SIMD vectorisation of the innermost loop:

```
s = te.create_schedule(C.op)
i, j = C.op.axis
k = C.op.reduce_axis[0]
io, ii = s[C].split(i, factor=32)
jo, ji = s[C].split(j, factor=32)
s[C].reorder(io, jo, k, ii, ji)
s[C].parallel(io)
s[C].vectorize(ji)
```

For a GPU target, the same computation is instead mapped to CUDA thread blocks and warps:

```
bx, tx = s[C].split(i, factor=16)
by, ty = s[C].split(j, factor=16)
s[C].bind(bx, te.thread_axis("blockIdx.x"))
s[C].bind(by, te.thread_axis("blockIdx.y"))
s[C].bind(tx, te.thread_axis("threadIdx.x"))
s[C].bind(ty, te.thread_axis("threadIdx.y"))
```

In both cases the computation definition is unchanged; only the schedule differs. MetaSchedule automates exactly this search: it generates candidate schedules, measures their latency on the target device, and selects the best-performing one, producing a hardware-specific binary without manual kernel engineering.

## 2.5.2 TensorRT

TensorRT [40] is NVIDIA’s inference compiler and runtime for NVIDIA GPUs. Unlike TVM, it is not hardware-portable: it specializes a model for a concrete target platform, GPU architecture, precision regime, and software stack, and produces a serialized engine encoding a fixed execution strategy. A model is imported via ONNX; the builder then applies graph-level rewrites (constant folding, layer fusion, dead-layer elimination), propagates shape and precision constraints, and empirically benchmarks multiple implementation candidates (called tactics) for each layer or fused subgraph. The best-performing tactic on the target device is embedded into the final engine. This ahead-of-time empirical optimization is the central reason for TensorRT’s strong practical performance [41].

Effective performance depends on Tensor Cores being utilizable. For large convolutions and matrix multiplications, TensorRT selects implementations backed by cuDNN

or cuBLASLt that use Tensor Core instructions when shapes, data layouts, and precision satisfy the hardware constraints. Operators not supported by TensorRT must be rewritten, decomposed, or replaced through the plugin interface. The `IPluginV3` interface allows a hand-written CUDA or CUTLASS kernel to replace a subgraph while keeping all surrounding layers in the standard TensorRT pipeline. In the typical workflow, the target subgraph is replaced in the ONNX export using ONNX GraphSurgeon, and TensorRT embeds the plugin as a first-class component during engine building. This mechanism enables targeted replacement of persistent bottlenecks without abandoning TensorRT’s graph-level optimizations for the rest of the model.

Compared with TVM, TensorRT is narrower but more mature: it prioritises deployment effectiveness on NVIDIA hardware over portability or research transparency. Compared with ONNX Runtime, it is more aggressive on supported NVIDIA targets, performing deep kernel selection and tactic benchmarking, rather than routing execution through interchangeable execution providers. Its main limitations are GPU vendor lock-in, limited introspection of tactic selection, and performance degradation at boundaries with unsupported operators where reformatting or data-type conversions break the fusion chain. In this thesis, TensorRT serves two roles: it is the strongest latency baseline on the Jetson Orin Nano GPU, and its plugin interface is used to study the overhead boundary of the standard engine by replacing the C2f subgraph with a custom kernel (Section 5.9).

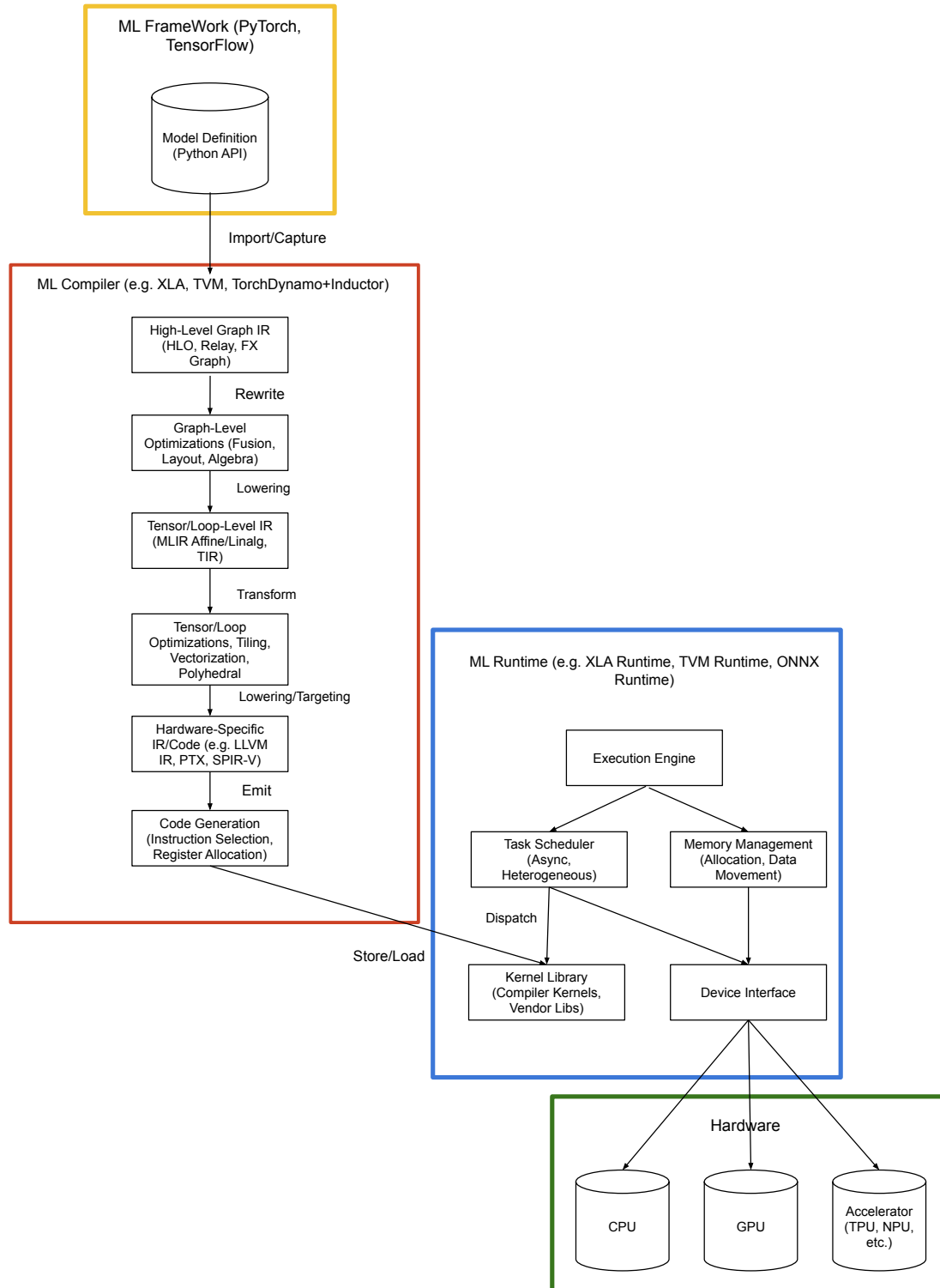


Figure 2.4: Flow of an ML model from definition through compilation and runtime execution on hardware.

## 2.6 Quantization and Mixed Precision

Quantization is one of the main techniques used to reduce the cost of neural-network inference on resource-constrained hardware. A trained neural network is usually represented using floating-point parameters and activations. Let's assume that a network has  $L$  layers with learnable parameters  $\{W_1, W_2, \dots, W_L\}$ , and let  $\theta$  denote the collection of all trainable parameters. For a supervised learning problem, the model is trained by minimizing the empirical risk

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N l(x_i, y_i; \theta), \quad (2.2)$$

where  $(x_i, y_i)$  denotes an input sample and its corresponding label,  $l(x_i, y_i; \theta)$  is the loss function, and  $N$  is the number of training samples. After training, the parameters  $\theta$  are typically stored in floating-point precision, and inference also produces intermediate activation tensors between layers.

The goal of quantization is to represent these floating-point weights and activations using lower-precision numerical formats, most commonly 8-bit integers. This can reduce model size, memory bandwidth requirements, cache pressure, and arithmetic cost, which is especially important for edge devices and accelerators with limited memory and power budgets. In practical terms, quantization defines a mapping from a real-valued tensor to a discrete set of representable values. If this mapping is chosen carefully, the model can run with substantially lower computational and memory cost while preserving most of its original accuracy.

Surveys by Gholami et al. [10] and Rokh et al. [63] provide broad taxonomies of quantization methods, including post-training quantization, quantization-aware training, uniform and non-uniform quantization, and mixed-precision approaches. Practical integer-only inference is commonly associated with the work of Jacob et al. [19], which describes how neural networks can be executed using integer arithmetic during inference. Later post-training methods, such as AdaRound [37], further improve quantized model accuracy by optimizing the rounding decisions used when mapping floating-point weights to low-precision values.

Gholami et al. [10] defines a function that can quantize NN weight and activations to a finite set of values:

$$Q(r) = \text{Int}(r/S) - Z, \quad (2.3)$$

where  $Q$  is the quantization operator,  $r$  is a real valued input (activation or weight),

$S$  is a real valued scaling factor, and  $Z$  is an integer zero point. So the function  $Q(r)$  is mapping a real value to an integer value through a rounding operation. This method is a uniform quantization, as quantization levels are uniformly spaced.

Real values  $r$  can be recovered from the quantized values  $Q(r)$  through an operation that is referred to as dequantization:

$$\tilde{r} = S(Q(r) + Z) \quad (2.4)$$

Because the operator  $\text{Int}(\cdot)$  involves rounding, the dequantized value  $\tilde{r}$  is generally only an approximation of the original floating-point value  $r$ . The difference between them is the quantization error. For uniform quantization, the quantization levels are equally spaced, so the error is controlled mainly by the scale  $S$ , the zero point  $Z$ , and the range of values that is mapped into the available integer domain. A smaller scale gives finer resolution, but only over a smaller real-valued range; a larger scale covers a wider range, but with coarser quantization steps. Thus, quantization introduces a trade-off between clipping error and rounding error.

A central design choice is the clipping range  $[\alpha, \beta]$ , which defines the range of real values represented by the quantizer. For a  $b$ -bit integer representation, a common scale choice is

$$S = \frac{\beta - \alpha}{2^b - 1}, \quad (2.5)$$

where  $\alpha$  and  $\beta$  are the lower and upper clipping bounds. Values outside this interval are clipped before quantization, while values inside the interval are rounded to the nearest available quantization level. Choosing the clipping range is often called calibration. A simple calibration method uses the minimum and maximum observed values, but this can be sensitive to outliers: a few extreme activation values may enlarge the range and reduce the effective resolution for the majority of values. Alternative calibration methods therefore use percentiles, mean-squared-error minimization, or divergence-based criteria to choose a range that better preserves the distribution of the tensor [10].

**Symmetric and asymmetric quantization** Uniform quantization can be either symmetric or asymmetric. In symmetric quantization, the clipping range is centered around zero, so  $\alpha = -\beta$ . This usually allows the zero point to be removed:

$$Q(r) = \text{Int} \left( \frac{r}{S} \right). \quad (2.6)$$

This form is attractive because it simplifies the arithmetic and is especially common

for weight quantization. For example, matrix multiplication and convolution can be implemented more efficiently when the zero point is zero, since fewer correction terms are needed during integer accumulation. However, symmetric quantization may waste representational levels when the tensor distribution is strongly skewed. This often happens for activations after ReLU, where the values are non-negative. In such cases, asymmetric quantization can use the available integer range more efficiently by allowing a non-zero zero point  $Z$  and a clipping interval that is not centered around zero. Figure 2.5 illustrates the difference between symmetric and asymmetric quantization.

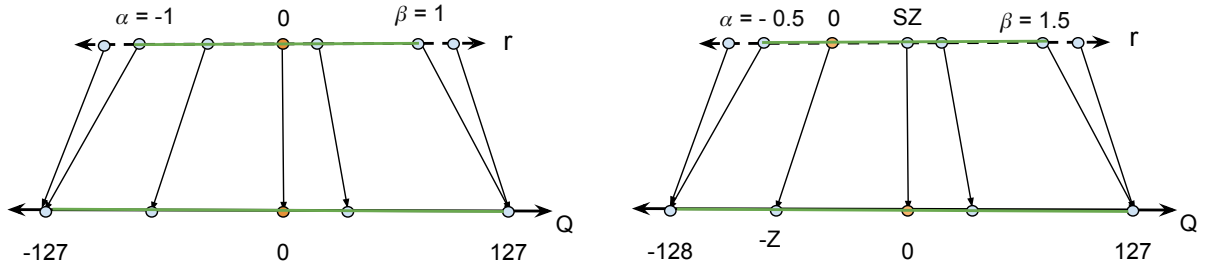


Figure 2.5: Illustration of symmetric and asymmetric quantization. In symmetric quantization, the real-valued clipping range is centered around zero, so the zero point can be set to  $Z = 0$ . In asymmetric quantization, the clipping range is shifted to better match an imbalanced tensor distribution, such as non-negative activations after ReLU. Adapted from Gholami et al. [10].

For symmetric quantization, there are also different conventions for mapping the real-valued range into the integer range. A full-range INT8 scheme uses the interval  $[-128, 127]$ , while a restricted-range scheme uses  $[-127, 127]$ . The corresponding scale can be written as

$$S_{\text{full}} = \frac{2 \max(|r|)}{2^n - 1}, \quad (2.7)$$

or, for the restricted symmetric range,

$$S_{\text{restricted}} = \frac{\max(|r|)}{2^{n-1} - 1}. \quad (2.8)$$

The full-range scheme uses one additional representable value, while the restricted scheme keeps the positive and negative ranges balanced. In practice, the best convention depends on the hardware backend, the runtime format, and the kernels used by the inference engine.

**Static and dynamic quantization** Another important distinction is when the quantization parameters are determined. For weights, the range can usually be computed statically, because trained parameters are fixed during inference. Activations are more difficult because their distributions depend on the input sample. In dynamic quantization, activation ranges are computed at runtime for each input or batch. This can improve accuracy because each tensor is quantized using its actual observed range, but it adds overhead from computing statistics such as minimum and maximum values during inference. In static quantization, activation ranges are estimated before deployment using a small calibration dataset and then kept fixed during inference. This avoids runtime calibration overhead and is therefore more attractive for low-latency deployment, but it can reduce accuracy if the calibration set does not represent the true deployment distribution well. Gholami et al. summarize this trade-off: dynamic quantization often provides better accuracy, while static quantization is more commonly used in practice because it avoids the cost of computing activation ranges during inference [10].

**Quantization granularity** The scale and zero point may be shared over different parts of a tensor. In layer-wise quantization, one scale is used for all weights in a layer. This is simple and efficient, but it can be inaccurate if different channels have very different value ranges. A channel with a narrow distribution may lose resolution because another channel with a wider distribution determines the shared scale. Channel-wise quantization assigns a separate scale to each output channel or filter, which usually improves accuracy for convolutional layers because each kernel receives its own clipping range. Group-wise quantization lies between these two extremes by assigning scales to groups of channels. The main disadvantage of finer granularity is implementation overhead: additional scales must be stored and applied, and not all hardware kernels support arbitrary granularities efficiently. For convolutional weights, channel-wise quantization is widely used because it often gives a good accuracy improvement with modest overhead, whereas very fine sub-channel quantization can be harder to deploy efficiently [10].

**Uniform and non-uniform quantization** The formulas above describe uniform quantization, where adjacent quantization levels are equally spaced. Non-uniform quantization relaxes this constraint and allows the quantization levels and thresholds to be distributed unevenly. A general form is

$$Q(r) = X_i, \quad \text{if } r \in [\Delta_i, \Delta_{i+1}), \quad (2.9)$$

where  $X_i$  is a quantization level and  $\Delta_i$  are the corresponding thresholds. Non-uniform quantization can better match non-uniform tensor distributions, for example by

allocating more levels to dense regions of the distribution and fewer levels to rarely occurring values. This can reduce quantization error for a fixed bit width. However, it is less convenient for general-purpose hardware because integer arithmetic units usually assume uniformly spaced numerical values. As a result, non-uniform quantization can require lookup tables, custom kernels, or additional decode logic. For this reason, uniform quantization remains the dominant choice in practical CPU, GPU, and edge inference pipelines, even though non-uniform methods can be attractive from an accuracy perspective [10].

A related form of non-uniform quantization is clustering-based quantization. In this approach, weights are grouped into clusters and all weights in the same cluster are represented by a shared centroid. Instead of storing every full-precision weight, the model stores a compact cluster index and a codebook of centroids. This can reduce model size and adapt the quantization levels to the actual weight distribution. However, clustering-based quantization is less attractive for efficient deployment because codebook reconstruction adds overhead and weights belonging to the same cluster are not necessarily contiguous in memory, which can lead to irregular memory accesses. For this reason, clustering-based methods are mainly used for weight compression and are less suitable for activation quantization, where values change at runtime [63].

**Post-training quantization and quantization-aware training** Quantization methods are commonly divided into post-training quantization (PTQ) and quantization-aware training (QAT). In PTQ, a pretrained floating-point model is quantized after training. The calibration data is used to estimate clipping ranges, scales, and zero points, but the model parameters are not retrained. PTQ is attractive because it is fast, easy to apply, and can often be performed with only a small calibration set. Its main limitation is that it may lead to noticeable accuracy degradation, especially at low bit widths, for sensitive layers, or when the calibration data is not representative.

QAT addresses this by simulating quantization during training or fine-tuning. The forward pass includes fake-quantization operations so that the model learns parameters that are robust to the rounding and clipping effects of low-precision inference. Since the rounding operator is non-differentiable and has zero gradient almost everywhere, QAT usually relies on the straight-through estimator (STE), which approximates the gradient of the quantizer as if it were the identity function. QAT usually gives better accuracy than PTQ, particularly for aggressive low-bit quantization, but it requires training data, additional computation, and careful hyperparameter tuning. Therefore, PTQ is often preferred for rapid deployment, while QAT is used when accuracy is critical and retraining cost is acceptable. Figure 2.6 summarizes the difference between QAT and PTQ workflows.

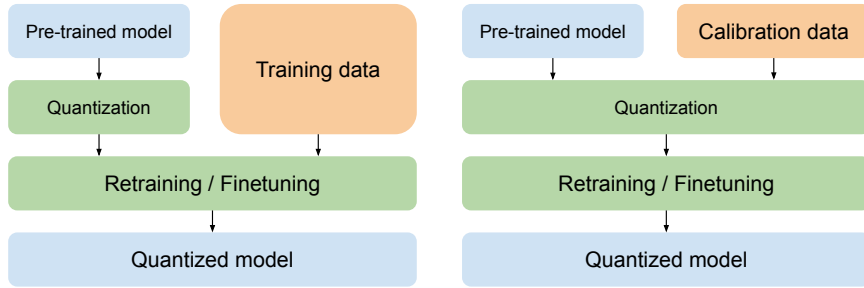


Figure 2.6: Comparison between quantization-aware training (QAT, Left) and post-training quantization (PTQ, Right). In QAT, quantization is simulated during training or fine-tuning so that the model can adapt to low-precision weights and activations. In PTQ, a pretrained model is quantized after training, typically using a calibration dataset to determine quantization ranges and scaling factors. Cropped from Gholami et al. [10].

**Simulated and integer-only quantization** It is also important to distinguish simulated quantization from integer-only inference. In simulated or fake quantization, tensors may be stored in low precision, but the actual arithmetic is still performed in floating point after dequantization. This can reduce memory footprint, but it does not fully exploit low-precision arithmetic units. In integer-only quantization, operations such as convolution and matrix multiplication are performed using integer arithmetic, often with INT8 multiplication and INT32 accumulation. This is the form of quantization most relevant for deployment on edge accelerators and mobile inference libraries, because it can reduce both memory traffic and arithmetic cost. However, integer-only inference imposes stricter constraints on operator support. Operations such as residual additions, normalization, softmax, GELU, or other nonlinear functions may require special integer approximations, rescaling steps, or fallback to floating-point execution. If unsupported operators fall back to floating point, the practical speedup can be much smaller than expected.

**Mixed precision** Uniformly assigning the same low precision to every layer is often suboptimal. Different layers have different sensitivity to quantization: some layers can be reduced to INT8, INT4, or even lower precision with little loss, while others cause large accuracy degradation if quantized too aggressively. Mixed-precision quantization assigns different bit widths to different layers or operators. Sensitive layers are kept at higher precision, while less sensitive or more expensive layers are quantized more aggressively. This can provide a better trade-off between accuracy, latency, energy, and model size.

The main difficulty is that choosing the bit width of each layer creates a large combinatorial search problem. For a network with  $L$  layers and  $k$  possible bit widths per layer, the number of possible assignments is  $k^L$ . Existing methods therefore use heuristics, sensitivity analysis, reinforcement learning, differentiable search, or hardware-aware opti-

mization to choose the precision configuration. The optimal assignment is also hardware-dependent: a lower bit width is only useful if the target device and inference backend provide efficient kernels for that precision. For example, INT8 may be well supported on CPUs and edge accelerators, while INT4 may require specialized kernels or hardware support. Thus, mixed precision should be evaluated not only by model accuracy, but also by real latency, memory bandwidth, and power consumption on the target platform.

**Summary** Quantization reduces inference cost by replacing floating-point weights and activations with lower-precision representations. Its practical effectiveness depends on several interacting design choices: the quantization function, clipping range, calibration method, symmetric or asymmetric mapping, quantization granularity, static or dynamic activation ranges, and whether inference is simulated or integer-only. PTQ provides a simple and fast deployment path, but may reduce accuracy, while QAT usually improves accuracy at the cost of retraining. Mixed precision further extends this trade-off by assigning different bit widths to different layers according to their sensitivity and hardware efficiency. Therefore, quantization should be understood not only as a numerical compression method, but also as a hardware-aware deployment technique whose benefits depend strongly on compiler support, backend kernels, and the target device.

## 2.7 Pruning and Structural Simplification

Pruning is a model-compression technique that reduces the number of parameters, filters, channels, or other computational structures in a neural network after or during training. The basic motivation is that modern deep neural networks are often over-parameterized: many weights or channels can be removed with little effect on accuracy. A smaller network can reduce memory usage, arithmetic cost, and energy consumption, which is especially relevant for deployment on edge devices. However, the practical benefit of pruning depends strongly on the type of sparsity introduced. Removing individual weights can produce a highly sparse model, but such unstructured sparsity is difficult to accelerate on general-purpose CPUs and GPUs without specialized sparse kernels or hardware support. In contrast, structured pruning removes entire filters, channels, neurons, or blocks, producing dense smaller tensors that are easier for existing compilers and inference runtimes to exploit.

Vadera and Ameen [70] categorize studies on neural-network pruning into eight broad groups: magnitude-based pruning, similarity and clustering methods, sensitivity analysis methods, knowledge distillation, low-rank methods, quantization methods, architectural design methods, and hybrid methods. In a strict sense, not all of these are pruning methods. Knowledge distillation, low-rank approximation, quantization, and architectural

search are better understood as related model-compression or model-efficiency techniques. Nevertheless, they are often discussed together with pruning because they pursue the same goal: reducing the memory, latency, or energy cost of neural-network deployment.

**Magnitude-based pruning** Magnitude-based pruning removes parameters or structures according to simple local measures, most commonly the absolute value or norm of weights, filters, or activations. The intuition is that weights or filters with small magnitude contribute less to the network output and can therefore be removed. Early work by Han et al. [14] demonstrated that iterative magnitude pruning followed by fine-tuning can remove a large fraction of weights from networks such as LeNet, AlexNet, and VGG without significant loss in accuracy. Their results also showed two important practical lessons: iterative pruning is usually more effective than one-shot pruning, and fine-tuning the remaining weights is preferable to simply reinitializing the pruned network.

Later work observed that pure magnitude pruning can remove weights too early. Guo et al. introduced Dynamic Network Surgery [13], where pruned weights are not permanently discarded but can be restored if they later become important. This addresses the problem that the importance of a weight can change after other weights have been removed.

Magnitude-based pruning also extends naturally to structured pruning. Instead of removing individual weights, a method can remove filters or channels whose norms are small. For example, Li et al. [27] prune convolutional filters based on the sum of absolute filter weights. This form of pruning is more deployment-friendly than unstructured weight pruning because removing an entire filter or channel reduces tensor dimensions and produces a smaller dense network. Such structural changes are more likely to translate into real speedups in standard deep-learning frameworks and compiler backends.

**The lottery ticket hypothesis** The Lottery Ticket Hypothesis introduced by Frankle and Carbin [7] reframed pruning as a way to identify trainable subnetworks inside a larger network. The hypothesis states that a randomly initialized dense network contains a sparse subnetwork, a “winning ticket”, that can be trained to match the accuracy of the original network in no more training iterations than the full model. Their procedure uses magnitude pruning to identify the subnetwork, resets the remaining weights to their initial values, and retrain.

This work is important because it suggests that pruning is not only a post-hoc compression technique, but also a tool for studying over-parameterization and trainability. However, later studies showed that the behavior depends on model size, dataset, pruning level, and whether the remaining weights are reset to their original initialization or to

a later training checkpoint. For larger networks and harder datasets, resetting to later training iterations can be more stable than resetting to the initial weights [8]. From a deployment perspective, the lottery-ticket literature is conceptually important, but its direct usefulness is limited unless the resulting sparse subnetworks can be efficiently executed.

**Structured channel and filter pruning** Structured pruning removes complete channels, filters, neurons, or blocks. This is often more important for real inference acceleration than unstructured sparsity. If an entire output channel is removed from a convolutional layer, the corresponding input channel in the following layer can also be removed. This reduces both the number of parameters and the number of dense multiply-accumulate operations.

Vadera and Ameen [70] identify three main directions in structured channel and filter pruning: data-dependent methods, data-independent methods, and optimization-based channel approximation methods.

**Data-dependent channel pruning** Data-dependent methods estimate the usefulness of a channel by observing its behavior on a sample of input data. The underlying assumption is that an informative channel should respond differently to different inputs. Polyak and Wolf [50] proposed pruning methods based on feature-map variance. Channels or filters whose activations vary little across samples are considered less informative and can be pruned. Their Reduce and Reuse method also addresses a key difficulty in output-channel pruning: removed channels may still be expected by subsequent layers. To compensate, removed channels are approximated as linear combinations of the remaining channels, implemented using an additional  $1 \times 1$  convolution.

Entropy-based pruning follows a similar idea but measures the information content of feature maps rather than their variance. If a feature map has low entropy, it is assumed to carry less discriminative information and its corresponding filter may be removed. Such methods can be more meaningful than pure magnitude criteria, but they require representative data and additional forward passes to estimate channel statistics.

**Data-independent channel pruning** Data-independent methods use properties of the learned weights or activations without explicitly solving a reconstruction problem. Hu et al. [17] introduced Average Percentage of Zeros (APoZ), where filters producing many zero activations are considered weak and are removed. This is motivated by ReLU-based networks, where inactive channels may contribute little to the final prediction. After pruning, the network is fine-tuned to recover accuracy.

Another simple data-independent approach is filter-norm pruning, where filters with

small  $\ell_1$  or  $\ell_2$  norms are removed. These methods are attractive because they are cheap and easy to implement. Their limitation is that small magnitude does not always imply low importance: a filter can have small weights but still be useful, while a large filter may be redundant if its function is duplicated by other filters.

**Optimization-based channel approximation** Optimization-based channel pruning methods remove channels while explicitly trying to preserve the layer output. Rather than asking which channels are individually small or inactive, these methods ask which subset of channels best reconstructs the original feature maps. ThiNet [31], for example, formulates channel selection as an optimization problem and uses a greedy strategy to choose channels whose removal minimally affects the next layer. After the subset of channels is chosen, the remaining weights are re-optimized using least squares.

Related methods use LASSO regression or other approximations to select channels and refit the remaining weights. These approaches are often more accurate than simple magnitude pruning because they account for the interaction between channels. Their disadvantage is additional computational cost: estimating reconstruction error and solving optimization problems requires data samples and extra pruning-time computation.

**Similarity and clustering methods** Similarity and clustering methods remove redundant structures rather than individually weak ones. The main assumption is that over-parameterized networks may contain filters or channels that perform similar functions. If two filters are highly similar, one may be removed or replaced with an approximation based on the other. Representative approaches include geometric median filter pruning [15], ThiNet [31], and FilterSketch [29].

Geometric median pruning removes filters that are close to the geometric median of the filter set, because such filters are assumed to contain information already represented by other filters. FilterSketch instead aims to reduce the number of filters while preserving covariance information among them. These methods are useful because they target redundancy directly. However, their success depends on the assumption that similarity in weight or feature space corresponds to functional redundancy, which may not always hold across architectures and tasks.

**Sensitivity-analysis methods** Sensitivity-analysis methods attempt to estimate the effect of removing a parameter, channel, or filter on the loss function. These methods are more directly tied to the learning objective than simple magnitude criteria. Classical examples include Optimal Brain Damage and Optimal Brain Surgeon, which use second-order Taylor approximations of the loss to estimate weight saliency. More recent methods

use first-order or second-order approximations to prune channels and feature maps in convolutional networks.

Taylor pruning [36] estimates the change in loss caused by removing a feature map using a first-order Taylor approximation. SNIP [25] performs single-shot pruning before training by estimating the saliency of each connection at initialization. GraSP [71] instead aims to preserve gradient flow after pruning. These methods can achieve strong accuracy–compression trade-offs because they prune according to estimated task sensitivity. However, they are often more expensive than magnitude-based pruning and may require gradient computation, data samples, or careful tuning.

**Knowledge distillation and pruning** Knowledge distillation is not pruning in the strict sense, because it does not necessarily remove weights from an existing model. Instead, a large teacher model transfers predictive behavior to a smaller student model [16]. The student may be a manually designed compact architecture, a pruned network, or a structurally simplified model. Distillation is especially useful after structured pruning, because the pruned model can be trained to match the outputs or intermediate representations of the original model. Methods such as discrimination-aware channel pruning [75] combine pruning with auxiliary losses to preserve discriminative features.

The advantage of distillation is that it can recover accuracy that would otherwise be lost during compression. The limitation is that it introduces additional training complexity and usually requires access to training data and a strong teacher model. Therefore, distillation is powerful when accuracy is critical, but less convenient for fast, hardware-agnostic compression.

**Low-rank approximation, quantization, and architectural search** Low-rank approximation, quantization, and architectural search are related to pruning but should be treated as broader model-efficiency techniques. Low-rank methods factor large weight tensors or matrices into smaller components, reducing computation in convolution-heavy models. Their effectiveness depends on whether the learned weights admit a good low-rank approximation. At high compression levels, accuracy may degrade quickly.

Quantization reduces numerical precision rather than removing structures. It is commonly combined with pruning because the two methods attack different sources of cost: pruning reduces the number of parameters or operations, while quantization reduces the cost of storing and executing the remaining ones. Quantization-aware training [19] and binary or ternary networks such as XNOR-Net [59] are therefore better discussed as complementary compression methods rather than pruning methods.

Architectural design methods, including NAS- and reinforcement-learning-based ap-

proaches such as AMC, AutoSlim, and Once-for-All, search directly for efficient architectures. These methods may produce networks that resemble pruned architectures, but they differ from classical pruning because they optimize the architecture itself rather than removing parts from a fixed trained network.

**Deployment considerations** The main practical distinction is between unstructured and structured pruning. Unstructured pruning can remove a very large number of individual weights and therefore achieves high nominal compression. However, unless the target hardware, compiler, and runtime support sparse tensor operations efficiently, the model may not become faster. Sparse weight matrices can introduce irregular memory access, index-storage overhead, and poor utilization of dense SIMD, GPU, or NPU execution units.

Structured pruning is usually more useful for deployment because it changes tensor shapes and reduces dense computation. Removing channels, filters, or blocks can reduce FLOPs and memory traffic in a way that standard inference engines can exploit. This makes structured pruning more relevant for compiler-aware model optimization, where the goal is not only to reduce the number of parameters, but to produce a model that maps efficiently to the target backend. For edge deployment, pruning should therefore be evaluated using real latency, memory consumption, and power measurements, not only parameter count or theoretical FLOPs.

**Summary** Pruning methods differ in what they remove and how they estimate importance. Magnitude-based methods are simple and cheap, but may remove parameters that are locally small yet globally important. Sensitivity-analysis methods are more task-aware but computationally more expensive. Similarity and clustering methods target redundancy between filters or channels, while optimization-based methods explicitly preserve layer outputs after channel removal. For practical deployment, structured pruning is generally more useful than unstructured pruning because it produces smaller dense models that are easier for existing compilers, libraries, and hardware backends to accelerate. In the context of this thesis, pruning is therefore most relevant when it leads to structural simplification of the model, such as reducing channels, filters, or blocks in a way that improves actual inference latency and energy efficiency on edge hardware.

# 3 Theory

## 3.1 Metrics Description

To evaluate the performance of object detection models, a set of standard metrics is used throughout this report. These metrics assess both localization accuracy and classification quality, providing a comprehensive view of model behavior.

- **Intersection over Union (IoU):**

For a predicted bounding box  $p$  and a ground-truth box  $g$ , the Intersection over Union is defined as:

$$\text{IoU}(p, g) = \frac{|p \cap g|}{|p \cup g|}.$$

IoU measures localization accuracy by quantifying the overlap between predicted and ground-truth boxes. It is computed as the ratio of the intersection area to the union area of the two boxes, resulting in a value between 0 and 1. A score of 1 indicates perfect alignment, while 0 indicates no overlap.

In practice, an *IoU threshold* (e.g., 0.5) is used to determine whether a prediction is correct. Predictions with  $\text{IoU} \geq \tau$  are considered true positives (TP), while those below the threshold are treated as false positives (FP). IoU serves as a fundamental building block for higher-level evaluation metrics such as precision, recall, and mAP.

- **Precision and Recall:**

Precision and recall evaluate the quality of classification and detection results. Precision measures the proportion of correct positive predictions:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}.$$

Recall measures the proportion of correctly detected instances among all ground-truth objects:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

Here, TP, FP, TN, and FN denote true positive, false positive, true negative, and false negative predictions, respectively. Precision reflects the ability to avoid false detections, while recall captures the ability to detect all relevant objects.

- **Average Precision (AP):**

Average Precision summarizes the precision-recall trade-off by computing the area

under the precision-recall (P–R) curve [12]:

$$\text{AP}_\tau = \int_0^1 p(r; \tau) dr.$$

The P–R curve is constructed by varying the detection confidence threshold:

- Each detection is assigned a confidence score  $s \in [0, 1]$ .
- Detections are sorted in descending order of confidence.
- Predictions are progressively included, and precision and recall are recomputed.
- This process generates the precision-recall curve.

AP corresponds to the area under this curve and provides a single scalar measure of detection performance.

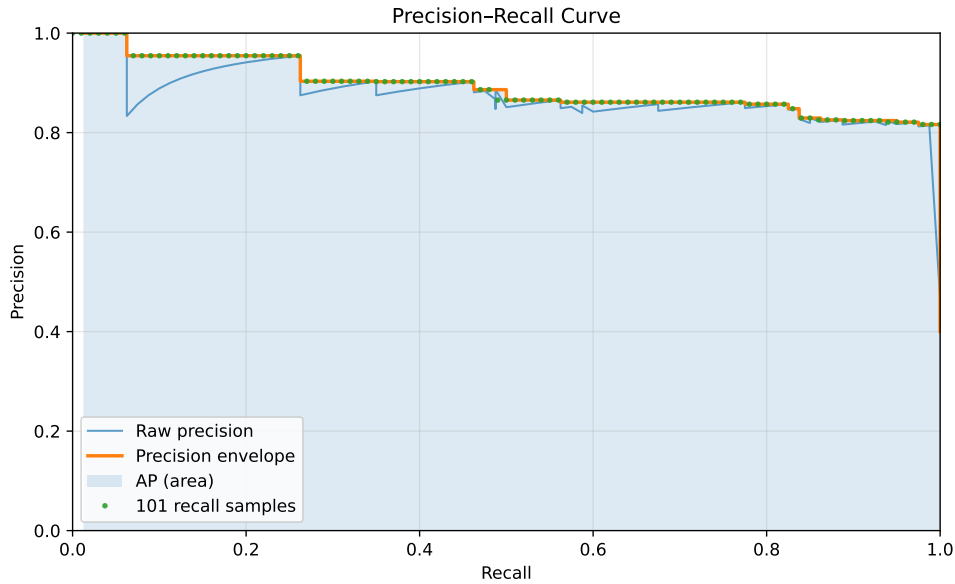


Figure 3.1: Precision–Recall curve.

• **Mean Average Precision (mAP):**

Mean Average Precision extends AP to multi-class settings by averaging AP over all object classes:

$$\text{mAP}_{0.50} = \frac{1}{C} \sum_{c=1}^C \text{AP}_{c, 0.50},$$

$$\text{mAP}_{[0.50:0.95]} = \frac{1}{C} \sum_{c=1}^C \frac{1}{10} \sum_{k=0}^9 \text{AP}_{c, (0.50+0.05k)}.$$

Here,  $C$  denotes the number of classes. The notation indicates the IoU threshold(s) used to determine true positives. For example,  $\text{mAP}_{0.50}$  considers predictions with

$\text{IoU} \geq 0.5$ , while  $\text{mAP}_{[0.50:0.95]}$  averages performance across multiple thresholds, providing a more comprehensive evaluation.

## 3.2 Roofline Model Concept

The Roofline model, introduced by Williams et al. [72], provides an intuitive way to analyze whether a workload is limited by memory bandwidth or computational throughput. It plots performance (e.g., GigaFlops per second (GFLOP/s)) versus arithmetic intensity (FLOPs per byte of memory traffic). The model defines two limits: a sloped memory roof determined by peak memory bandwidth, and a flat compute roof determined by peak floating-point throughput. Any kernel's performance is bounded by the minimum of these two ceilings. If a point lies on the sloped region, it is memory-bound; if it lies on the flat region, it is compute-bound. The Roofline model is widely used to analyze whether a workload is limited by memory movement or computation and to guide optimization efforts.

Optimization strategy of a neural network model is dependent on whether our device limits us in the speed of reading/writing to/from memory or the computational speed overall.

- **If the workload is Memory-Bound:**

This means performance is limited by memory bandwidth, not by compute units. The arithmetic intensity (FLOPs/byte) is low, so the kernel spends most of its time moving data. The goal is then to either increase arithmetic intensity, or reduce memory traffic.

**Optimization strategies:**

- Improve data locality: reuse data in caches (tiling/blocking); Keep tensors in shared memory (GPU);
- Fuse operations to avoid writing intermediate tensors to memory (Conv  $\rightarrow$  BatchNorm  $\rightarrow$  Silu)  $\rightarrow$  (Fused Conv-BN-Activation)
- Reduce Precision (FP32  $\rightarrow$  FP16  $\rightarrow$  INT8  $\rightarrow$  INT4)
- Reduce tensor size: pruning, structured sparsity, channel reduction, smaller input resolution
- Improve memory layout: avoid unnecessary copies, channel-last format (NHWC)

- **If the workload is Compute-Bound:**

This means arithmetic units are saturated. Arithmetic intensity is high enough that bandwidth is no longer limiting. The goal is then to increase compute efficiency.

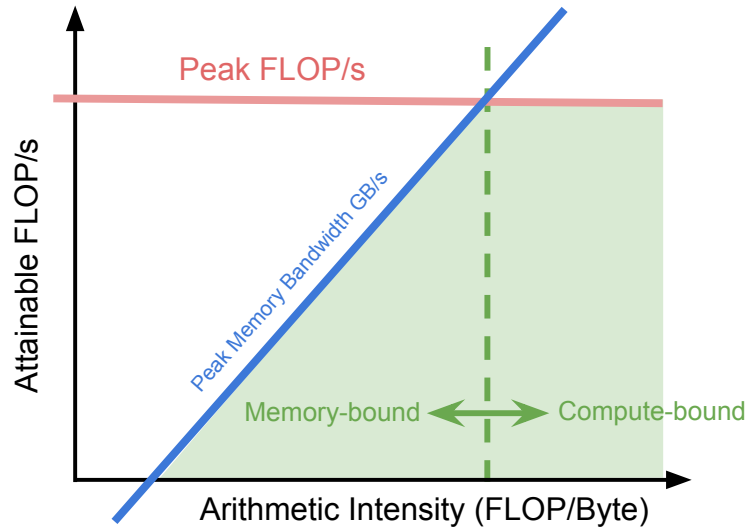


Figure 3.2: A Roofline model illustrating performance limits based on compute capability (horizontal line) and memory bandwidth (sloped line) versus the arithmetic intensity of operations (dots). Operations below the “roof” are limited by either memory bandwidth or compute peak.

This case is harder, as it requires the work with scheduling, parallelization, kernel optimization and algorithmic improvements.

- **Peak computational performance (in FLOP/s)** is obtained by benchmarking a compute-bound kernel that saturates the floating-point execution units. In practice, this can be achieved using large dense matrix multiplications of size  $m \times k$  and  $k \times n$ , for which the floating-point operation count is given by:

$$\text{FLOPs} = 2 \times m \times k \times n. \quad (3.1)$$

The sustained peak performance is then computed as

$$P_{\text{peak}} = \frac{\text{FLOPs}}{t_{\text{compute}}}, \quad (3.2)$$

where  $t_{\text{compute}}$  denotes the measured execution time.

- **Peak memory bandwidth (in GB/s)** is determined using a streaming memory benchmark. A large device-to-device copy or STREAM-like triad kernel is executed over a sufficiently large buffer to avoid cache effects. The sustained memory

bandwidth is computed as

$$B_{\text{peak}} = \frac{\text{Bytes transferred}}{t_{\text{memory}}}. \quad (3.3)$$

- **Arithmetic intensity (AI)** is defined as the ratio between the number of floating-point operations and the volume of data transferred to and from main memory:

$$\text{AI} = \frac{\text{FLOPs}}{\text{Bytes}}. \quad (3.4)$$

Since exact Dynamic Random-Access Memory (DRAM) traffic is generally unavailable without hardware performance counters, a lower-bound estimate of memory traffic is commonly used. For convolutional or linear layers, this lower bound consists of the input tensor, weight parameters, and output tensor:

$$\text{Bytes}_{\text{LB}} = \text{Bytes}_{\text{input}} + \text{Bytes}_{\text{weights}} + \text{Bytes}_{\text{output}}. \quad (3.5)$$

This estimate ignores cache reuse and intermediate buffering, thereby providing an upper bound on arithmetic intensity.

- **The achieved performance of a workload** is computed as

$$P_{\text{achieved}} = \frac{\text{FLOPs}}{t_{\text{execution}}}. \quad (3.6)$$

The Roofline bound is then defined as

$$P_{\text{roof}}(\text{AI}) = \min(P_{\text{peak}}, B_{\text{peak}} \times \text{AI}). \quad (3.7)$$

If  $P_{\text{achieved}}$  approaches  $B_{\text{peak}} \times \text{AI}$ , the workload is memory-bound, if it approaches  $P_{\text{peak}}$ , the workload is compute-bound. This methodology enables systematic classification of performance bottlenecks and guides optimization strategies such as improving data locality, increasing arithmetic intensity, or enhancing computational parallelism.

### 3.3 Activation Functions: ReLU and SiLU

Activation functions play a critical role in introducing non-linearity into neural networks, enabling them to learn complex representations. In this work, we consider two commonly used activation functions: Rectified Linear Unit (ReLU) and Sigmoid Linear Unit (SiLU).

The ReLU activation function was introduced by [38] and is widely used due to its computational simplicity. More recently, smooth activation functions such as SiLU (or Swish) [55] have been proposed, offering improved accuracy at the cost of increased computational complexity.

The ReLU activation is defined as:

$$\text{ReLU}(x) = \max(0, x) \quad (3.8)$$

ReLU is computationally simple, requiring only a comparison and selection operation. This makes it highly efficient on modern hardware, as it can be implemented with minimal arithmetic and memory overhead. Due to its simplicity, ReLU is widely used in performance-critical applications and is well-supported by hardware accelerators.

In contrast, the SiLU activation (also known as the Swish function) is defined as:

$$\text{SiLU}(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}} \quad (3.9)$$

where  $\sigma(x)$  denotes the sigmoid function. Compared to ReLU, SiLU introduces additional non-linearity and has been shown to improve model accuracy in many cases. However, it is computationally more expensive, as it requires evaluation of the exponential function, a division, and a multiplication.

From a computational perspective, the key difference between the two functions lies in their complexity. ReLU involves only a simple thresholding operation, which is typically memory-bound and incurs minimal computational cost. In contrast, SiLU requires multiple floating-point operations, including transcendental functions, making it more compute-intensive and potentially slower on hardware with limited support for such operations.

This difference has practical implications for model optimization. Replacing SiLU with ReLU can reduce inference latency and improve hardware efficiency, especially on edge devices. However, this simplification may lead to a degradation in model accuracy. Therefore, the choice of activation function represents a trade-off between computational efficiency and predictive performance.

While smooth activation functions such as SiLU have been shown to improve accuracy, recent work suggests that simpler functions like ReLU can offer advantages in terms of computational efficiency and activation sparsity [34]. This sparsity can be exploited by hardware and compilers to reduce the number of effective computations, leading to improved inference performance.

## 4 Methodology

This chapter describes the experimental design and procedures used to evaluate inference optimization of the YOLOv8n object detection model on two edge hardware platforms: the NVIDIA Jetson Orin Nano (GPU-class edge accelerator) and the Raspberry Pi 5 (ARM CPU-class single-board computer). The central research question is how different compiler-aware optimization strategies, spanning framework selection, numerical precision, structural model simplification, and low-level kernel customization, affect inference latency, energy consumption, and detection accuracy on resource-constrained hardware.

The chapter covers model selection and justification, hardware and software platform configuration, baseline establishment, bottleneck characterization via Roofline analysis and profiling, and the design of each optimization experiment. The evaluated backends include TensorRT, Apache TVM with MetaSchedule auto-tuning, ONNX Runtime, and LiteRT (formerly TensorFlow Lite), applied at FP32, FP16, and INT8 precision where supported by the target platform.

### 4.1 Specification of the Scientific Work

The Experimental Evaluation chapter is organized as a progressive sequence of optimization steps applied to the YOLOv8n model across two target platforms: the Jetson Orin Nano GPU and the Raspberry Pi 5 CPU.

The experiments begin with a Roofline analysis, which characterizes the computational and memory bandwidth limits of each platform and determines whether the unoptimized model is bottlenecked by arithmetic throughput or memory access. This establishes a theoretical bound on achievable inference performance and motivates the subsequent optimization directions.

The second stage compares inference compiler performance in FP32 precision. TensorRT, ONNX Runtime, and Apache TVM are evaluated on both platforms. For TVM, this includes a full MetaSchedule auto-tuning campaign to ensure that measured latency reflects hardware-optimized kernel implementations rather than generic fallback schedules. Nsight Systems profiling is used to decompose GPU execution time by kernel category, revealing which compilation and scheduling strategies account for performance differences between frameworks.

Precision reduction is then evaluated as a standalone optimization axis. FP16 and INT8 inference are measured for all applicable backends, with INT8 quantization performed using post-training calibration on a representative subset of the COCO training data. Results are reported not only in terms of latency but also in terms of detection accuracy (mAP) and power consumption, capturing the full accuracy–efficiency–energy trade-off.

The fourth stage evaluates structural pruning of the model. A channel-level sensitivity analysis is performed to determine which layers tolerate channel reduction without disproportionate accuracy loss, followed by a full-model pruning experiment to quantify the achievable speedup from structural simplification.

The fifth stage investigates whether manually replacing a bottleneck subgraph with a custom TensorRT plugin can reduce the overhead identified during profiling. The C2f block at model layer 2 is reimplemented as a hand-written cuDNN-based plugin, and the resulting latency and kernel execution profile are compared against the standard TensorRT engine.

Finally, the effect of activation function choice is examined. SiLU activations are replaced with ReLU across the full model, and the impact on fusion quality, latency, and detection accuracy is evaluated across all backends and both platforms.

## 4.2 Model Selection

YOLOv8n is selected as the reference model for this study. The model represents a practical and relevant target for edge deployment because it combines a compact architecture with competitive detection accuracy. It is widely used in real-world object detection tasks, is actively supported by the research and engineering community, and can be exported to multiple deployment backends. These properties make it suitable for comparative analysis across inference frameworks.

From a methodological perspective, YOLOv8n is large enough to expose non-trivial optimization challenges, including operator fusion, memory movement overhead, kernel launch inefficiencies, and sensitivity to reduced precision. At the same time, it remains sufficiently lightweight to allow repeated experimentation, retraining, export, and validation on commonly available development hardware. This enables a controlled and reproducible study of both high-level and low-level optimization strategies.

## 4.3 Overall Experimental Procedure

The experimental workflow follows five main stages:

1. Establish a baseline using the original FP32 PyTorch model.
2. Identify performance bottlenecks through roofline analysis and runtime profiling.
3. Apply optimization techniques at the framework, numerical, and architectural levels.
4. Evaluate the optimized models on target hardware using identical benchmarking conditions.
5. Compare the resulting variants in terms of latency, throughput, and accuracy.

This procedure ensures that each optimization step is assessed relative to a common reference and that conclusions are based on measurable changes in both performance and predictive quality.

## 4.4 Baseline Establishment

The starting point of the study is the pretrained FP32 YOLOv8n model provided by Ultralytics and executed in the original PyTorch runtime. This baseline serves two purposes. First, it provides a reference for all later comparisons. Second, it makes it possible to determine whether the observed performance is already close to the limits of the target hardware or whether substantial optimization opportunities remain.

The baseline is evaluated on the selected hardware platforms under controlled conditions. For each platform, inference latency and throughput are measured using repeated runs with warm-up iterations to reduce initialization effects. Accuracy is evaluated on the validation dataset using standard object detection metrics. These baseline results form the reference against which all optimized variants are compared.

## 4.5 Bottleneck Analysis

After establishing the baseline, the next step is to determine the dominant performance limitations of the model on each platform. Two complementary approaches are used for this purpose: roofline-based reasoning and runtime profiling.

The roofline model is used as a coarse analytical framework to determine whether performance is likely to be limited by memory bandwidth or by available compute throughput. This provides a hardware-aware interpretation of the measured performance and helps identify whether the main limitation arises from insufficient arithmetic intensity, inefficient memory access, or underutilization of the compute units.

To refine this analysis, profiling tools are used to inspect the execution of individual operators and kernels. This makes it possible to identify layers with high latency, excessive memory movement, frequent kernel launches, or inefficient execution patterns. In particular, the profiling stage is used to answer the following questions:

- Which layers dominate end-to-end inference time?
- Whether the execution is primarily memory-bound or compute-bound.
- Whether the framework introduces avoidable overhead through layout conversions, intermediate buffers, or host-device synchronization.
- Whether operator fusion and backend-specific optimizations are effectively applied.

The outcome of this stage determines which optimization direction is most relevant for a given backend and hardware target.

## 4.6 Optimization Directions

Based on the bottleneck analysis, the study investigates several classes of optimization methods.

### 4.6.1 Compiler and Runtime Optimization

The first optimization direction consists of evaluating how different inference frameworks transform and execute the same model. Starting from the original PyTorch implementation, the model is exported or compiled for alternative backends such as TensorRT and TVM. The purpose of this comparison is to assess how much performance can be gained from compiler-level graph transformations, backend-specific kernel selection, operator fusion, and scheduling optimizations.

This stage is not limited to reporting end-to-end latency. It also examines how the internal execution differs between backends, for example in the number of kernels launched, the use of fused operators, the presence of data layout transformations, and the distribution of execution time across layers. In this way, the comparison provides insight not only into which backend is faster, but also into why it is faster.

### 4.6.2 Numerical Precision Reduction

The second optimization direction focuses on reducing numerical precision in order to decrease memory traffic and accelerate arithmetic operations. Two forms of precision reduction are considered in this work.

First, inference in FP16 is evaluated against the FP32 baseline. This comparison isolates the effect of reduced floating-point precision and shows whether the target hardware benefits from half-precision execution. This step is performed through ONNX Framework. It provides a dedicated conversion utility, that transforms a floating-point model into a half-precision one without retraining. The converter traverses every node and tensor initializer in the graph: weight tensors stored as `float32` NumPy arrays are recast to `float16` in place, and constant nodes are rewritten to carry 16-bit values. Where the original graph boundary expects 32-bit input or produces 32-bit output, explicit `Cast` nodes are inserted to handle the precision transition transparently. Certain operators whose numerical stability is sensitive to reduced range, such as `LayerNormalization` or some reduction ops, can be placed on a blocklist; the converter keeps those nodes in `float32` and surrounds them with `Cast` nodes so the rest of the graph remains in FP16. The result is a structurally identical ONNX graph in which most weights, activations, and intermediate tensors occupy half the memory and can be executed with FP16 arithmetic on any backend that supports it, including TensorRT and ONNX Runtime with CUDA execution provider.

Second, quantization to INT8 is explored where supported by the backend. Quantization has the potential to reduce model size, memory bandwidth requirements, and arithmetic cost, but it may also introduce accuracy degradation. Therefore, each quantized model is evaluated not only in terms of runtime performance but also in terms of detection quality. The purpose of this stage is to determine whether the performance gains justify the loss, if any, in predictive accuracy. Quantization is explored via framework, which actively support it. Reducing precision from `float32` to `int8` is more involved than the FP16 cast because 8-bit integers cannot represent the same dynamic range as floating-point values. The mapping requires a learned scale factor and a zero point per tensor, so that a floating-point value is approximated as:

$$x \approx s \cdot (x_{\text{int8}} - z), \quad x_{\text{int8}} = \text{clamp}\left(\left\lfloor \frac{x}{s} \right\rfloor + z, -128, 127\right) \quad (4.1)$$

The scale and zero point are not free parameters, and they must be determined from the actual distribution of values the tensor takes during inference. This calibration step was performed by gathering and running a small representative dataset through the unquantized network and collecting per-tensor activation statistics. Weights are quantized

per-channel (one scale per output channel) to preserve accuracy; activations are quantized per-tensor for runtime efficiency, but it can vary by the choice made by framework developers and chosen setting. The calibration dataset was drawn from the COCO training split, the same distribution on which YOLOv8n was originally trained. Three subset sizes were evaluated: 0.1%, 1%, and 10% of the training set, corresponding to approximately 118, 1,180, and 11,800 images respectively. The 1% subset was used as the primary calibration set throughout the report, in line with framework documentation recommendations and confirmed by preliminary accuracy tests. Subset size mainly affects the accuracy of the post-quantization network: a larger set provides a more faithful estimate of the activation distribution, while a smaller set reduces calibration time with diminishing accuracy cost beyond a certain threshold. The 1% subset was found to represent the activation distribution sufficiently for the purposes of this evaluation.

### 4.6.3 Model Compression and Structural Simplification

The third optimization direction targets the structure of the model itself. Rather than relying only on framework-level improvements, this part of the study investigates whether the network can be simplified in ways that improve runtime characteristics.

Pruning approach used in experiments was implemented in `manual_yolov8_pruner.py`, it performs targeted structural pruning directly on the YOLOv8n PyTorch model graph. The script accepts an explicit list of layer targets (e.g. `model.2.cv1`, `model.4.m.0.cv2`) and applies output-channel pruning to each nominated convolutional block. Channel importance is estimated by the L2 norm of each output filter:

$$s_k = \|W_{k,:,:,}\|_2, \quad k = 1, \dots, C_{\text{out}} \quad (4.2)$$

where  $W_k$  is the  $k$ -th output filter of the convolution weight tensor. The top- $\lfloor (1-r) \cdot C_{\text{out}} \rfloor$  channels ranked by this score are retained, with  $r$  being the per-block prune ratio and an optional rounding constraint (e.g. to a multiple of 8 for hardware alignment). After each selected block is pruned, the script performs a full graph realignment pass: it propagates the updated channel indices through all downstream consumers, slicing input weight matrices accordingly and handling Concat, Upsample, and Detect nodes transparently. Special handlers manage the internal structure of C2f blocks (cv1 hidden dimension, bottleneck cv1 and cv2 paths) and the SPPF hidden channels. The pruned model is saved as a `.pt` checkpoint suitable for fine-tuning and subsequent ONNX export. The other researched optimization is a replacement of computationally expensive operations with simpler alternatives. In particular, the effect of replacing SiLU activations with ReLU is evaluated. This substitution is motivated by the fact that simpler activations may improve fusion opportunities and reduce kernel complexity, even if they slightly affect model

accuracy. The SiLU activation substitution is applied directly to the PyTorch module tree before the model is exported to ONNX. Every `Conv` wrapper in the Ultralytics model carries an `.act` attribute; by iterating over all named modules and replacing each `nn.SiLU()` instance with `nn.ReLU()`, the entire network is converted in-place without altering any weight tensors. The modified model is then saved as a `.pt` file and passed to the standard Ultralytics `model.export(format="onnx")` pipeline. Because the substitution happens at the PyTorch level, the resulting ONNX graph contains `Relu` nodes in all positions where `Sigmoid-Mul` pairs (the ONNX decomposition of SiLU) would otherwise appear, allowing downstream compilers and runtimes to fuse the activation with the preceding convolution more readily.

The purpose of this stage is to understand whether modest architectural simplifications can produce a favorable trade-off between accuracy and efficiency, especially on hardware with limited compute capability or reduced support for complex fused kernels.

#### 4.6.4 Low-Level TensorRT Customization

The final optimization direction extends beyond standard framework usage and examines low-level customization within TensorRT. In this stage, custom plugins are developed to replace selected subgraphs with tailored implementations. The main case considered in this work is a custom plugin for the C2f block.

The objective of this step is to investigate whether performance can be improved further by reducing backend overhead, controlling kernel behavior more explicitly, or eliminating inefficient execution patterns that are not resolved by the default TensorRT conversion pipeline. This stage complements the framework-level comparison by showing the extent to which manual low-level intervention can outperform standard automated optimization.

This optimization is discussed in more details in the corresponding results section.

### 4.7 Evaluation Criteria

All model variants are evaluated using the same set of criteria in order to ensure consistent comparison.

The primary performance metrics are inference latency and throughput. Latency is used to measure the response time of a single inference pass, while throughput indicates the sustained processing rate under repeated execution. Where relevant, profiling data is also used to analyze kernel-level execution time and the contribution of individual layers to the total runtime.

The primary quality metric is object detection accuracy on the validation dataset. Standard detection metrics are used to verify whether a given optimization preserves acceptable predictive performance. This is particularly important for quantization, pruning, and activation replacement, where improvements in runtime may come at the cost of degraded detection quality.

The final analysis therefore considers both efficiency and accuracy, with emphasis on the trade-off between them. An optimization is not treated as beneficial solely because it accelerates execution; it must also maintain sufficient model quality for practical deployment.

## 4.8 Methodological Summary

The methodology combines baseline benchmarking, bottleneck analysis, framework comparison, numerical optimization, structural simplification, and low-level backend customization. The goal is not only to identify the fastest deployment configuration for YOLOv8n, but also to understand which classes of optimization are most effective on edge-oriented hardware and under which conditions they provide measurable benefit.

The full implementation, including pruning scripts, quantization pipelines, TVM compilation workflows, and the custom TensorRT plugin, is available as an open-source repository [5].

Figure 4.1 summarizes the scope of the study and the main experimental directions considered in this thesis.

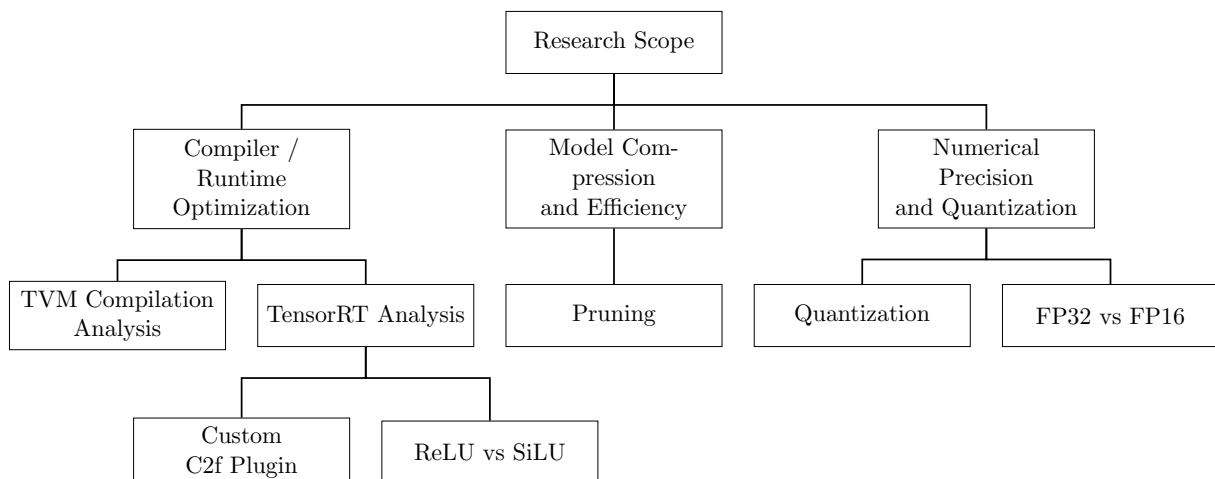


Figure 4.1: Overview of the thesis scope and main experimental directions.

# 5 Experimental Evaluation

## 5.1 Experimental Setup

The experimental evaluation is conducted on two edge-oriented target platforms: the NVIDIA Jetson Orin Nano and the Raspberry Pi 5. These devices represent two different deployment scenarios for embedded inference: a GPU-accelerated AI platform with dedicated support for mixed-precision computation, and a general-purpose low-power ARM platform without a dedicated on-chip neural network accelerator.

Model training, export, and selected modification experiments, such as pruning and operator substitution, are performed on a host workstation equipped with an NVIDIA GeForce RTX 3070 GPU. This system is used as the primary development and experimentation environment, while all deployment-oriented inference benchmarks are executed on the target edge devices.

The Jetson Orin Nano is designed for edge AI and robotics workloads. Its system-on-chip integrates a 6-core Arm CPU, an Ampere-based GPU, and shared LPDDR5 memory accessible by both CPU and GPU. This architecture is particularly relevant for this study because it supports CUDA execution, Tensor Cores, mixed-precision inference, and INT8 acceleration, making it suitable for evaluating framework-level and low-level optimization strategies.

In contrast, the Raspberry Pi 5 represents a resource-constrained edge platform without a comparable integrated AI accelerator. It is therefore used to assess how optimization techniques behave on a more conventional embedded CPU-based system, where memory bandwidth and general-purpose compute resources are more limited.

Only the hardware characteristics most relevant to performance analysis are reported in this section, namely compute capability, memory capacity, and memory bandwidth.

Table 5.1: Host GPU specifications used for training and model development

| Property         | Value                   |
|------------------|-------------------------|
| Device           | NVIDIA GeForce RTX 3070 |
| Architecture     | Ampere                  |
| FP32 Performance | ~20 TFLOPs              |
| Memory           | 8 GB GDDR6              |
| Memory Bandwidth | ~448 GB/s               |

Table 5.2: Specifications of the target edge devices

| Property         | Jetson Orin Nano         | Raspberry Pi 5        |
|------------------|--------------------------|-----------------------|
| CPU              | 6-core Arm Cortex-A78AE  | 4-core Arm Cortex-A76 |
| GPU              | Ampere (1024 CUDA cores) | VideoCore VII         |
| FP32 Performance | ~0.5–1 TFLOPs (GPU)      | —                     |
| Memory           | 8 GB LPDDR5              | 8 GB LPDDR4X          |
| Memory Bandwidth | ~102 GB/s                | ~17 GB/s              |
| Accelerators     | Tensor Cores             | None*                 |

\* The Raspberry Pi 5 can be extended with external AI accelerators, such as the Hailo-8, via PCIe. However, such accelerators are not used in the experiments reported in this thesis.

## 5.2 Roofline Model

Constructing an accurate Roofline model for real-world workloads presents several challenges. First, the performance characteristics reported by hardware manufacturers often differ from measured values under practical conditions. Second, the effective behavior of a model depends on the execution framework, including factors such as cache utilization, memory reuse, and implicit optimizations (e.g., operator fusion), which are typically not visible at the model level.

Official YOLOv8 model characteristics are summarized in Table 5.3. The smallest variant, YOLOv8n, contains 3.2M parameters and requires 8.7B floating-point operations for a single forward pass at  $640 \times 640$  resolution.

While vendor specifications indicate that the Raspberry Pi 5 provides up to 17 GB/s of memory bandwidth [58], and the Jetson Orin Nano offers up to 102 GB/s [39], measured performance is often significantly lower in practice.

Table 5.3: Comparison of YOLO models (trained on COCO) at 640×640 resolution [69].

| Model | CPU ONNX (ms) | A100 TensorRT (ms) | Params (M) | FLOPs (B) |
|-------|---------------|--------------------|------------|-----------|
| v8n   | 80.4          | 0.99               | 3.2        | 8.7       |
| v8s   | 128.4         | 1.20               | 11.2       | 28.6      |
| v8m   | 234.7         | 1.83               | 25.9       | 78.9      |
| v8l   | 375.2         | 2.39               | 43.7       | 165.2     |
| v8x   | 479.1         | 3.53               | 68.2       | 257.8     |

### 5.2.1 Roofline Analysis on Raspberry Pi 5

Using the methodology described in Chapter 3, the sustained memory bandwidth of the Raspberry Pi 5 was measured at 5.67 GB/s, and peak compute throughput at 73.95 GFLOP/s. All measurements were obtained with PyTorch configured to use 4 CPU threads, and layer timings were reported as the median over 20 runs after warmup.

These values define the device Roofline:

$$P(I) = \min(73.95, 5.67 \cdot I),$$

with a ridge point at approximately 13.04 FLOPs/byte, separating memory-bound and compute-bound regimes.

Table 5.4 summarizes representative per-layer performance characteristics of YOLOv8n executed on the Raspberry Pi 5 CPU using eager-mode PyTorch.

Layers with relatively low arithmetic intensity ( $AI \approx 7\text{--}10$  FLOPs/byte) achieve only modest throughput, typically in the range of 4.8–8.0 GFLOP/s, indicating memory-bound or memory-sensitive behavior under the measured Roofline parameters. In contrast, deeper convolutional layers with higher arithmetic intensity ( $AI > 70$  FLOPs/byte) achieve substantially higher effective throughput, often in the range of 40–60 GFLOP/s. Some layers with very high arithmetic intensity exceed 50 GFLOP/s, showing that smaller deep feature maps benefit from improved locality and higher compute utilization.

The distribution focal loss (DFL) convolution in the detection head has extremely low arithmetic intensity (0.47 FLOPs/byte) and achieves only 0.6 GFLOP/s, confirming strong memory limitation.

Although several individual convolutional layers exhibit relatively high effective throughput, the overall model achieves only 13.54 GFLOP/s, corresponding to approximately 18.3% of the measured device-level peak compute throughput of 73.95 GFLOP/s.

This discrepancy indicates that end-to-end inference performance is not determined solely by the efficiency of individual convolution kernels. Instead, it is also affected by

Table 5.4: Per-layer performance statistics of YOLOv8n on Raspberry Pi 5 (CPU, FP32).

| Layer           | Time (ms) | AI (FLOPs/byte) | Perf (GFLOP/s) | Regime                     |
|-----------------|-----------|-----------------|----------------|----------------------------|
| 0.conv          | 18.39     | 7.71            | 4.8            | Memory-bound               |
| 2.cv1.conv      | 8.84      | 7.99            | 5.9            | Memory-bound               |
| 2.cv2.conv      | 9.84      | 9.59            | 8.0            | Memory-bound               |
| 4.m.0.cv2.conv  | 2.74      | 70.42           | 43.1           | Compute-favored            |
| 6.m.1.cv2.conv  | 1.93      | 122.03          | 61.1           | Compute-favored            |
| 18.m.0.cv2.conv | 2.02      | 122.03          | 58.5           | Compute-favored            |
| 22.cv3.0.1.conv | 13.53     | 170.41          | 54.5           | Compute-favored            |
| 22.dfl.conv     | 1.89      | 0.47            | 0.6            | Strongly memory-bound      |
| Full Model      | 327.00    | 157.76          | 13.54          | Mixed / overhead-dominated |

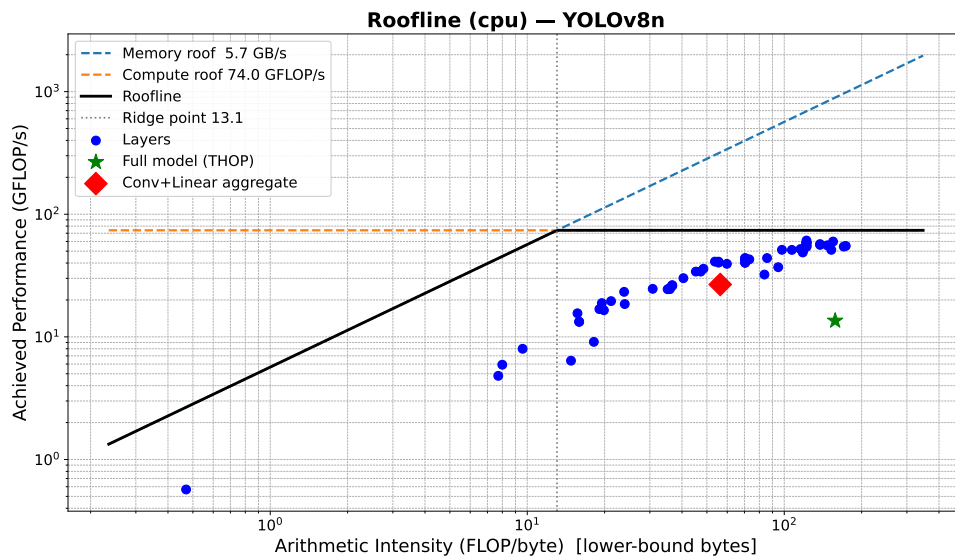


Figure 5.1: Roofline model of YOLOv8n on Raspberry Pi 5 (CPU, FP32).

system-level and framework-level overheads, including kernel dispatch overhead, synchronization, non-convolutional operations, tensor reshaping and concatenation, inter-layer memory traffic, and limited fusion between adjacent operators. Thus, the Roofline serves primarily as an upper bound and an interpretive tool rather than a directly attainable end-to-end target.

From an optimization perspective, these results suggest that further improvements should focus not only on reducing memory traffic, but also on improving execution efficiency at the graph and kernel level. Promising directions include operator fusion, graph-

level optimization, backend-specific kernel tuning, quantization, and reducing framework overhead through more optimized runtimes.

The observed performance behavior can be further explained by the relationship between layer activation sizes and the cache hierarchy:

- L1: 64 KB per core
- L2: 512 KB per core
- L3: 2 MB shared
- DRAM bandwidth:  $\approx 5.67$  GB/s (measured)

In particular, the shared L3 cache of the Raspberry Pi 5 is limited to 2 MB, which constrains the amount of feature-map data that can be reused without returning to main memory.

Early layers of YOLOv8n operate on high-resolution feature maps (e.g.,  $320 \times 320$ ), resulting in activation tensors that far exceed the available cache capacity. Consequently, these layers experience frequent DRAM accesses and comparatively low effective performance, which is consistent with their lower arithmetic intensity and more memory-bound behavior.

In contrast, deeper layers operate on progressively smaller feature maps (e.g.,  $40 \times 40$  or  $20 \times 20$ ), allowing a larger fraction of their working set to remain in cache. This improves locality and data reuse, enabling substantially higher effective throughput. Thus, even though the full model remains far below the ideal Roofline bound, the layerwise results still reveal a clear transition from memory-bound early layers to more compute-favored deeper layers.

### 5.2.2 Roofline Analysis on Jetson Orin Nano

The same analysis was performed on the NVIDIA Jetson Orin Nano GPU. With PyTorch configured to use a single host thread and using median timings after warmup, the measured sustained memory bandwidth was 55.68 GB/s, while peak compute throughput reached 810.68 GFLOP/s.

The resulting Roofline model is defined as:

$$P(I) = \min(810.68, 55.68 \cdot I),$$

with a ridge point at:

$$I^* = \frac{810.68}{55.68} \approx 14.56 \text{ FLOPs/byte.}$$

Compared to the Raspberry Pi 5, the Orin Nano exhibits both substantially higher compute throughput and much higher memory bandwidth. At the same time, the ridge point remains in a moderate arithmetic-intensity range, which means that many YOLOv8n convolution layers lie to the right of the knee and can benefit from the GPU’s high compute capability.

Table 5.5: Representative per-layer performance statistics of YOLOv8n on Jetson Orin Nano (GPU, FP32).

| Layer                 | Time (ms) | AI (FLOPs/byte) | Perf (GFLOP/s) | Regime                     |
|-----------------------|-----------|-----------------|----------------|----------------------------|
| model.0.conv          | 0.78      | 7.71            | 114.1          | Memory-bound               |
| model.2.cv1.conv      | 0.26      | 7.99            | 204.2          | Memory-bound               |
| model.4.m.0.cv2.conv  | 0.56      | 70.42           | 211.8          | Compute-bound              |
| model.6.m.1.cv2.conv  | 0.23      | 122.03          | 502.4          | Compute-bound              |
| model.12.m.0.cv2.conv | 0.24      | 122.03          | 499.3          | Compute-bound              |
| model.22.cv2.0.1.conv | 0.56      | 137.80          | 842.6          | Compute-bound              |
| model.22.cv3.0.1.conv | 0.83      | 170.41          | 892.5          | Compute-bound              |
| model.22.dfl.conv     | 0.27      | 0.47            | 3.9            | Strongly memory-bound      |
| Full Model            | 26.02     | 157.76          | 170.22         | Mixed / Overhead-dominated |

The full model achieves 170.22 GFLOP/s, corresponding to approximately 21.0% of the measured peak compute throughput. This is substantially higher than on the Raspberry Pi 5, but still far below the theoretical device peak.

Per-layer analysis shows that many convolutional layers achieve several hundred GFLOP/s, with some layers approaching or even exceeding 800 GFLOP/s. In particular, layers with high arithmetic intensity ( $AI > 70$  FLOPs/byte) clearly operate in the compute-bound regime and make effective use of the GPU’s compute resources.

However, low-intensity layers remain limited by memory behavior. The DFL convolution in the detection head, with  $AI \approx 0.47$  FLOPs/byte, achieves only 3.9 GFLOP/s and remains strongly memory-bound even on this more capable platform.

Table 5.5 highlights that, compared to the Raspberry Pi 5, the majority of convolutional layers on the Orin Nano operate in the compute-bound regime and achieve

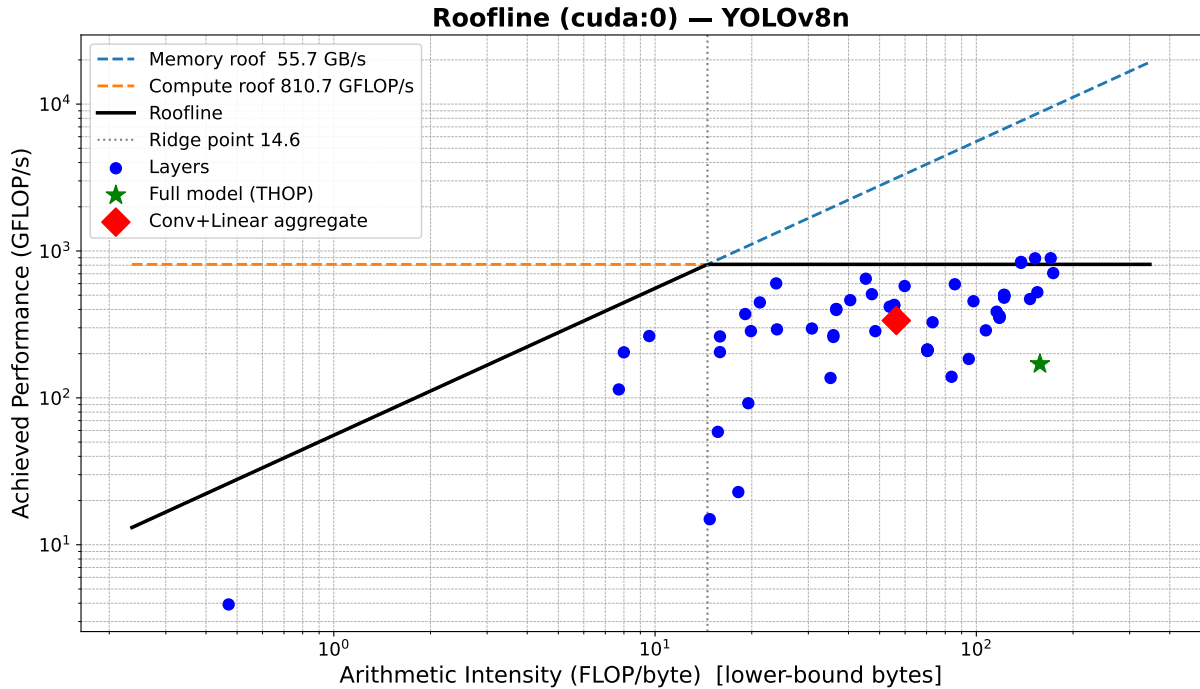


Figure 5.2: Roofline model of YOLOv8n on Jetson Orin Nano (GPU, FP32).

substantially higher throughput due to the GPU’s much larger compute capacity and memory bandwidth.

At the same time, the results also show that high per-layer kernel efficiency does not directly translate into equally high end-to-end model efficiency. Although several layers approach the measured peak throughput, the overall model reaches only 170.22 GFLOP/s. This indicates the continued presence of system-level overhead, including kernel launch overhead, intermediate tensor movement, non-convolutional operations, and synchronization costs between layers.

Thus, the Orin Nano provides a much more favorable platform for YOLOv8n inference than the Raspberry Pi 5, but the Roofline analysis still reveals that end-to-end performance remains constrained by factors beyond raw compute throughput alone.

Detailed per-layer benchmarking results are provided in Appendix A.

### 5.3 Compiler Performance Comparison

We evaluated the performance of YOLOv8n models on two hardware platforms: the NVIDIA Orin Nano and the Raspberry Pi 5. Following this evaluation, we investigated different model compilation approaches tailored to each target platform.

Due to substantial differences in hardware architectures, the set of available inference frameworks also differs. Hardware-specific frameworks such as TensorRT are limited to

NVIDIA GPUs and therefore cannot be used for a direct cross-platform comparison. Instead, we assess the performance improvements provided by each framework relative to a common baseline on the same hardware.

The baseline execution is performed in PyTorch eager mode. In this mode, operations are executed immediately as Python code is interpreted line by line. The computational graph is constructed dynamically at runtime, which makes debugging straightforward and enables inspection of intermediate results, such as per-layer performance metrics presented in the previous section. However, eager execution introduces several limitations: Python-level overhead for each operation, lack of global graph-level optimization, and limited hardware-specific tuning. As a result, the model is effectively executed as a sequence of independent operators rather than as an optimized computational graph.

To address these limitations, the model can be treated as a static graph and optimized using compiler-based frameworks. A variety of such compilers have been developed to enable hardware-aware optimizations, including operator fusion, memory planning, and kernel selection. In this section, we evaluate these approaches on the target hardware platforms.

A standard step in this workflow is converting the model to the ONNX (Open Neural Network Exchange) format. In this work, we use ONNX export with opset version 17 and enable graph simplification. The opset defines the set of available operators (e.g., Conv, Add, Resize) and their semantics. Choosing an appropriate opset version is important: newer opsets may not be fully supported by all backends, while older versions may lack required operators or optimizations. The opset selection can therefore impact operator decomposition, kernel count, memory traffic, and opportunities for fusion.

Graph simplification is applied to reduce computational redundancy and streamline the graph structure. This includes:

- Constant folding: precomputing constant expressions (e.g.,  $x + (2 \times 3) \rightarrow x + 6$ )
- Node elimination: removing redundant operations such as identity mappings
- Subgraph simplification: collapsing sequences of operations into simpler equivalents when possible
- Dead code elimination: removing unused branches of the graph

While these transformations simplify the graph, backend-specific compilers (e.g., TensorRT, TVM) may perform additional, more aggressive optimizations such as kernel fusion and layout transformations.

Once converted, the ONNX model can be executed using a variety of inference frameworks. On CPU-based platforms, commonly used frameworks include PyTorch (via Torch-

Script or TorchInductor), ONNX Runtime, and TensorFlow Lite. These frameworks provide different levels of optimization, ranging from general-purpose execution to lightweight deployment on resource-constrained devices.

On GPU-based platforms, available options include PyTorch with CUDA execution, ONNX Runtime with GPU or TensorRT execution providers, and TensorRT as a dedicated compiler and runtime for NVIDIA hardware. TensorRT performs aggressive graph-level and kernel-level optimizations, such as operator fusion, precision calibration (e.g., FP16/INT8), and memory layout transformations, making it particularly effective for high-performance inference on NVIDIA devices.

Another notable framework that supports both CPU and GPU targets is Apache TVM. Unlike runtime-focused frameworks, TVM operates as a deep learning compiler stack that performs end-to-end optimization, including graph transformations, operator scheduling, and hardware-specific code generation. This approach provides a high degree of flexibility and control over low-level execution compared to more API-driven frameworks, although it typically requires more manual tuning and expertise.

### 5.3.1 TVM MetaSchedule Tuning

Unlike other evaluated frameworks, TVM requires an explicit tuning stage before deployment: MetaSchedule generates candidate schedules for each extracted operator task, evaluates them on the target device, and compiles the best-performing implementations into a final binary. The following subsections report the tuning outcome for each platform and precision, covering which operators benefited most and where the main performance limits lie.

### 5.3.2 TVM MetaSchedule Tuning Results on Raspberry Pi 5 - FP32

TVM MetaSchedule was run on the Raspberry Pi 5 (4-core Cortex-A76, AArch64, 2.4 GHz) for approximately 6 hours and 15 minutes, evaluating 7 221 valid schedules across 65 operator tasks extracted from the YOLOv8n Relax graph. The tuned model achieved a summed task latency of 173.6 ms and an end-to-end latency of approximately 185–188 ms, with the small gap attributable to runtime overhead outside the tuned kernels (input preparation, NMS postprocessing). Against the 290.1 ms PyTorch eager baseline, the end-to-end result of approximately 192 ms corresponds to a  $1.5\times$  improvement.

FP32 tuning on this platform did not enable tensorisation: no task could be mapped to a dedicated ARM intrinsic, so all performance gains came from cache-blocking, loop re-

ordering, and LLVM auto-vectorisation.  $1\times 1$  convolutions in the neck and detection head performed best, reaching above 80 GFLOPS because their simpler structure maps naturally to vectorised execution. Larger  $3\times 3$  convolutions in the mid-network also improved once channel dimensions were wide enough to sustain cache-resident working sets.

The main bottleneck was the initial stem convolution, which processes a  $640\times 640$  spatial input with only three input channels. This combination yields very low arithmetic intensity that schedule search cannot fundamentally overcome. When operator multiplicity is accounted for, repeated  $3\times 3$  convolutions inside C2f blocks contributed more to total runtime than any individually slow operator, confirming that tuning effort is best directed at frequently executed kernels rather than the single worst-performing layer. The overall conclusion is that FP32 TVM tuning is beneficial on the Raspberry Pi 5, but gains are bounded by the absence of tensorisation; a qualitative improvement requires FP16.

### 5.3.3 TVM MetaSchedule Tuning Results on the Jetson Orin Nano GPU - FP32

On the Jetson Orin Nano GPU, MetaSchedule explored 7 257 valid schedules across the same 65 operator tasks. The tuned model achieved a summed task latency of 22 920  $\mu\text{s}$  and an end-to-end latency of 23.38 ms, with tuned kernels accounting for almost all of the observed runtime. Compared with the Raspberry Pi 5 CPU result, this corresponds to a  $7.6\times$  kernel-level speedup, confirming the expected advantage of GPU execution for convolution-dominated inference.

As in the FP32 CPU case, no task could be tensorised: the GPU benefited from CUDA-specific schedule parameters (thread-block structure, tiling, memory movement patterns) rather than from Tensor Core mapping. Unlike the CPU, where  $1\times 1$  convolutions were most efficient, the GPU performed best on large mid-network  $3\times 3$  convolutions with high channel counts: the best-performing task reached 585.6 GFLOPS, and several others exceeded 500 GFLOPS, because the larger working sets gave the GPU enough parallel work to amortise memory movement.

The stem layer and early low-channel convolutions remained the weakest operators for the same structural reason as on the CPU: three input channels over a large spatial resolution limits arithmetic intensity. When multiplicity is taken into account, task 50 (`conv2d24`) and task 16 (`conv2d11`, appearing seven times in the graph) were the dominant weighted bottlenecks. The final end-to-end result of 23.38 ms was only a modest improvement over the PyTorch eager baseline (25.12 ms), and TensorRT remained substantially faster at 11.38 ms. This confirms that FP32 GPU tuning without tensorisation leaves significant performance on the table.

### 5.3.4 Raspberry Pi 5 Benchmark Overview - FP32

All Raspberry Pi 5 experiments were conducted on a Linux system based on the 6.12.62+rpt-rpi-2712 kernel using the AArch64 architecture. The CPU frequency governor was fixed to `performance`, and the CPU was configured to 2.4GHz during the measurements. To reduce thermal and scheduling-related variation, the benchmarking procedure included a cooldown phase before each repetition and checks for frequency throttling. Each backend was evaluated over five repeated runs, with 200 timed inferences per repetition after 30 warm-up iterations. In addition, each repetition was executed in a fresh subprocess in order to reduce cross-framework interference, such as cache contamination or residual runtime state. The comparison was performed using all four available CPU cores on the Raspberry Pi 5 in order to reflect the maximum CPU throughput available in this deployment setting. This setup was chosen to reduce short-term variance and to provide a stable estimate of sustained inference performance under controlled CPU conditions.

The evaluated CPU backends were TVM, ONNX Runtime, TorchCompile, Google AI Edge LiteRT, and PyTorch eager mode used as a baseline. ONNX Runtime was evaluated with the `CPUExecutionProvider`. Table 5.6 summarises the measured latency and throughput, while Table 5.7 summarises power results. These measurements are further visualised in Figure 5.3, which compares latency, power consumption, the latency–power trade-off, and throughput per watt across the evaluated CPU backends.

Table 5.6: End-to-end YOLOv8n inference performance on Raspberry Pi 5 CPU. Values are reported as mean latency  $\pm$  half-width of the 95 % confidence interval across 5 isolated repetitions, with 200 timed runs per repetition after 30 warm-up iterations.

| Backend       | Latency (ms)    | FPS mean | Median (ms) |
|---------------|-----------------|----------|-------------|
| TVM           | 192.0 $\pm$ 7.5 | 5.21     | 188.8       |
| ONNX Runtime  | 189.2 $\pm$ 7.0 | 5.28     | 187.2       |
| TorchCompile  | 202.9 $\pm$ 8.5 | 4.93     | 202.2       |
| LiteRT        | 154.1 $\pm$ 2.5 | 6.49     | 155.2       |
| PyTorch eager | 290.1 $\pm$ 2.3 | 3.45     | 289.8       |

**Overall Performance.** Among all evaluated frameworks, LiteRT achieved the best overall latency on the Raspberry Pi 5, with a mean inference time of 154.1ms and a throughput of 6.49 FPS. ONNX Runtime and TVM formed the next performance group, with mean latencies of 189.2ms and 192.0ms respectively. TorchCompile was slower than both compiled deployment backends, reaching a mean latency of 202.9ms and a throughput of 4.93 FPS. PyTorch eager mode remained the slowest configuration, with a mean latency of 290.1ms and a throughput of 3.45 FPS. The difference between ONNX

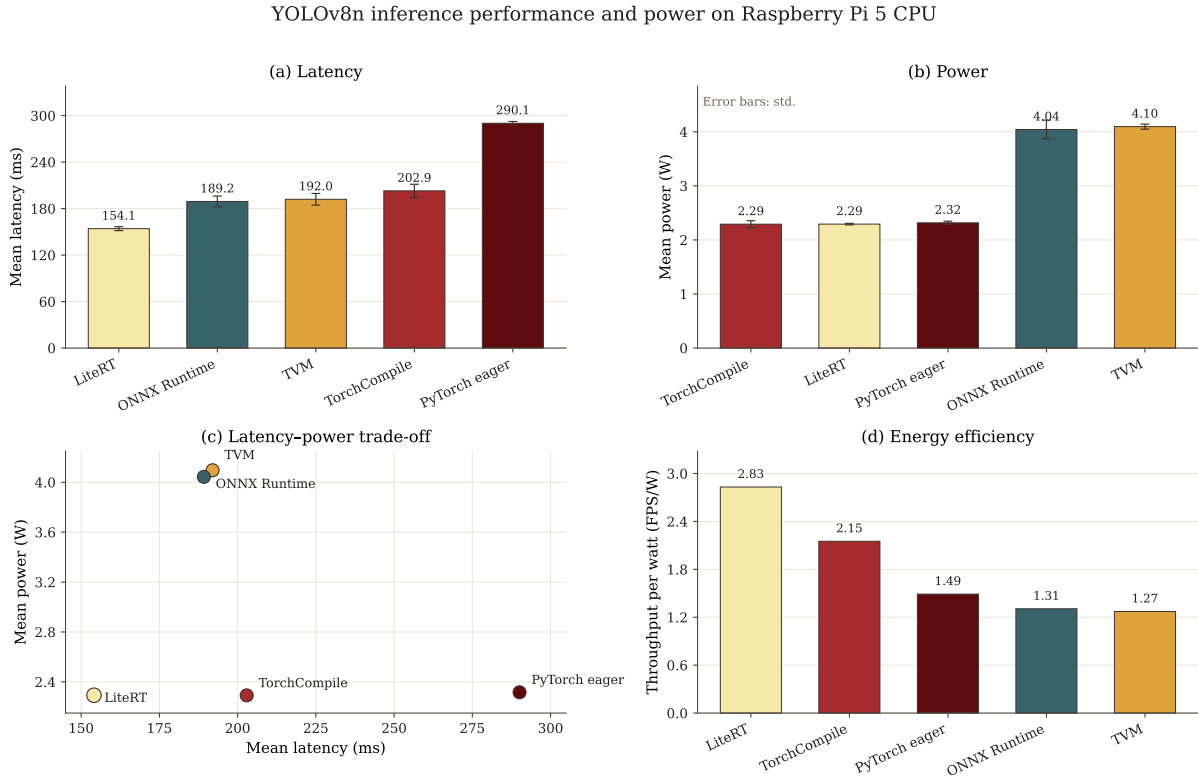


Figure 5.3: YOLOv8n inference performance and power consumption on the Raspberry Pi 5 CPU across the evaluated FP32 backends. Subfigure (a) shows the mean end-to-end inference latency with 95% confidence interval error bars. Subfigure (b) reports mean board power consumption, with error bars indicating standard deviation. Subfigure (c) summarizes the latency–power trade-off, where lower latency and lower power are preferable. Subfigure (d) shows energy efficiency in terms of throughput per watt. LiteRT achieves the lowest latency and the highest energy efficiency, while TVM and ONNX Runtime exhibit similar latency but substantially higher power consumption.

Runtime and TVM was only 2.8 ms, whereas TorchCompile lagged behind ONNX Runtime by 13.6 ms and behind TVM by 10.8 ms.

**Stability of Measurements.** The repeated-run statistics show that TVM and ONNX Runtime had comparable variability, with standard deviations of 6.03 ms and 5.65 ms respectively. TorchCompile showed a similar but slightly higher variability of 6.86 ms. In contrast, PyTorch eager mode and LiteRT were more stable, with standard deviations of 1.84 ms and 2.05 ms. LiteRT therefore combined the best average performance in this benchmark with comparatively consistent execution.

**Power Behaviour.** TVM and ONNX Runtime showed similar average power draw, measuring 4.10 W and 4.04 W respectively. In contrast, TorchCompile, LiteRT, and PyTorch eager mode all operated near 2.3 W average board power. LiteRT exhibited the lowest average power consumption at 2.29 W, although the difference relative to Torch-

Table 5.7: Average power consumption during YOLOv8n inference on Raspberry Pi 5 CPU.

| Backend       | Mean Power (mW) | Max Power (mW) | Std (mW) |
|---------------|-----------------|----------------|----------|
| TVM           | 4096.2          | 4783.1         | 46.1     |
| ONNX Runtime  | 4043.2          | 5415.1         | 172.5    |
| TorchCompile  | 2291.8          | 2742.5         | 63.2     |
| LiteRT        | 2292.8          | 2591.5         | 14.2     |
| PyTorch eager | 2316.4          | 2759.8         | 30.3     |

Compile was very small. This result is notable because LiteRT was both the fastest backend in this experiment and the one with the lowest measured average power consumption.

**Interpretation of the Results.** Although TVM did not outperform LiteRT, it remained competitive with ONNX Runtime and outperformed TorchCompile in this end-to-end benchmark. TVM achieved slightly lower latency than TorchCompile while relying on a fully compiler-driven deployment pipeline based on Relax and MetaSchedule tuning. This suggests that ahead-of-time compilation through a dedicated inference compiler remains more effective on this workload than applying `torch.compile` to the original PyTorch model on CPU. At the same time, the results also show that LiteRT provided the most efficient execution path among the tested frameworks on Raspberry Pi 5 for this FP32 model.

**Conclusion.** The benchmark results show four practical performance tiers on the Raspberry Pi 5 CPU. LiteRT delivered the best end-to-end result, combining the lowest latency, the highest throughput, and the lowest average power consumption. ONNX Runtime and TVM achieved similar performance, with ONNX Runtime being slightly faster in the measured configuration. TorchCompile improved substantially over PyTorch eager mode, but remained slower than both ONNX Runtime and TVM. PyTorch eager mode was the slowest baseline by a large margin.

Relative to the PyTorch eager baseline, TVM achieved a  $1.51\times$  speedup, corresponding to a 33.8% reduction in mean inference latency. ONNX Runtime achieved a  $1.53\times$  speedup, corresponding to a 34.8% latency reduction. TorchCompile achieved a  $1.43\times$  speedup, corresponding to a 30.1% latency reduction. LiteRT provided the largest improvement, reaching a  $1.88\times$  speedup and a 46.9% reduction in mean latency. Overall, these results indicate that TVM is capable of producing competitive CPU inference performance on Raspberry Pi 5, but in this experiment the most efficient backend was LiteRT rather than TVM.

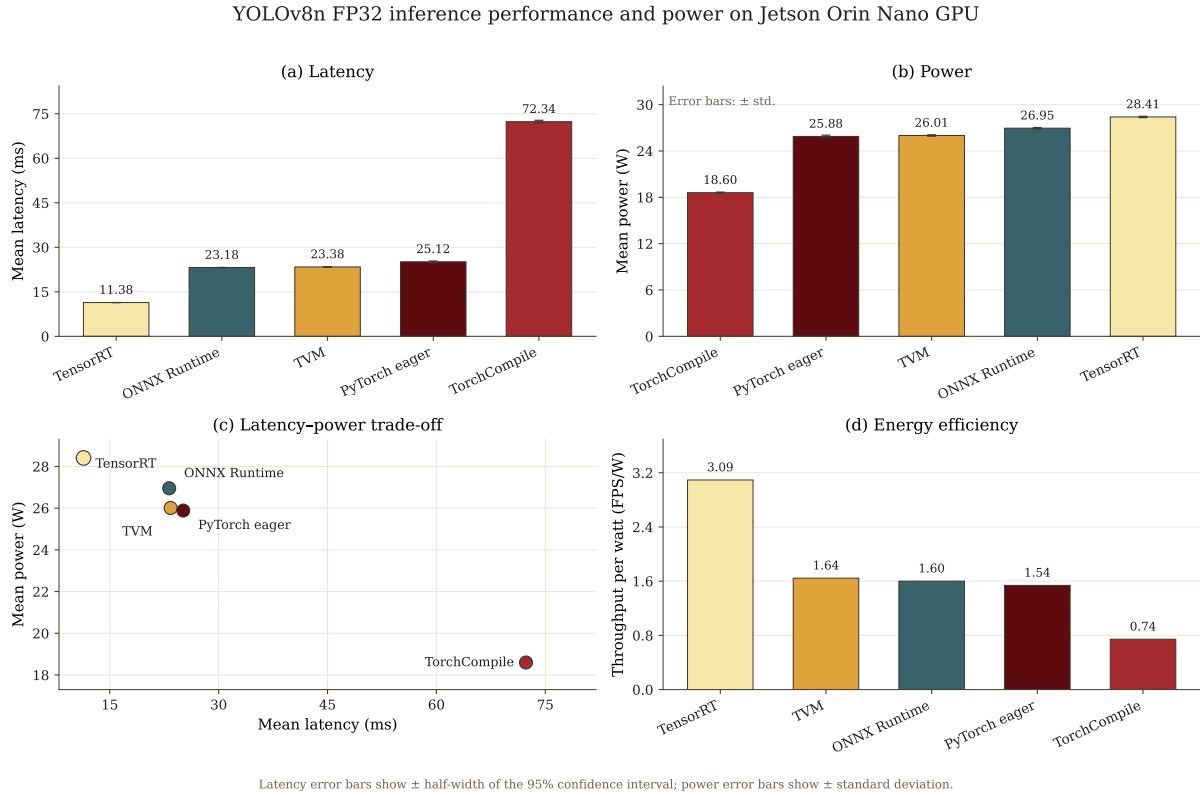


Figure 5.4: YOLOv8n FP32 inference performance and power consumption on the NVIDIA Jetson Orin Nano GPU. Subfigure (a) shows mean end-to-end inference latency with 95% confidence interval error bars. Subfigure (b) reports mean board power consumption, with error bars indicating standard deviation. Subfigure (c) summarizes the latency–power trade-off, where lower latency and lower power are preferable. Subfigure (d) shows energy efficiency in terms of throughput per watt. TensorRT achieves the lowest latency and the highest energy efficiency among the evaluated FP32 GPU backends, while torch.compile is substantially slower despite lower power draw.

### 5.3.5 Jetson Orin Nano GPU Benchmark Overview - FP32

GPU experiments were conducted on the NVIDIA Jetson Orin Nano using the full YOLOv8n model in FP32 precision with batch size 1. The evaluated backends were TVM, ONNX Runtime, TensorRT, PyTorch eager mode, and torch.compile. As in the CPU experiments, the benchmark procedure used isolated repetitions in order to reduce interference between backends. Table 5.8 summarises the measured latency and throughput, while Table 5.9 summarises power results. These measurements are further visualised in Figure 5.4, which compares latency, power consumption, the latency–power trade-off, and throughput per watt across the evaluated GPU backends.

**Overall Performance.** TensorRT achieved by far the best end-to-end latency on the Jetson Orin Nano, reaching 11.38 ms, which corresponds to 87.9 FPS. ONNX Runtime and TVM formed a second performance group with nearly identical results, at 23.18 ms

Table 5.8: End-to-end YOLOv8n inference performance on the NVIDIA Jetson Orin Nano GPU. Values are reported as mean latency  $\pm$  half-width of the 95 % confidence interval across 5 isolated repetitions, with 200 timed runs per repetition after 30 warm-up iterations.

| Backend       | Latency (ms)     | FPS mean | Median (ms) |
|---------------|------------------|----------|-------------|
| TVM           | $23.38 \pm 0.02$ | 42.77    | 23.39       |
| ONNX Runtime  | $23.18 \pm 0.04$ | 43.14    | 23.17       |
| TorchCompile  | $72.34 \pm 0.47$ | 13.82    | 72.26       |
| TensorRT      | $11.38 \pm 0.02$ | 87.87    | 11.39       |
| PyTorch eager | $25.12 \pm 0.27$ | 39.81    | 25.12       |

and 23.38ms respectively. PyTorch eager mode was slightly slower at 25.12ms, while `torch.compile` performed substantially worse than all other GPU backends, reaching 72.34ms.

Relative to the PyTorch eager baseline, TVM achieved a  $1.07\times$  speedup, corresponding to a 6.9 % reduction in mean latency. ONNX Runtime achieved a  $1.08\times$  speedup and a 7.7 % latency reduction. TensorRT provided the largest improvement, reaching a  $2.21\times$  speedup and a 54.7 % reduction in mean latency. In contrast, `torch.compile` was slower than PyTorch eager mode by a factor of  $2.88\times$ .

Table 5.9: Average power consumption during YOLOv8n inference on the NVIDIA Jetson Orin Nano GPU.

| Backend       | Mean Power (mW) | Max Power (mW) | Std (mW) |
|---------------|-----------------|----------------|----------|
| TVM           | 26007.9         | 26781.1        | 88.7     |
| ONNX Runtime  | 26952.4         | 29632.6        | 84.8     |
| TorchCompile  | 18599.2         | 19967.0        | 76.7     |
| TensorRT      | 28405.8         | 30936.2        | 86.4     |
| PyTorch eager | 25884.4         | 29186.0        | 168.4    |

**Power and Efficiency.** In absolute power draw, the GPU backends operated within a relatively narrow range, from 18.6W for `torch.compile` to 28.4W for TensorRT. However, when throughput is taken into account, TensorRT was also the most power-efficient backend, achieving 3.10FPS/W. This is approximately 1.9 times higher than TVM and ONNX Runtime, which both reached about 1.6FPS/W, and about twice the efficiency of PyTorch eager mode. Although `torch.compile` consumed the least power, its very low throughput made it the least efficient backend overall.

**Interpretation of the TVM Result.** The TVM GPU result is notable for two reasons. First, its end-to-end latency was essentially identical to ONNX Runtime, with a difference

of only 0.20 ms. Second, TVM still outperformed PyTorch eager mode, although the margin was much smaller than on the CPU. This suggests that, on the Orin Nano GPU, the main convolution-heavy parts of the model are already executed efficiently by mature CUDA libraries across multiple frameworks, which reduces the practical advantage of compiler-level scheduling alone. In other words, TVM remained competitive, but did not establish a clear lead over ONNX Runtime on this platform.

**Conclusion.** The GPU benchmark results reveal three practical performance tiers. TensorRT forms the clear top tier, achieving both the lowest latency and the highest power efficiency. ONNX Runtime and TVM form a second tier with nearly identical performance, while PyTorch eager mode is only slightly slower. In contrast, `torch.compile` was not competitive for this workload in the tested configuration. Overall, these results show that TensorRT was the most effective deployment backend for YOLOv8n on the Jetson Orin Nano GPU, whereas TVM and ONNX Runtime delivered similar and competitive FP32 performance.

Relative to the PyTorch eager baseline, TVM achieved a  $1.07\times$  speedup, equivalent to a 6.9% reduction in mean inference latency, and ONNX Runtime achieved a  $1.08\times$  speedup with a 7.7% latency reduction. TensorRT provided the largest improvement, reaching a  $2.21\times$  speedup and a 54.7% reduction in mean latency. In contrast, `torch.compile` failed to improve upon the baseline and instead increased latency by a factor of  $2.88\times$ . These results indicate that TVM can provide competitive GPU inference performance on the Jetson Orin Nano, but in this experiment the largest practical advantage was obtained with TensorRT.

## 5.4 FP16 Inference Results

Before discussing the measured FP16 results, it is important to clarify that conversion to FP16 is not the same as quantization. In FP16 inference, weights and activations remain in floating-point representation, but with reduced numerical precision compared to FP32. By contrast, quantization usually refers to mapping tensors into lower-bit integer or fixed-point formats, such as INT8, which changes not only the storage format but also the arithmetic model used during execution. FP16 should therefore be understood as reduced-precision floating-point inference rather than true quantization.

This distinction is important for interpreting the following measurements. Reducing precision from FP32 to FP16 can decrease memory traffic and improve throughput on hardware with native half-precision support, often with only a small impact on model accuracy. However, the resulting performance behaviour is different from that of integer

quantization, since the model still operates in a floating-point domain. The following subsection therefore evaluates FP16 as a separate optimisation path and analyses its effect on latency, throughput, and overall deployment efficiency.

### 5.4.1 TVM MetaSchedule Tuning Results on the Jetson Orin Nano GPU - FP16

TVM MetaSchedule was re-run on the Jetson Orin Nano GPU using a FP16 version of the YOLOv8n graph, converted from the FP32 ONNX export using `onnxconverter-common` with `keep_io_types=False`. The tuning campaign ran for approximately 18 hours and explored 6 477 valid schedules across 66 operator tasks. The tuned model achieved a summed task latency of 9 872  $\mu$ s and an end-to-end latency of 12.45 ms, corresponding to a  $2.3\times$  kernel-level improvement and a  $1.88\times$  end-to-end speedup over the FP32 result.

The critical difference from FP32 was that FP16 enabled tensorised execution via Warp Matrix Multiply-Accumulate (WMMA) on the Orin Nano GPU. In FP32, no task could be mapped to a hardware intrinsic and all schedules used standard CUDA thread-block tiling. In FP16, a large fraction of convolutional tasks were tensorised, raising peak operator throughput from 585.6 GFLOPS to 1 872.7 GFLOPS – a  $3.2\times$  gain in peak kernel efficiency. The speedup therefore arose not only from halved data movement but from a qualitatively different optimisation path: MetaSchedule could now search over implementations that directly exploit the GPU’s matrix-multiplication hardware.

The main bottleneck pattern remained unchanged: the stem layer (3 input channels,  $640\times 640$  spatial resolution) could not benefit from tensorisation and remained the least efficient individual operator even in FP16. When multiplicity is included, repeated backbone and neck convolutions – notably task 50 (`conv2d24`) and task 16 (`conv2d11`) – again dominated total runtime. TensorRT remained faster at 6.31 ms, indicating that vendor-specific optimisations still provide an additional margin beyond what MetaSchedule found. Nevertheless, FP16 tuning establishes TVM as a genuinely competitive deployment backend: it cut end-to-end latency from 23.38 ms to 12.45 ms, confirming FP16 as the appropriate precision target for TVM-based deployment on Orin-class hardware.

### 5.4.2 Jetson Orin Nano GPU Benchmark Overview - FP16

GPU experiments were also conducted on the NVIDIA Jetson Orin Nano using the full YOLOv8n model in FP16 precision with batch size 1. The evaluated backends were TVM, ONNX Runtime, TensorRT, PyTorch eager mode, and `torch.compile`. As in the previous benchmark sections, the comparison was performed using isolated repetitions in order to reduce interference between frameworks. Table 5.10 summarises the measured

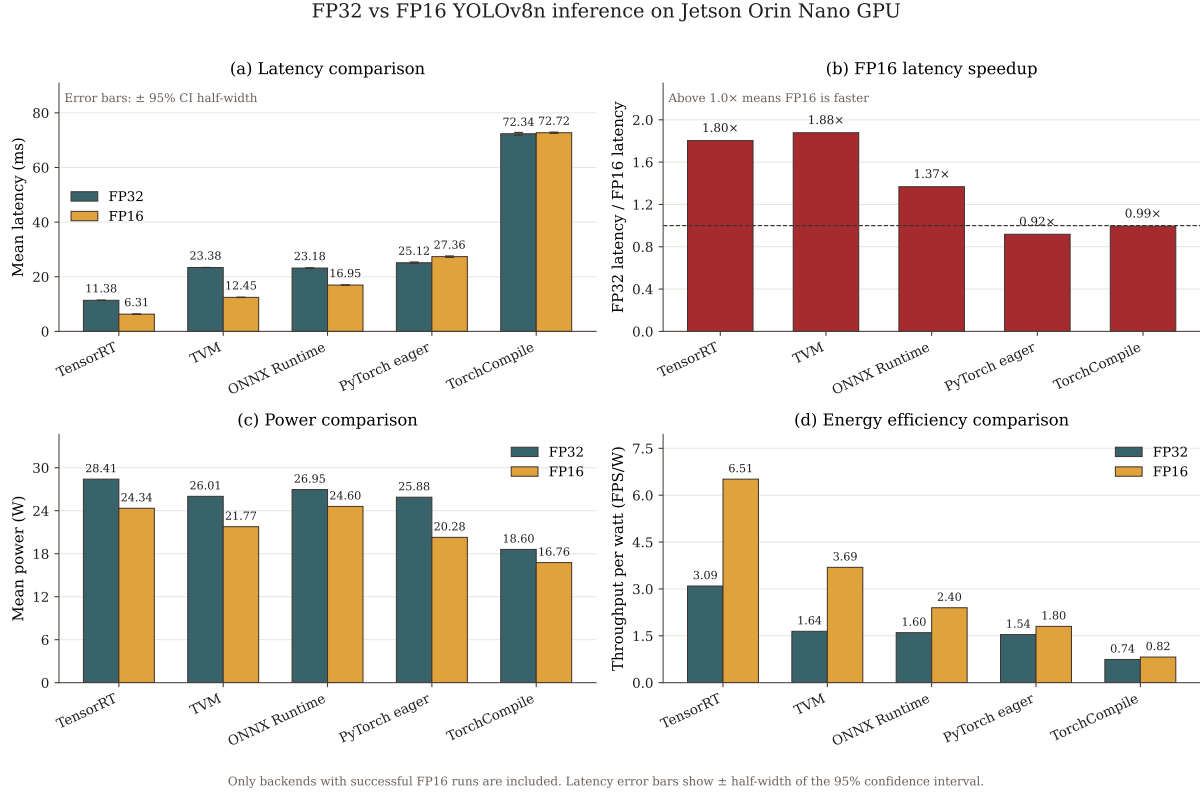


Figure 5.5: FP32 versus FP16 YOLOv8n inference on the NVIDIA Jetson Orin Nano GPU. Subfigure (a) compares mean end-to-end latency for each backend, with error bars indicating the half-width of the 95% confidence interval. Subfigure (b) reports the FP16 latency speedup relative to FP32 for the same backend, where values above  $1.0\times$  indicate that FP16 is faster. Subfigure (c) compares mean board power consumption, and Subfigure (d) compares energy efficiency in terms of throughput per watt. TensorRT and TVM obtain the strongest FP16 latency improvements, while `torch.compile` and PyTorch eager mode do not benefit from FP16 in this experiment.

latency and throughput, while Table 5.11 summarises power results. Figure 5.5 further compares the FP16 measurements against the corresponding FP32 results, showing the latency, FP16 speedup, power consumption, and energy efficiency for each backend.

**Overall Performance.** TensorRT achieved the best end-to-end latency in the FP16 setting, reaching 6.31 ms, which corresponds to 158.6 FPS. TVM formed the second-best result at 12.45 ms and substantially outperformed ONNX Runtime, which reached 16.95 ms. PyTorch eager mode was considerably slower at 27.36 ms, while `torch.compile` again performed substantially worse than all other backends, reaching 72.72 ms.

Relative to the PyTorch eager baseline, TVM achieved a  $2.20\times$  speedup, corresponding to a 54.5% reduction in mean latency. ONNX Runtime achieved a  $1.61\times$  speedup and a 38.1% latency reduction. TensorRT provided the largest improvement, reaching a  $4.34\times$  speedup and a 76.9% reduction in mean latency. In contrast, `torch.compile` was

Table 5.10: End-to-end YOLOv8n inference performance on the NVIDIA Jetson Orin Nano GPU in FP16 precision. Values are reported as mean latency  $\pm$  half-width of the 95 % confidence interval across 5 isolated repetitions, with 200 timed runs per repetition after 30 warm-up iterations.

| Backend       | Latency (ms)     | FPS mean | Median (ms) |
|---------------|------------------|----------|-------------|
| TVM           | $12.45 \pm 0.05$ | 80.35    | 12.43       |
| ONNX Runtime  | $16.95 \pm 0.02$ | 59.00    | 16.96       |
| TorchCompile  | $72.72 \pm 0.21$ | 13.75    | 72.76       |
| TensorRT      | $6.31 \pm 0.01$  | 158.55   | 6.31        |
| PyTorch eager | $27.36 \pm 0.24$ | 36.55    | 27.34       |

slower than PyTorch eager mode by a factor of  $2.66\times$ .

Table 5.11: Average power consumption during YOLOv8n inference on the NVIDIA Jetson Orin Nano GPU in FP16 precision.

| Backend       | Mean Power (mW) | Max Power (mW) | Std (mW) |
|---------------|-----------------|----------------|----------|
| TVM           | 21770.0         | 24514.2        | 122.6    |
| ONNX Runtime  | 24601.7         | 29104.0        | 88.3     |
| TorchCompile  | 16760.5         | 17874.5        | 33.9     |
| TensorRT      | 24342.7         | 26500.5        | 19.0     |
| PyTorch eager | 20282.6         | 20852.0        | 79.9     |

**Power and Efficiency.** In absolute power draw, the FP16 GPU backends again operated within a relatively narrow range, from 16.8 W for `torch.compile` to 24.6 W for ONNX Runtime. However, throughput differed much more strongly than power, which made efficiency comparisons especially informative. TensorRT achieved the best efficiency at approximately 6.51 FPS/W, followed by TVM at about 3.69 FPS/W. PyTorch eager mode reached about 1.80 FPS/W, and ONNX Runtime about 2.40 FPS/W. Although `torch.compile` consumed the least power, its low throughput again made it the least competitive backend in practice.

**Interpretation of the TVM Result.** The TVM FP16 result is particularly notable because, unlike in FP32, TVM no longer merely matched ONNX Runtime but clearly outperformed it. Its latency was 4.50 ms lower than ONNX Runtime, corresponding to an additional reduction of approximately 26.6 %. This indicates that FP16 compilation gave TVM access to optimisation opportunities that were not available in the FP32 case, most importantly tensorised execution for a large fraction of the convolutional operators. As a result, TVM moved from being merely competitive in FP32 to becoming the second-best backend overall in FP16.

At the same time, TensorRT still remained significantly faster, with a latency of 6.31 ms compared with 12.45 ms for TVM. This shows that even though TVM benefited greatly from FP16 compilation, additional backend-specific optimisations still remain available beyond what MetaSchedule found in this work.

**Conclusion.** The FP16 benchmark results reveal a clearer separation between deployment backends than in the FP32 GPU case. TensorRT formed the top tier with the best latency, throughput, and efficiency. TVM formed a strong second tier and substantially outperformed ONNX Runtime, PyTorch eager mode, and `torch.compile`. ONNX Runtime remained competitive but was noticeably slower than TVM, while PyTorch eager mode was substantially slower than all three specialised deployment paths. As in the FP32 case, `torch.compile` was not competitive for this workload in the tested configuration.

Overall, these results indicate that FP16 is the most favourable precision setting for TVM deployment on the Jetson Orin Nano GPU. In this configuration, TVM no longer merely matched other general deployment runtimes, but delivered a clear practical advantage over ONNX Runtime and PyTorch eager mode. However, TensorRT still provided the fastest overall execution and remained the most effective deployment backend in the final comparison.

### 5.4.3 FP16 Inference on the Raspberry Pi 5 CPU

To investigate whether half-precision arithmetic provides a runtime benefit on the Raspberry Pi 5, the YOLOv8n model was re-evaluated at FP16 precision across all applicable backends. The FP16 ONNX model was generated from the FP32 export using `onnxconverter-common` with `keep\_io\_types=False`, producing a graph in which all weights, activations, and the input tensor are represented as 16-bit floats. TVM was re-tuned on this model, yielding a separately compiled `.so` library. All other backends received the same FP16 ONNX as input.

**TVM FP16 Tuning Session.** For TVM, the FP16 tuning campaign covered 65 operator tasks in approximately 4 hours and 6 minutes, compared with 6 hours and 15 minutes for FP32. The shorter run reflects the stronger latency signal from FP16 kernels: once LLVM finds a schedule that packs eight `float16` values into a 128-bit NEON (ARM Advanced SIMD extension) register rather than four `float32` values, the improvement is immediately visible and the tuner shifts to refinement instead of exploration. As in FP32, no task could be tensorised – the three-dimensional convolution reduction (input channels, kernel height, kernel width) cannot be mapped to NEON’s one-dimensional dot-product intrinsic – so all schedules relied on cache-blocked tiling

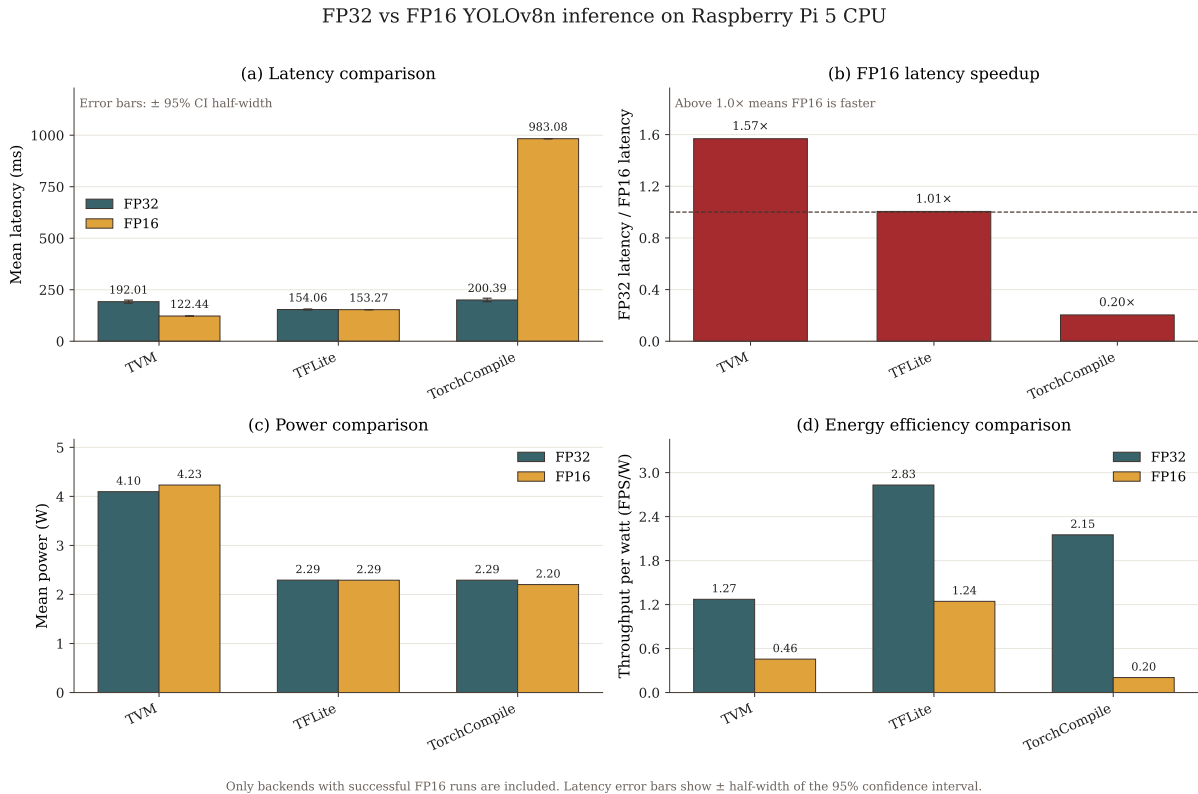


Figure 5.6: FP32 versus FP16 YOLOv8n inference on the Raspberry Pi 5 CPU for backends with successful FP16 runs. Subfigure (a) compares mean end-to-end latency, with error bars indicating the half-width of the 95% confidence interval where available. Subfigure (b) reports the FP16 latency speedup relative to FP32 for the same backend, where values above  $1.0\times$  indicate that FP16 is faster. Subfigure (c) compares mean board power consumption, and Subfigure (d) compares energy efficiency in terms of throughput per watt. TVM is the only backend that shows a clear FP16 latency improvement on the Raspberry Pi 5 CPU, while TFLite remains nearly unchanged and TorchCompile becomes substantially slower.

with `float16x8` auto-vectorisation. Peak operator throughput rose from 87.1 GFLOPS (FP32) to 169.2 GFLOPS (FP16), a  $1.94\times$  ratio consistent with the theoretical  $2\times$  gain from halving element width within a fixed 128-bit register.

**End-to-End Benchmark Results.** Figure 5.6 visualises the successful FP16 runs against their corresponding FP32 measurements on the Raspberry Pi 5 CPU, including latency, FP16 speedup, power consumption, and energy efficiency. Table 5.12 presents the full benchmark comparison between FP32 and FP16 precision across all backends on the Raspberry Pi 5, including the unsuccessful or fallback FP16 cases.

**TVM - the Only Backend with Real FP16 Acceleration.** TVM reduced its end-to-end latency from 192.01ms to 122.44ms, a speedup of  $1.57\times$ . This improvement stems from a qualitative change in the generated code: LLVM emits `VFMA.F16` NEON

Table 5.12: FP32 vs FP16 inference on the Raspberry Pi 5 CPU (5 repetitions  $\times$  200 runs, 30 warm-up runs, performance governor, batch size 1, 640 $\times$ 640 input). ORT and PyTorch FP16 results are estimated from single-repetition measurements; TorchCompile FP16 is from the rigorous run.

| Backend      | FP32 mean (ms) | FP16 mean (ms)   | Ratio                | FP16 status   |
|--------------|----------------|------------------|----------------------|---------------|
| TVM          | 192.01         | 122.44           | 1.57 $\times$ faster | real fp16     |
| ONNXRuntime  | 189.23         | >900 s           | timeout              | no fp16 Conv  |
| TFLite       | 154.06         | 153.27           | 1.00 $\times$        | weights only  |
| TorchCompile | 200.39         | 983.08           | 4.9 $\times$ slower  | GEMM fallback |
| PyTorch      | 290.13         | $\approx$ 11 000 | 38 $\times$ slower   | scalar loop   |

instructions that operate on eight `float16` values in parallel within the same 128-bit register that previously held four `float32` values. The gap between the measured 1.57 $\times$  end-to-end speedup and the theoretical 2 $\times$  kernel-level ceiling is explained by two factors. First, non-arithmetic operators - splits, concatenations, and the upsampling layers of the FPN neck - execute at the same speed regardless of element width. Second, the initial stem convolution, which has only three input channels, cannot fill the eight available FP16 NEON lanes; its latency therefore improves only marginally between precisions. Both of these effects are irreducible at the schedule level and represent genuine architectural constraints of the YOLOv8n graph on a scalar-SIMD processor.

**ONNXRuntime - Timeout Due to Missing FP16 Convolution Kernel.** The ONNXRuntime worker exceeded the 900-second subprocess timeout and produced no result. This is not a measurement artefact but a direct consequence of the MLAS math library not implementing a half-precision convolution kernel for ARM. When MLAS encounters a FP16 Conv operator, it inserts `Cast(float16$ to $float32)` nodes before and after every convolution and computes the operation in FP32. The resulting per-inference latency is estimated at approximately 7 100 ms - derived by scaling the FP32 ORT result by the same regression factor observed in PyTorch eager FP16. With 30 warm-up and 200 timed iterations, the total subprocess time would reach approximately 1 630 seconds, well above the timeout. This behaviour has been documented in the ONNX Runtime issue tracker and reflects a fundamental gap in MLAS coverage rather than a configuration error.

**TFLite - Storage Benefit Only.** TFLite produced virtually identical latency at both precisions: 154.06 ms in FP32 and 153.27 ms in FP16, a difference smaller than one standard deviation. This result confirms that TFLite’s FP16 model format stores weights as 16-bit values but promotes all arithmetic to FP32 before computation on this platform. The model occupies half the memory, which can reduce cache pressure and improve load

time, but no arithmetic throughput benefit accrues at inference time. The near-identical power consumption at both precisions (2 293 mW FP16 vs 2 293 mW FP32) corroborates this interpretation.

**TorchCompile and PyTorch - Regression Under FP16.** Both PyTorch-based backends degraded significantly at FP16. PyTorch eager mode regressed by a factor of  $38\times$ , from 290 ms to approximately 11 000 ms. When `model.half()` is called on a CPU device, the ATen kernel dispatcher finds no half-precision GEMM implementation and falls back to an element-wise scalar loop that processes each `float16` value individually, upcasting to `float32` for each multiply-accumulate step.

TorchCompile degraded by  $4.9\times$ , from 200 ms to 983 ms. This is less severe than PyTorch eager because the Inductor backend successfully generates vectorised C++ code for element-wise operations such as the SiLU activation and residual additions. However, the convolution operators - which dominate total runtime - are not covered by Inductor’s FP16 CPU code generation path and still fall through to the same ATen scalar fallback. The resulting graph alternates between fast Inductor-compiled segments and slow ATen scalar segments, with type-conversion overhead at every boundary. The net effect is that TorchCompile FP16 is slower than TorchCompile FP32 despite Inductor genuinely improving some of the individual operations.

**Power Consumption.** Table 5.13 reports the mean board power measured during inference for the backends that completed successfully.

Table 5.13: Power and energy-efficiency comparison on the Raspberry Pi 5 CPU at FP16 precision (successful runs only).

| Backend      | Mean latency (ms) | Power (mW) | Energy/frame (mJ) | FPS/W |
|--------------|-------------------|------------|-------------------|-------|
| TVM          | 122.44            | 4 232      | 517.7             | 1.93  |
| TFLite       | 153.27            | 2 291      | 351.1             | 2.85  |
| TorchCompile | 983.08            | 2 202      | 2 164.0           | 0.45  |

TFLite achieves the best energy efficiency at FP16 (2.85 FPS/W), primarily because its lightweight runtime draws substantially less board power than the TVM or Inductor runtimes while delivering competitive latency. TVM draws more power (4 232 mW) due to higher CPU utilisation from its densely vectorised kernels, but still outperforms TFLite in absolute latency. TorchCompile consumes the least power per unit time but has by far the highest energy per frame because of its poor throughput.

**Overall Assessment.** The Raspberry Pi 5 FP16 results demonstrate that half-precision acceleration on ARM CPU is a compiler problem rather than a hardware problem. The

Cortex-A76 implements ARMv8.2-A FEAT\_FP16, providing native `float16` arithmetic in its NEON units. The bottleneck is that no major CPU runtime library - MLAS, OpenBLAS, or the ATen dispatcher - implements a half-precision GEMM kernel for this target. Only TVM, which generates the convolution kernel directly as LLVM IR and lets the AArch64 backend select `float16x8` NEON instructions, can exploit the available hardware capability. The result is a  $1.57\times$  end-to-end speedup over FP32 TVM, achieved within a four-hour tuning session. All other frameworks either time out, regress, or show no improvement at FP16 precision. This finding has a direct practical implication: deploying FP16 models on edge CPU platforms without a compilation-based inference engine yields no latency benefit and may cause severe degradation.

## 5.5 Accuracy Comparison Across Inference Frameworks

To verify that different inference frameworks preserve the predictive quality of the original model, YOLOv8n was evaluated across several backends and compared against the default PyTorch eager FP32 implementation. Since large multi-backend evaluation is memory-intensive, especially when repeatedly loading different runtimes and exported artifacts, the comparison was performed on a 1000-image subset of the validation set rather than on the full validation split. Although smaller than the full COCO validation set, this subset is sufficient to determine whether export, compilation, or reduced precision introduce any meaningful degradation in detection quality.

All backends were evaluated using the same basic setup: image size  $640 \times 640$ , batch size 1, confidence threshold 0.001, IoU threshold 0.7, and maximum number of detections 300. Each backend was loaded and evaluated sequentially in isolation in order to reduce memory pressure and avoid interference between runtime environments.

Because the GPU-oriented and CPU-oriented backends were evaluated in separate comparison runs, they are presented separately below. The small difference between the corresponding PyTorch FP32 baseline values should therefore be interpreted as run-to-run variation rather than as a framework effect.

### GPU-Oriented Backends

The GPU-side comparison included PyTorch eager FP32/FP16, ONNX Runtime CUDA FP32/FP16, TensorRT FP32/FP16, and TVM GPU FP32/FP16. Absolute results are shown in Table 5.14, and relative changes with respect to the PyTorch eager FP32 baseline of the same run are given in Table 5.15.

Table 5.14: Absolute detection metrics for GPU-oriented backends on 1000 validation images.

| Backend            | Precision | Recall | mAP50  | mAP50–95 |
|--------------------|-----------|--------|--------|----------|
| PyTorch eager FP32 | 0.6299    | 0.4878 | 0.5384 | 0.3899   |
| PyTorch eager FP16 | 0.6301    | 0.4880 | 0.5385 | 0.3898   |
| ONNX Runtime FP32  | 0.6283    | 0.4905 | 0.5386 | 0.3899   |
| ONNX Runtime FP16  | 0.6300    | 0.4888 | 0.5387 | 0.3900   |
| TensorRT FP32      | 0.6282    | 0.4893 | 0.5385 | 0.3899   |
| TensorRT FP16      | 0.6248    | 0.4907 | 0.5385 | 0.3901   |
| TVM GPU FP32       | 0.6319    | 0.4889 | 0.5390 | 0.3899   |
| TVM GPU FP16       | 0.6290    | 0.4893 | 0.5381 | 0.3894   |

Table 5.15: Relative changes (%) for GPU-oriented backends with respect to PyTorch eager FP32.

| Backend            | $\Delta$ Precision (%) | $\Delta$ Recall (%) | $\Delta$ mAP50 (%) | $\Delta$ mAP50–95 (%) |
|--------------------|------------------------|---------------------|--------------------|-----------------------|
| PyTorch eager FP16 | +0.03                  | +0.03               | +0.02              | -0.01                 |
| ONNX Runtime FP32  | -0.25                  | +0.54               | +0.05              | +0.02                 |
| ONNX Runtime FP16  | +0.02                  | +0.19               | +0.06              | +0.02                 |
| TensorRT FP32      | -0.27                  | +0.30               | +0.02              | +0.01                 |
| TensorRT FP16      | -0.81                  | +0.58               | +0.03              | +0.07                 |
| TVM GPU FP32       | +0.32                  | +0.23               | +0.12              | +0.00                 |
| TVM GPU FP16       | -0.14                  | +0.29               | -0.05              | -0.12                 |

The GPU results show that all backends preserved accuracy very well. The observed changes in mAP50 and mAP50–95 were very small, indicating that export to ONNX Runtime, TensorRT, and TVM did not introduce any practically meaningful loss in detector quality. FP16 also remained stable overall. PyTorch FP16, ONNX Runtime CUDA FP16, and TensorRT FP16 all stayed extremely close to the FP32 baseline. TVM GPU FP16 showed the largest negative deviation on the GPU side, but the absolute change remained small.

### CPU-Oriented Backends

A separate CPU-side comparison included PyTorch eager FP32/FP16, ONNX Runtime CPU, TFLite CPU FP32/FP16, and TVM CPU FP32/FP16. Absolute results are shown in Table 5.16, and relative changes with respect to the PyTorch eager FP32 baseline of that run are shown in Table 5.17.

The CPU results are similarly stable. PyTorch FP16, ONNX Runtime CPU, TFLite FP32 CPU, and TVM CPU FP32 reproduced the PyTorch FP32 baseline almost exactly. TFLite FP16 and TVM CPU FP16 introduced small shifts in the balance between

Table 5.16: Absolute detection metrics for CPU-oriented backends on 1000 validation images.

| Backend            | Precision | Recall | mAP50  | mAP50–95 |
|--------------------|-----------|--------|--------|----------|
| PyTorch eager FP32 | 0.6291    | 0.4884 | 0.5384 | 0.3898   |
| PyTorch eager FP16 | 0.6291    | 0.4884 | 0.5384 | 0.3898   |
| ONNX Runtime CPU   | 0.6291    | 0.4884 | 0.5384 | 0.3898   |
| TFLite FP32 CPU    | 0.6291    | 0.4884 | 0.5384 | 0.3898   |
| TFLite FP16 CPU    | 0.6276    | 0.4905 | 0.5384 | 0.3899   |
| TVM CPU FP32       | 0.6291    | 0.4884 | 0.5384 | 0.3898   |
| TVM CPU FP16       | 0.6280    | 0.4894 | 0.5395 | 0.3900   |

Table 5.17: Relative changes (%) for CPU-oriented backends with respect to PyTorch eager FP32.

| Backend            | $\Delta$ Precision (%) | $\Delta$ Recall (%) | $\Delta$ mAP50 (%) | $\Delta$ mAP50–95 (%) |
|--------------------|------------------------|---------------------|--------------------|-----------------------|
| PyTorch eager FP16 | +0.00                  | +0.00               | +0.00              | +0.00                 |
| ONNX Runtime CPU   | -0.00                  | +0.00               | +0.00              | +0.00                 |
| TFLite FP32 CPU    | -0.00                  | -0.00               | +0.00              | +0.00                 |
| TFLite FP16 CPU    | -0.24                  | +0.43               | +0.00              | +0.03                 |
| TVM CPU FP32       | -0.00                  | -0.00               | +0.00              | +0.00                 |
| TVM CPU FP16       | -0.17                  | +0.20               | +0.20              | +0.06                 |

precision and recall, but the aggregate mAP values remained effectively unchanged. In particular, TVM CPU FP16 slightly improved mAP50 and mAP50–95 while slightly reducing precision, again indicating only minor numerical variation rather than a meaningful change in model behavior.

## Discussion

Across both GPU and CPU experiments, no framework caused a significant drop in detection quality. The most important result is that mAP50 and especially mAP50–95 remained very close to the corresponding PyTorch eager FP32 baselines in all runs. This shows that the predictive behavior of YOLOv8n is preserved well across the evaluated deployment frameworks.

Because all backends execute the same trained model, the remaining small differences should not be interpreted as one framework being inherently more accurate than another. Instead, they are better explained by runtime-dependent numerical variation. Different inference frameworks may use different kernel implementations, operator fusion patterns, accumulation orders, and precision conversion paths. In object detection, even tiny numerical differences can slightly shift confidence scores or bounding box coordinates, which may then change whether a detection is kept or removed during thresholding and non-

maximum suppression. This explains the recurring pattern that some FP16 configurations slightly reduce precision while slightly increasing recall, without materially changing the overall mAP.

Overall, these results indicate that framework conversion and mixed-precision inference do not meaningfully degrade YOLOv8n accuracy in the evaluated setups. This justifies focusing the later comparison primarily on latency, throughput, and hardware efficiency, since accuracy was effectively preserved across both GPU-oriented and CPU-oriented backends.

## 5.6 Influence of Quantization on YOLOv8n

Quantization is often presented as a direct way to improve inference efficiency by reducing numerical precision, model size, and memory bandwidth requirements. In practice, however, quantization remains a difficult deployment problem. Its final effect depends not only on the quantization method itself, but also on the maturity of the compiler or runtime stack, the target hardware, and the way quantized operators are represented in the exported graph.

### 5.6.1 INT8 Inference on NVIDIA Orin Nano GPU

For this reason, the quantization study in this thesis focuses on practical deployment behavior rather than on theoretical INT8 speedup alone. The experiments were carried out on the Jetson Orin Nano GPU and compared FP32 and INT8 inference across several deployment paths. The PyTorch FP32 model served as the reference baseline. ONNX Runtime was evaluated using statically quantized QuantizeLinear–DequantizeLinear (QDQ) graphs, while TensorRT was evaluated both with an implicit INT8 calibration path starting from an FP32 ONNX model and with a QDQ-based INT8 ONNX graph.

A first important observation is that efficient quantization support is not uniformly available across deployment frameworks. For example, while Apache TVM provides strong support for FP32 compilation and scheduling through MetaSchedule, the Relax-based quantization flow remains immature in the tested version. In practice, implementing a custom INT8 path would require calibration tooling, graph rewriting, requantization logic, and validated low-precision kernels, which is beyond the scope of the present thesis. By contrast, ONNX Runtime and TensorRT both provide deployable INT8 paths, but their practical outcomes differ substantially.

ONNX Runtime static quantization inserts explicit `QuantizeLinear` and `Dequantize`

**Linear** operators into the computational graph. Although this representation is portable, it may also introduce substantial overhead when these conversions are not fully fused away by the execution backend. In a convolution-heavy architecture such as YOLOv8n, this is especially problematic, since quantization boundaries may appear repeatedly throughout the network. TensorRT can ingest such QDQ graphs, but the resulting execution still contains additional quantization-related transformations, reformatting steps, and copy operations. An alternative TensorRT path is implicit INT8 calibration, where the engine is built from an FP32 ONNX model and calibrated directly for TensorRT’s internal low-precision execution. This remains one of the most accessible hardware-native INT8 deployment mechanisms on Jetson-class devices.

Table 5.18 summarizes the detection accuracy results. All measurements were performed on the same set of 1000 images from the COCO validation split. TensorRT FP32 reproduces the PyTorch FP32 baseline almost exactly, confirming that TensorRT export by itself does not meaningfully affect model quality. In contrast, all evaluated INT8 paths reduce accuracy, but the magnitude of the degradation varies strongly. The ONNX Runtime QDQ models lead to a moderate drop of approximately 6–9% in mAP, depending on the calibration subset. TensorRT implicit INT8 calibration performs worse in accuracy than the FP32 TensorRT engine and also remains less accurate than the previously evaluated FP16 TensorRT configuration, but it is still clearly more accurate than the TensorRT QDQ path.

The TensorRT QDQ engine exhibits the largest accuracy degradation, with mAP50 decreasing by more than 25% and mAP50–95 by nearly 42%, which makes this variant impractical for deployment in the tested setup. A likely reason is that the ONNX QDQ graph requires additional adjustment before efficient TensorRT execution becomes possible. In particular, TensorRT does not support the default UINT8 zero-point convention commonly used in ONNX graphs, while the exported QDQ graph used here also quantizes bias tensors and final network outputs. As a result, the graph is not directly aligned with TensorRT’s preferred low-precision execution path and requires extra transformations that can negatively affect both numerical behavior and final accuracy.

The same trend is visible in relative form in Table 5.19. Among the ONNX Runtime QDQ runs, the 0.1% calibration subset, corresponding to roughly 130 images, produced the least severe degradation, but the differences between the tested calibration splits are relatively small compared with the overall gap between INT8 and FP32 execution. This suggests that calibration-set size alone is not the dominant factor in the observed loss. Instead, the larger issue appears to be the deployment path itself and the extent to which low-precision conversions are propagated and fused efficiently.

Figure 5.7 visualises the INT8 benchmark results on the NVIDIA Jetson Orin Nano GPU, including latency, speedup relative to the PyTorch FP32 eager baseline, power con-

Table 5.18: Accuracy comparison of FP32 and INT8 YOLOv8n variants on the Jetson Orin Nano.

| Backend                      | Precision | Recall | mAP50  | mAP50-95 |
|------------------------------|-----------|--------|--------|----------|
| PyTorch FP32                 | 0.6299    | 0.4878 | 0.5384 | 0.3899   |
| ONNX Runtime INT8 QDQ (1%)   | 0.6049    | 0.4573 | 0.5053 | 0.3548   |
| ONNX Runtime INT8 QDQ (0.1%) | 0.6122    | 0.4694 | 0.5067 | 0.3591   |
| ONNX Runtime INT8 QDQ (10%)  | 0.5872    | 0.4682 | 0.5001 | 0.3533   |
| TensorRT FP32                | 0.6288    | 0.4905 | 0.5386 | 0.3899   |
| TensorRT INT8 implicit       | 0.5806    | 0.4418 | 0.4846 | 0.3481   |
| TensorRT INT8 QDQ            | 0.4781    | 0.4162 | 0.4016 | 0.2264   |

Table 5.19: Relative accuracy change (%) of INT8 and TensorRT backends with respect to the PyTorch FP32 baseline.

| Backend                      | $\Delta$ Precision | $\Delta$ Recall | $\Delta$ mAP50 | $\Delta$ mAP50-95 |
|------------------------------|--------------------|-----------------|----------------|-------------------|
| ONNX Runtime INT8 QDQ (0.1%) | -2.80              | -3.78           | -5.88          | -7.89             |
| ONNX Runtime INT8 QDQ (1%)   | -3.96              | -6.26           | -6.15          | -8.98             |
| ONNX Runtime INT8 QDQ (10%)  | -6.78              | -4.02           | -7.11          | -9.37             |
| TensorRT FP32                | -0.18              | +0.56           | +0.05          | +0.02             |
| TensorRT INT8 implicit       | -7.83              | -9.43           | -9.98          | -10.71            |
| TensorRT INT8 QDQ            | -24.10             | -14.68          | -25.40         | -41.93            |

sumption, and energy efficiency. Latency results, shown numerically in Table 5.20, reveal an equally important deployment trade-off. The fastest configuration in the measured benchmark was TensorRT implicit INT8 quantization, achieving approximately 5.34 ms mean latency and 187.23 FPS. This corresponds to a  $4.69\times$  speedup over the PyTorch FP32 eager baseline. ONNX Runtime INT8 QDQ, despite using an INT8 graph, was actually slower than the PyTorch FP32 baseline in the presented experiment, reaching approximately 39.82 ms per inference, or only  $0.63\times$  of the baseline speed. TensorRT FP16, discussed in the previous section, already provides a strong optimized reference at 6.31 ms, which means that the practical value of INT8 must be judged against both accuracy loss and the already highly optimized higher-precision TensorRT paths. The QDQ-based TensorRT engine reached 7.71 ms, which is still faster than PyTorch and ONNX Runtime, but notably slower than the best TensorRT implicit INT8 result.

The profiling data helps explain why these differences occur. For the implicit TensorRT INT8 engine, the NVIDIA Tools Extension (NVTX) summary shows that execution is dominated by the TensorRT enqueue stage, while the GPU kernel summary is largely composed of INT8 GEMM and IMMA-style convolution kernels. This indicates that Ten-

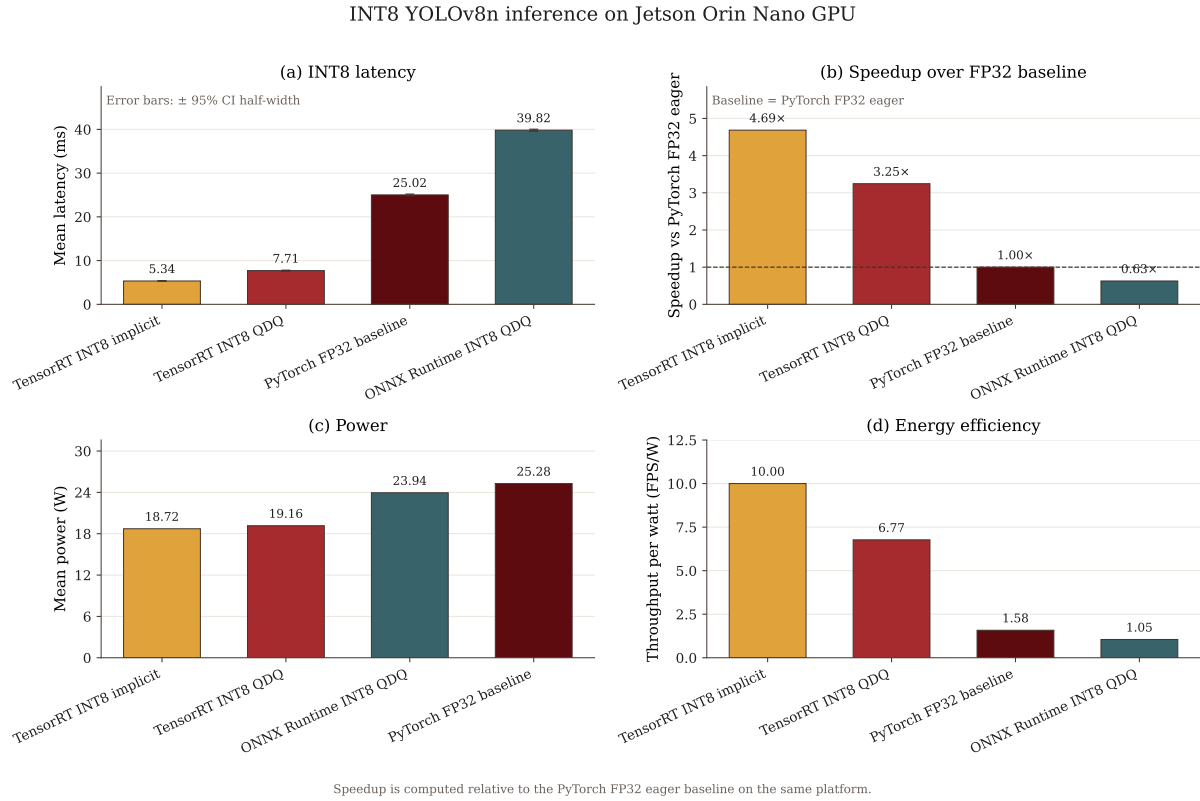


Figure 5.7: INT8 YOLOv8n inference performance on the NVIDIA Jetson Orin Nano GPU. Subfigure (a) shows mean end-to-end latency with error bars indicating the half-width of the 95% confidence interval. Subfigure (b) reports speedup relative to the PyTorch FP32 eager baseline, where values above 1.0 $\times$  indicate faster execution than the baseline. Subfigure (c) compares mean board power consumption, and Subfigure (d) shows energy efficiency in terms of throughput per watt. TensorRT implicit INT8 achieves the best result, reaching the lowest latency, the highest speedup, and the highest throughput per watt. ONNX Runtime INT8 QDQ is slower than the PyTorch FP32 baseline despite using INT8 quantization.

TensorRT is able to map a large fraction of the network to hardware-efficient INT8 kernels. At the same time, the profile still contains non-negligible reformat, copy, and permutation kernels, which shows that even the more successful implicit INT8 path does not achieve a fully clean low-precision execution pipeline.

For QDQ-based deployment, the profile is substantially more cluttered with quantization-related operations. Many NVTX ranges explicitly include QuantizeLinear/DequantizeLinear stages together with clone, copy, and tensor-reformat operations. This confirms that a significant portion of the runtime is spent not on useful convolution or detection work, but on managing low-precision data transformations between operators. In a model with many convolutional blocks and concatenation-heavy structures such as YOLOv8n, these additional graph boundaries accumulate quickly and can dominate the practical cost of the supposed INT8 path.

Table 5.20: End-to-end YOLOv8n inference performance on the NVIDIA Jetson Orin Nano GPU. Values are reported as mean latency  $\pm$  half-width of the 95% confidence interval across 5 isolated repetitions, with 200 timed runs per repetition after 30 warm-up iterations.

| Backend                 | Latency (ms)     | FPS mean | Median (ms) |
|-------------------------|------------------|----------|-------------|
| PyTorch FP32 (baseline) | $25.02 \pm 0.18$ | 39.96    | 25.10       |
| ONNX Runtime INT8 QDQ   | $39.82 \pm 0.21$ | 25.11    | 39.82       |
| TensorRT INT8 implicit  | $5.34 \pm 0.05$  | 187.23   | 5.34        |
| TensorRT INT8 QDQ       | $7.71 \pm 0.09$  | 129.66   | 7.69        |

Table 5.21: Power and energy-efficiency results for selected YOLOv8n backends on the NVIDIA Jetson Orin Nano GPU.

| Backend                 | Power mean (mW) | Power max (mW) | FPS/W |
|-------------------------|-----------------|----------------|-------|
| PyTorch FP32 (baseline) | 25276           | 27646          | 1.58  |
| ONNX Runtime INT8 QDQ   | 23941           | 27674          | 1.05  |
| TensorRT INT8 implicit  | 18720           | 20466          | 10.00 |
| TensorRT INT8 QDQ       | 19161           | 21678          | 6.77  |

These results show that quantization cannot be evaluated only by checking whether a graph nominally uses INT8 values. What matters is whether the compiler and runtime can preserve low-precision execution through fused kernels and avoid repeated format conversions. In the present study, ONNX Runtime QDQ quantization produced limited or even negative efficiency benefits while still causing noticeable accuracy degradation. TensorRT QDQ improved latency relative to the PyTorch baseline, but the loss in accuracy was severe and the profile showed substantial conversion overhead. TensorRT implicit INT8 calibration produced the best speed among the tested INT8 approaches, but still degraded accuracy by roughly 10% in mAP50 and mAP50–95 relative to the FP32 baseline.

Overall, the results suggest that quantization for YOLOv8n on Jetson Orin Nano is highly deployment-path dependent. A naive or graph-heavy INT8 representation does not guarantee a practically useful result. In this work, the most effective TensorRT deployment path remained the standard optimized TensorRT engine in higher precision, while INT8 became attractive only when the hardware-native implicit TensorRT calibration path was used. Even then, the speed gain came with a clear drop in predictive quality. Quantization therefore appears here as a deployment trade-off rather than as a universally beneficial optimization.

Statistical significance was evaluated with a paired bootstrap test on validation images using 1000 resamples, and confidence intervals were computed with the percentile method. The bootstrap analysis confirmed that the evaluated INT8 variants produced a statistically significant decrease in detection accuracy relative to the FP32 baseline, with

the strongest degradation observed for the TensorRT QDQ path.

### 5.6.2 INT8 Inference on Raspberry Pi 5 CPU

To evaluate the practical impact of integer quantization on an embedded CPU platform, additional benchmarks were performed on the Raspberry Pi 5 using INT8 YOLOv8n models. The comparison focused on two deployment frameworks that provided a usable CPU INT8 path in the present study: TFLite and ONNX Runtime. The experiments were executed with the same rigorous benchmarking procedure used in the previous sections: 5 isolated repetitions, 200 timed runs per repetition, 30 warm-up iterations, performance governor enabled, and active cooldown control between repetitions.

The resulting latency and power measurements are summarized in Table 5.22. Accuracy was evaluated on a 1000-image subset of the COCO validation split and is summarized in Table 5.23. The PyTorch FP32 model is used as the accuracy baseline.

Table 5.22: End-to-end INT8 YOLOv8n inference results on Raspberry Pi 5 CPU. Values are reported as mean latency  $\pm$  half-width of the 95% confidence interval across 5 isolated repetitions, with 200 timed runs per repetition after 30 warm-up iterations.

| Backend           | Latency (ms)      | FPS mean | Power (mW) | FPS/W |
|-------------------|-------------------|----------|------------|-------|
| TFLite INT8       | 31.05 $\pm$ 0.17  | 32.20    | 2340       | 13.76 |
| ONNX Runtime INT8 | 133.81 $\pm$ 2.31 | 7.47     | 4394       | 1.70  |

Table 5.23: Accuracy of INT8 YOLOv8n variants on Raspberry Pi 5 CPU. Relative changes are computed with respect to the PyTorch FP32 baseline on the same 1000-image validation subset.

| Backend                 | Precision | Recall | mAP50  | mAP50–95 |
|-------------------------|-----------|--------|--------|----------|
| PyTorch FP32 (baseline) | 0.6291    | 0.4884 | 0.5384 | 0.3898   |
| ONNX Runtime INT8 QDQ   | 0.5729    | 0.4801 | 0.5052 | 0.3557   |
| TFLite INT8             | 0.6172    | 0.4450 | 0.5042 | 0.3486   |

The corresponding relative accuracy changes are shown in Table 5.24.

Table 5.24: Relative accuracy change of INT8 backends on Raspberry Pi 5 CPU with respect to the PyTorch FP32 baseline.

| Backend               | $\Delta$ Prec. (%) | $\Delta$ Rec. (%) | $\Delta$ mAP50 (%) | $\Delta$ mAP50–95 (%) |
|-----------------------|--------------------|-------------------|--------------------|-----------------------|
| ONNX Runtime INT8 QDQ | -8.93              | -1.70             | -6.16              | -8.74                 |
| TFLite INT8           | -1.89              | -8.88             | -6.35              | -10.56                |

A striking result is the large performance advantage of TFLite INT8. With a mean latency of only 31.05 ms, it is more than four times faster than ONNX Runtime INT8 and

substantially faster than all previously evaluated FP32 and FP16 CPU configurations. Compared with the earlier TFLite FP32/FP16 results (approximately 154 ms), the INT8 TFLite model achieves roughly a  $5\times$  speedup. At the same time, it also delivers by far the best energy efficiency, reaching 13.76 FPS/W.

This behavior is plausible on the Raspberry Pi 5 CPU. TFLite is strongly optimized for mobile and embedded processors and is able to execute quantized models using an efficient integer arithmetic path for a large portion of the network. In this case, both weights and activations are represented with lower precision, reducing memory traffic and improving cache utilization. On a CPU-bound platform such as Raspberry Pi 5, this can produce a much larger practical gain than would be expected from arithmetic reduction alone.

By contrast, ONNX Runtime uses a QDQ representation in which explicit **QuantizeLinear** and **DequantizeLinear** operators are inserted into the graph. If these conversions are not fully eliminated or fused during execution, a significant part of the runtime may be spent on tensor conversions rather than on useful convolution work. In a model such as YOLOv8n, which contains many convolution blocks, concatenation operations, and detection-head transformations, these graph boundaries can accumulate and reduce the practical benefit of quantization. This explains why ONNX Runtime INT8 still improves over its FP32 CPU path, but only modestly compared with TFLite.

From the accuracy perspective, both INT8 variants introduce a noticeable degradation relative to the PyTorch FP32 baseline. The mAP50 drop is similar for the two backends, around 6%, while the mAP50–95 reduction reaches 8.74% for ONNX Runtime and 10.56% for TFLite. The error pattern is not identical: ONNX Runtime loses more precision, whereas TFLite preserves precision better but shows a larger drop in recall. This suggests that the two deployment paths affect confidence scores and box regression somewhat differently, even though both are based on INT8 inference.

On Raspberry Pi 5 CPU, Linux perf profiling showed that TFLite INT8 inference spent most active execution time inside clearly identifiable Accelerated Neural Network (XNNPACK) quantized microkernels, such as Integer GEMM (IGEMM)/GEMM and LUT-based kernels. In contrast, ONNX Runtime INT8 required substantially more total cycles and instructions, achieved a lower Instructions Per Cycle (IPC), and exhibited a higher Last-Level Cache (LLC) miss rate. This indicates that, for this model and deployment path, TFLite provided a more efficient CPU execution path, likely due to better-optimized low-level kernels and/or lower runtime overhead.

Overall, the Raspberry Pi 5 experiments show that quantization can deliver very large practical benefits on CPU, but only when the runtime can preserve a genuinely efficient integer execution path. In the evaluated setup, TFLite INT8 was the most effective

deployment option by a large margin in both latency and energy efficiency. However, this gain came with a measurable reduction in detection accuracy, which means that INT8 deployment remains a trade-off between efficiency and predictive quality rather than a universally beneficial optimization.

## 5.7 Evaluation of YOLOv8n Pruning

Structured pruning was evaluated as a strategy to reduce the computational cost of YOLOv8n on edge hardware without requiring a change of inference framework or precision format. The motivation was to investigate whether selectively removing output channels from convolutional layers could yield measurable latency reductions while preserving acceptable detection accuracy. The investigation was split into two stages: a systematic per-layer sensitivity sweep to identify which layers are most and least sensitive to channel reduction, followed by a full-model pruning experiment with fine-tuning.

### 5.7.1 Structural Constraints of YOLOv8n

Applying structured pruning to YOLOv8n requires handling several architectural constraints. When an output channel is removed from a convolutional layer, the corresponding input channels of all downstream layers that consume that feature map must be removed as well. This dependency chain must be resolved correctly for every targeted layer.

The most complex case is the **C2f** block illustrated in Figure 5.8 (used in `model.2`, `model.4`, `model.6`, `model.8`, and the detection neck). Each **C2f** block contains two boundary convolutions (`cv1`, `cv2`) and  $n$  bottleneck units whose outputs are concatenated along the channel dimension before being passed to `cv2`. Pruning the output channels of `cv1` affects the inputs of every bottleneck; pruning a bottleneck’s output affects the concatenated tensor feeding into `cv2`. This means that channel reduction in any **C2f** constituent requires consistent propagation through all branches of the block simultaneously. The shortcut connections within the bottleneck units impose a further constraint: the two branches of each residual addition must maintain equal channel counts, which limits the granularity at which individual components can be pruned independently.

Handling these dependencies required a custom propagation pass that, for each targeted layer, identified all dependent downstream layers and adjusted their weight tensors accordingly.

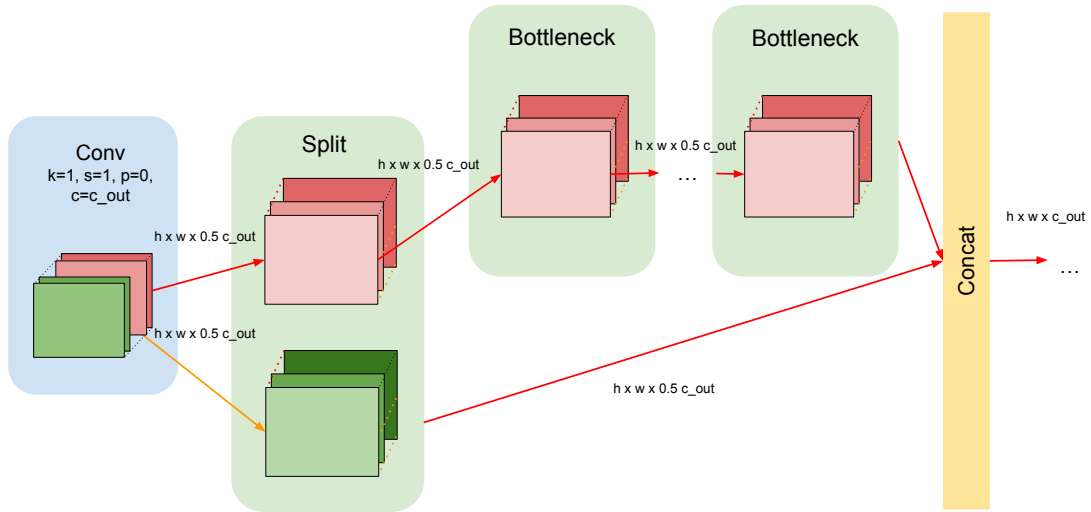


Figure 5.8: Data flow and channel dependencies within the C2f block.

### 5.7.2 Experimental Setup

Pruning experiments were split across two machines depending on whether retraining was required. The per-layer sensitivity sweep, which involves no gradient computation, was run on the Jetson Orin Nano using PyTorch FP32 with the CUDA backend. The full-model pruning with fine-tuning and the final benchmarks were run on a server equipped with an NVIDIA GeForce RTX 3070 GPU (`cuda:0`). All inference measurements used PyTorch FP32, batch size 1. The target model in all cases was the standard YOLOv8n checkpoint trained on COCO, with 3 157 200 parameters and 8.7 GFLOPs. Detection accuracy was evaluated on the COCO 2017 validation set; the unmodified baseline achieved mAP50-95 of 0.3715 and mAP50 of 0.5211.

**Per-Layer Sensitivity Sweep.** A dedicated script (`layer\_speed\_sensitivity.py`) swept 38 convolutional layers at sparsity levels from 10% to 99% in roughly 10% steps, covering every `cv1/cv2` boundary and all bottleneck internals across the backbone and detection neck. The sweep was executed on the Jetson Orin Nano using PyTorch FP32 with the CUDA backend and `--no-preload` to avoid saturating the device’s unified CPU/GPU memory pool. The baseline inference latency on this platform was 25.1 ms (batch size 1, 300 timed runs after 100 warm-up iterations), consistent with the PyTorch FP32 baseline reported in the backend comparison (Table 5.8). Accuracy was evaluated on up to 1 000 COCO validation images per configuration. For each configuration, output channels were selected for removal by ranking them according to their L2 weight norm (magnitude-based criterion); channels with the smallest norms were discarded first. Only

the **any** parity mode (nearest integer to the target) was used, yielding approximately 418 configurations in total. Each configuration was evaluated without any subsequent fine-tuning so that the latency change and accuracy drop could be attributed purely to the structural modification rather than to retraining.

**Full-Model Pruning with Fine-Tuning.** A separate experiment, conducted on the RTX 3070 server, applied magnitude-based channel pruning across multiple layers simultaneously, selecting the combination of layers that appeared most resilient in the sensitivity sweep. The pruned model was then fine-tuned for 50 epochs on a 1 % subset of the COCO training set using the standard Ultralytics training pipeline with half-precision gradients (`amp=true`). The fine-tuned checkpoint was benchmarked against both the same model before pruning and the stock YOLOv8n baseline, with 200 timed runs after 30 warm-up iterations.

### 5.7.3 Per-Layer Sensitivity Results

Table 5.25 summarises the speedup and accuracy impact for selected sparsity levels on three representative layers from the Jetson Orin Nano sweep. The full sweep covered 418 configurations; only the most informative points are shown.

Table 5.25: Per-layer pruning sensitivity on Jetson Orin Nano (PyTorch FP32, CUDA, batch 1, no fine-tuning). Speedup is relative to the unpruned baseline (25.1 ms). Accuracy drop is absolute change in mAP50-95 relative to baseline (0.392).

| Layer                 | Sparsity | Kept ch | Speedup | Latency (ms) | $\Delta$ mAP50-95 |
|-----------------------|----------|---------|---------|--------------|-------------------|
| model.21.cv2 (256 ch) | 0 %      | 256     | 1.016×  | 24.66        | +0.00             |
|                       | 20 %     | 205     | 1.010×  | 24.75        | −0.023            |
|                       | 50 %     | 128     | 1.011×  | 24.74        | −0.057            |
|                       | 70 %     | 77      | 1.010×  | 24.77        | −0.125            |
|                       | 90 %     | 26      | 1.013×  | 24.77        | −0.176            |
| model.18.cv2 (128 ch) | 0 %      | 128     | 1.011×  | 24.74        | +0.00             |
|                       | 20 %     | 102     | 1.016×  | 24.60        | −0.012            |
|                       | 50 %     | 64      | 1.015×  | 24.64        | −0.114            |
|                       | 70 %     | 38      | 1.002×  | 24.96        | −0.197            |
|                       | 90 %     | 13      | 1.015×  | 24.65        | −0.260            |
| model.2.cv1 (16 ch)   | 12.5 %   | 14      | 1.005×  | 25.00        | −0.362            |
|                       | 50.0 %   | 8       | 1.009×  | 24.93        | −0.389            |
|                       | 68.75 %  | 5       | 1.011×  | 24.87        | −0.392            |

The results reveal a consistent pattern across all 38 targeted layers: channel reduction provides essentially no latency benefit. Across all 418 configurations, the speedup

ratio ranged from 0.975 to 1.021, with a mean of approximately 1.005. No configuration exceeded a  $1.05\times$  speedup, and the distribution was centred near 1.0, indicating that latency is statistically unchanged by channel reduction.

The accuracy side of the tradeoff was more consistent: pruning aggressively degrades detection quality. Removing 50 % of the channels from `model.18.cv2` caused a 0.114 point drop in mAP50-95 (29 % relative degradation) while yielding only a 1.5 % latency improvement. Removing 70 % of channels from the same layer reduced mAP50-95 by 0.197 points (50 % relative) with essentially no latency gain. The earliest backbone layer (`model.2.cv1`) proved particularly fragile: even removing just 12.5 % of its channels caused a 0.362 point mAP50-95 collapse, consistent with the role of early layers in forming the low-level feature representations that all subsequent computations depend upon.

**Most Resilient Layers.** The layers with the smallest accuracy impact at moderate sparsity were concentrated in the detection neck and later backbone stages. Table 5.26 lists the ten most resilient layers identified in the Orin Nano sweep.

Table 5.26: Ten most accuracy-resilient layers in YOLOv8n under single-layer magnitude pruning without fine-tuning (Jetson Orin Nano, COCO val 1 000-image subset, mAP50-95 baseline 0.3921, evaluated at the closest available sparsity step to 12.5 %).

| Layer                         | Pruned mAP50-95 | $\Delta$ mAP50-95 |
|-------------------------------|-----------------|-------------------|
| <code>model.12.m.0.cv1</code> | 0.3915          | -0.0006 (-0.14 %) |
| <code>model.15.m.0.cv2</code> | 0.3906          | -0.0015 (-0.38 %) |
| <code>model.12.m.0.cv2</code> | 0.3905          | -0.0015 (-0.39 %) |
| <code>model.16</code>         | 0.3902          | -0.0018 (-0.47 %) |
| <code>model.21.cv2</code>     | 0.3901          | -0.0020 (-0.51 %) |
| <code>model.15.m.0.cv1</code> | 0.3901          | -0.0020 (-0.51 %) |
| <code>model.19</code>         | 0.3892          | -0.0028 (-0.72 %) |
| <code>model.18.m.0.cv2</code> | 0.3892          | -0.0028 (-0.73 %) |
| <code>model.6.m.1.cv1</code>  | 0.3891          | -0.0029 (-0.75 %) |
| <code>model.21.m.0.cv2</code> | 0.3889          | -0.0032 (-0.82 %) |

These are predominantly the bottleneck internals and upsample-path convolutions of the detection neck. By contrast, the backbone’s early layers (`model.2.cv1`, `model.4.cv1`) incurred accuracy drops exceeding 90 % of baseline mAP at comparable sparsity levels, confirming that sensitivity decreases with network depth.

One notable observation on the Orin is the exceptional tolerance of `model.21.cv1` (128 ch): even at 99 % channel removal (keeping a single output channel), mAP50-95 settled at 0.211 - a 46 % relative drop, but far less than the near-total collapse seen in backbone layers at comparable sparsity. This layer is a `cv1` projection inside the last C2f block of the FPN and contributes less directly to the final feature representation than

earlier stages; its redundancy is unusually high. Despite this, the corresponding latency speedup at 99% sparsity was only  $1.014\times$  (24.67 ms vs. 25.1 ms baseline), confirming that even an extreme structural reduction does not translate to a meaningful runtime improvement.

### 5.7.4 Full-Model Pruning Results

The multi-layer pruned and fine-tuned model was benchmarked against both its pre-pruning counterpart and the stock YOLOv8n baseline. Table 5.27 presents the three-way comparison.

Table 5.27: Full-model pruning results on NVIDIA GeForce RTX 3070 (PyTorch FP32, batch 1). Benchmark: 200 timed runs, 30 warm-up iterations. Accuracy: COCO 2017 val. *Stock baseline*: official Ultralytics YOLOv8n pretrained weights. *Fine-tune base (unpruned)*: full-width YOLOv8n fine-tuned for 50 epochs on 1% of COCO, same training procedure as the pruned model but without channel removal; used as a fair paired comparison. *Pruned + fine-tuned*: structurally pruned model (selected channels removed) then fine-tuned for 50 epochs under the same schedule.

| Model                                | mAP50-95        | mAP50  | Mean lat. (ms)    | FPS   |
|--------------------------------------|-----------------|--------|-------------------|-------|
| Stock YOLOv8n (pretrained)           | 0.3714          | 0.5212 | 6.18              | 161.9 |
| Fine-tune base (unpruned, 50 ep)     | 0.3588          | 0.5067 | 6.00              | 166.6 |
| Pruned + fine-tuned (50 ep)          | 0.3446          | 0.4897 | 5.85              | 170.9 |
| $\Delta$ pruned vs. stock pretrained | -0.027 (-7.2 %) | -0.032 | +0.28 ms (+4.6 %) | -7.1  |
| $\Delta$ pruned vs. fine-tune base   | -0.014 (-4.0 %) | -0.017 | -0.15 ms (-2.3 %) | +4.3  |

Compared to its direct paired counterpart - the same fine-tuned model without channel removal - the pruned model was 2.3% faster (5.85 ms vs. 6.00 ms) and lost 0.014 mAP50-95 (4.0% relative). Compared to the stock YOLOv8n baseline, the pruned model was slower by 4.6% (5.85 ms vs. 6.18 ms) while losing 7.2% of mAP50-95. The higher absolute latency of the stock baseline in this benchmark (6.18 ms) compared to the Orin Nano sensitivity sweep (25.1 ms, different device) reflects both the different hardware target and the benchmarking methodology: the RTX 3070 comparison used 200 timed runs with 30 warm-up iterations under higher scheduling variance, whereas the Orin sweep used 300 runs after 100 warm-up iterations in a more controlled per-layer environment.

### 5.7.5 Summary

The pruning experiments produced a clear and consistent negative result: structured channel pruning of YOLOv8n does not yield meaningful latency reductions, whether measured on the Jetson Orin Nano (CUDA sensitivity sweep) or the RTX 3070 (fine-tuned

model benchmarks), when operating in PyTorch FP32. The multi-platform agreement rules out the possibility that the null result is an artefact of a particular runtime or instruction-set alignment, and instead points to a hardware-agnostic root cause.

The fundamental reason lies in how cuDNN dispatches convolution kernels. As shown by the Roofline analysis in Section 5.2, the majority of convolutional layers on the Orin Nano GPU operate in the compute-bound regime, achieving several hundred GFLOP/s with high arithmetic intensity. In principle, reducing the channel count lowers the FLOP count and should bring a compute-bound layer closer to its throughput ceiling. In practice, however, cuDNN selects tiled GEMM or Winograd kernels based on the original tensor dimensions, and those kernels use fixed warp-level tiling strategies that do not shrink proportionally with modest channel reductions. A layer with 10 % fewer channels launches the same kernel, with the same grid geometry and shared-memory footprint, but simply leaves a fraction of the warp lanes idle. The actual reduction in execution time only manifests when the channel count drops far enough to trigger a different kernel configuration - a threshold that the sparsity levels explored here (up to  $\sim 90\%$ ) rarely cross cleanly. This is confirmed by the sweep data: speedups cluster at  $1.00 \pm 0.02$  across all 418 configurations, regardless of sparsity level.

A secondary issue is that YOLOv8n is already a compact model (3.15 M parameters, 8.7 GFLOPs). Compact models leave less redundancy to remove, and the accuracy-speed tradeoff degrades rapidly. The most resilient layers (the detection-neck bottlenecks) offer the smallest accuracy drop at moderate sparsity, but their absolute channel counts are already modest and the channel reductions applied here fall below the threshold at which cuDNN would select a meaningfully different kernel, so removing channels from them provides essentially no latency benefit.

In summary, structured channel pruning applied to the PyTorch FP32 representation of YOLOv8n is not an effective standalone optimization on either hardware target evaluated. The per-layer sweep on the Jetson Orin Nano confirmed a speedup ceiling of approximately  $1.02\times$  across 418 configurations. The full-model pruning on the RTX 3070 achieved at most 2.3 % relative speedup against its paired fine-tuned counterpart, and that gain came at a disproportionate accuracy cost (4.0 % mAP50-95 drop). The comparison against the stock baseline shows a net regression in both metrics. The quantization results reported in Section 5.6.2 - where LiteRT INT8 on the Raspberry Pi 5 achieved a  $9.4\times$  speedup over the PyTorch eager baseline - illustrate that precision reduction is a more effective lever than structural pruning for this class of model and hardware.

It should be noted that all pruning experiments reported here used PyTorch FP32 inference. It is possible that TensorRT’s layer-fusion and kernel-selection passes could extract a modest additional benefit from a reduced channel count, since the engine builder operates on the pruned graph directly, not relying on the runtime to skip channels. How-

ever, given the cuDNN kernel-granularity limitation and the already negligible speedups observed in PyTorch, a substantial gain from that path is not expected.

## 5.8 GPU Kernel-Level Performance Analysis

The TensorRT FP32 engine for YOLOv8n was profiled with Nsight Systems on the Jetson Orin Nano (batch = 1, 640×640 input, without NMS post-processing). The purpose of this session is to decompose GPU execution time by kernel category and identify overhead sources, not to measure absolute end-to-end latency. The authoritative latency figure (11.38 ms mean) comes from the main benchmark in Table 5.8, which used 200 timed runs across 5 isolated repetitions; the profiling session used only 5 timed runs to keep the trace file manageable.

Each inference cycle launches 170 GPU kernels in total: 167 TensorRT inference kernels plus 3 device-to-host copy kernels that transfer the output tensor back to the CPU. Three warm-up passes preceded the timed runs because the very first inference takes 44.4 ms – an 8.8× overhead caused by CUDA compiling all kernel variants on their first call.

### 5.8.1 Kernel Category Breakdown

GPU kernel execution time was grouped into five functional categories from the Nsight Systems kernel summary, covering all 8 profiled runs (3 warm-up + 5 timed). Table 5.28 shows the result.

Table 5.28: GPU kernel time breakdown for TensorRT FP32 YOLOv8n on Orin Nano. Percentages are relative to total GPU kernel time summed across all 8 runs.

| Kernel category                                                                            | Total (ms, 8 runs) | %    |
|--------------------------------------------------------------------------------------------|--------------------|------|
| NHWC convolutions ( <code>sm80_xmma_fprop...nhwc</code> )                                  | 34.5               | 39.5 |
| Pointwise / activations ( <code>generatedNativePointwise</code> )                          | 17.1               | 19.6 |
| NCHW-path convolutions ( <code>sm80_xmma_fprop...nchw</code> )                             | 15.0               | 17.2 |
| Format conversion ( <code>copyPackedKernel</code> , <code>nchwToNhwckKernel</code> , etc.) | 14.2               | 16.3 |
| Pooling, resize, softmax                                                                   | 3.5                | 4.0  |
| Detection head GEMM                                                                        | 3.0                | 3.4  |
| Total                                                                                      | 87.3               | 100  |

From the table we can see that format conversion kernels, which perform no neural network computation, consume 16.3 % of all GPU kernel time. Second, NCHW-path convolutions account for a further 17.2 %: these are bottleneck 3×3 convolutions whose input tensors were described as NCHW at engine-build time, causing cuDNN to perform an internal

NCHW→NHWC conversion on every call instead of using the faster native NHWC path. Together, these two categories represent 33.5 % of GPU time spent on reformatting.

### 5.8.2 C2f Split Copy Overhead

A specific and quantifiable source of format-conversion cost is the Split copy node that TensorRT inserts at every C2f block.

The profiling trace makes the internal tensor format directly observable: the `copyPackedKernel` calls appearing throughout the trace (see Table 5.29) are TensorRT’s kCHW32 pack-unpack kernel - the kernel name itself reflects the “packed” channel layout. TensorRT defines `TensorFormat::kCHW32` as  $[N, C/32, H, W, 32]$ , where channels are grouped into contiguous blocks of 32; this layout allows Ampere Tensor Core convolution kernels to operate on 32-channel-wide blocks without any intra-kernel reformatting [42]. The C2f block splits the cv1 output ( $C=64$  channels) along the channel axis into two halves of 32 channels each. In kCHW32 this is structurally free: the two halves already exist as separate contiguous blocks (block 0 = channels 0–31, block 1 = channels 32–63), so no data movement is needed for the split itself.

The copy arises after the split, not from it. The downstream  $3\times 3$  bottleneck convolution (`m0.cv1`) was built with NCHW tensor descriptors, it expects its input as  $[N, 32, H, W]$ . A kCHW32 sub-block  $[N, 1, H, W, 32]$  has the channel and spatial axes in a different order and cannot be passed directly to an NCHW-expecting kernel without transposition. TensorRT therefore inserts a `copyPackedKernel` to reformat kCHW32 to NCHW, and this copy appears at the Split boundary because that is where the data first enters the path leading to the NCHW conv. If the bottleneck convolutions used NHWC or kCHW32 descriptors instead, the split output could be passed directly to the kernel, eliminating the copy entirely.

The kernel execution trace makes this pattern directly visible. Table 5.29 shows a representative 15-kernel excerpt from a steady-state inference pass covering the C2f model.4 block: two convolution groups are separated by a burst of three `copyPackedKernel` calls (indices 27-29) at the Split boundary, followed immediately by the bottleneck convolutions.

The same burst pattern recurs at every C2f block in the 170-kernel sequence, with copy duration scaling with spatial resolution as shown in Table 5.30.

Table 5.30 shows the per-block cost from the TensorRT layer profiler (median GPU time across 278 profiling iterations; the `engine%` column is taken directly from the profiler’s own time-fraction accounting).

The cost scales with spatial resolution: model.2 at  $160\times 160$  accounts for 0.249 ms alone -

Table 5.29: Kernel execution trace excerpt (one inference pass, steady state) showing the `copyPackedKernel` cluster at the model.4 C2f Split boundary. Durations are raw GPU timestamps from the Nsight Systems kernel trace.

| #         | Kernel (abbreviated)                                     | Duration ( $\mu$ s) |
|-----------|----------------------------------------------------------|---------------------|
| 25        | <code>sm80_xmma_fprop...nhwc_128x32x64 (cv1 conv)</code> | 112.5               |
| 26        | <code>generatedNativePointwise (SiLU)</code>             | 59.5                |
| <b>27</b> | <code>copyPackedKernel (Split output 0)</code>           | <b>62.0</b>         |
| <b>28</b> | <code>copyPackedKernel (Split output 1)</code>           | <b>60.5</b>         |
| <b>29</b> | <code>copyPackedKernel (Add output copy)</code>          | <b>54.5</b>         |
| 30        | <code>sm80_xmma_fprop...nhwc_128x64x32 (m.cv1)</code>    | 114.2               |
| 31        | <code>generatedNativePointwise (SiLU)</code>             | 60.1                |
| 32        | <code>sm80_xmma_fprop...nhwc_256x128x32 (m.cv2)</code>   | 98.1                |
| 33        | <code>generatedNativePointwise (SiLU+Add)</code>         | 21.9                |
| 34        | <code>sm80_xmma_gemm...tn_n (cv2 1×1)</code>             | 40.8                |
| 35        | <code>generatedNativePointwise (SiLU)</code>             | 19.4                |
| 36–42     | (5 further conv+activation pairs)                        | 59–69               |
| 43        | <code>generatedNativePointwise</code>                    | 18.4                |
| <b>44</b> | <code>copyPackedKernel (next block Split)</code>         | <b>32.5</b>         |
| <b>45</b> | <code>copyPackedKernel (next block Split)</code>         | <b>33.6</b>         |

1.7× that of model.4 at 80×80, and 7.5× that of model.21 at 20×20 - because the number of elements that must be copied grows quadratically with the feature-map side length. Across the full network, C2f Split copies consume **6.7 % of total engine time**.

### 5.8.3 Root Cause: Mixed Tensor Layout at C2f Boundaries

Both the Split copy overhead and the NCHW-path convolution overhead share a single root cause: TensorRT selects NHWC-native Tensor Core kernels (`nhwckrsc_nhwc`) for most layers, but the 3×3 bottleneck convolutions inside each C2f block have their cuDNN tensor descriptors built with NCHW, causing two cascading penalties:

1. **Graph-level Split copy nodes.** The C2f channel split itself is free in kCHW32 (the two halves are already contiguous sub-blocks). The copy is inserted because the destination, the bottleneck 3×3 convolution, has NCHW descriptors and cannot accept kCHW32 input directly. TensorRT inserts a `copyPackedKernel` at every split to transpose from kCHW32 to NCHW, totalling 6.7% of engine time across the network.
2. **cuDNN-internal format conversion inside 3×3 convolutions.** When the tensor descriptor specifies NCHW, cuDNN selects the `nhwckrsc_nchw` kernel variant, which performs an NCHW→NHWC conversion internally before executing

Table 5.30: C2f Split copy node overhead per backbone and neck block. Costs include both `Split_output_0 copy` and `Split_output_1 copy`, plus the intermediate `Add_output copy` for blocks with two bottleneck stages (model.4, 6).

| Block               | Spatial resolution | Copy cost (ms) | Engine % |
|---------------------|--------------------|----------------|----------|
| model.2 (backbone)  | 160×160            | 0.249          | 2.23     |
| model.4 (backbone)  | 80×80              | 0.148          | 1.32     |
| model.15 (FPN neck) | 80×80              | 0.087          | 0.78     |
| model.6 (backbone)  | 40×40              | 0.083          | 0.74     |
| model.12 (FPN neck) | 40×40              | 0.049          | 0.44     |
| model.8 (backbone)  | 20×20              | 0.048          | 0.43     |
| model.18 (FPN neck) | 40×40              | 0.048          | 0.43     |
| model.21 (FPN neck) | 20×20              | 0.033          | 0.29     |
| Total               |                    | 0.745          | 6.67     |

the Tensor Core instruction. This variant is slower than the purely NHWC path (`nhwckrsc_nhwc`) and accounts for 17.2 % of all GPU kernel time.

Both problems can be resolved simultaneously for any given C2f block by replacing it with a custom TensorRT plugin that accepts the engine’s native kCHW32 input, keeps all intermediate buffers in NHWC layout, and returns NHWC output - eliminating the Split copy nodes and switching the 3×3 convolutions to NHWC descriptors. Section 5.9 describes this approach targeting model.2, which carries the highest per-block overhead (0.249 ms, 2.23 % of engine time).

#### 5.8.4 Unfused SiLU Activations

The kernel trace reveals a second class of overhead that is independent of format conversion. Every SiLU non-linearity ( $\sigma(x) \cdot x$ ) in the network is dispatched as a separate `generatedNativePointwise` GPU kernel - 57 launches per inference pass totalling 2.147 ms (19.3 %) of engine time. TensorRT does fuse the Sigmoid and the element-wise multiply into a single Pointwise Node (PWN), avoiding one kernel launch per activation, but it does not merge the activation into the preceding convolution the way it fuses `Conv\,+\,ReLU`. On Ampere, cuDNN exposes a `CUDNN\_ACTIVATION\_SWISH` epilogue that would let the convolution kernel apply the non-linearity in-register without writing an intermediate activation tensor back to memory, however, here TensorRT did not select this path for these layer shapes in FP32 mode.

Each activation kernel is purely memory-bandwidth-bound: it reads the full output tensor from DRAM, applies the non-linearity element-wise, and writes the result back. At 57 launches, unfused SiLU collectively consumes as much time as the entire detection head

convolutions (21.3%), making it the second-largest category after convolutions. Switching to FP16 precision or replacing SiLU with ReLU, which TensorRT always fuses as a conv epilogue, would eliminate this overhead.

### 5.8.5 Summary

Table 5.31 shows where GPU time is spent across a single FP32 inference pass, using the TensorRT layer profiler (278 profiling iterations; engine total 11.154 ms).

Table 5.31: Full GPU time breakdown for TensorRT FP32 YOLOv8n inference on Orin Nano, batch = 1. Sources marked † are structural; sources marked ★ are directly targeted by the plugin described in Section 5.9.

| Category                                               | Time (ms)     | Engine (%)   |
|--------------------------------------------------------|---------------|--------------|
| Backbone/neck convolutions★                            | 4.229         | 37.9         |
| (of which NCHW-path variants, internal cuDNN reformat) | (1.920)       | (17.2)       |
| Detection head convolutions (cv2, cv3, DFL)            | 2.378         | 21.3         |
| SiLU activations (unfused PWN)                         | 2.147         | 19.3         |
| Format copies & reformats (total)                      | 1.683         | 15.1         |
| C2f Split copy nodes (8 blocks)★                       | 0.744         | 6.7          |
| Input NCHW→kCHW32 reformat†                            | 0.269         | 2.4          |
| Resize output copies (FPN)†                            | 0.177         | 1.6          |
| MaxPool / other copies                                 | 0.493         | 4.4          |
| Resize / upsample ops (FPN)                            | 0.193         | 1.7          |
| Detection head ops (PWN, arithmetic)                   | 0.281         | 2.5          |
| MaxPool ops (SPPF)                                     | 0.132         | 1.2          |
| Softmax (DFL)                                          | 0.110         | 1.0          |
| <b>Total</b>                                           | <b>11.154</b> | <b>100.0</b> |

Convolutions (backbone, neck, and detection head) account for **59.2%** of engine time and represent useful computation. The backbone/neck row includes the 17.2% from NCHW-path variants (convolutions whose cuDNN descriptors specify NCHW, causing an internal format conversion on every call); together with the C2f Split copies (6.7%), these are the overhead sources directly addressed by plugin replacement. The remaining 40.8% is overhead: unfused SiLU activations (19.3%), format copies and reformats (15.1%), and other non-compute operations (6.4%). Of the remaining overhead, unfused SiLU (19.3%) is the largest single target not addressed by the plugin; eliminating it would require FP16 precision or a change of activation function.

## 5.9 Design and Implementation of a TensorRT Plugin for the C2f Block

The kernel-level analysis in Section 5.8 identified two overlapping sources of avoidable overhead at every C2f block: Split copy nodes (6.7% of engine time) and NCHW-path bottleneck convolutions that trigger an internal format conversion on each call (17.2%). This section investigates whether a custom TensorRT plugin can eliminate this overhead by replacing the entire C2f subgraph with a single opaque node.

Model.2 was selected as the target because it carries the largest per-block Split copy cost (0.249 ms) due to its  $160 \times 160$  spatial resolution. The plugin was implemented using the `IPluginV2DynamicExt` API; the build workflow is illustrated in Figure 5.9.

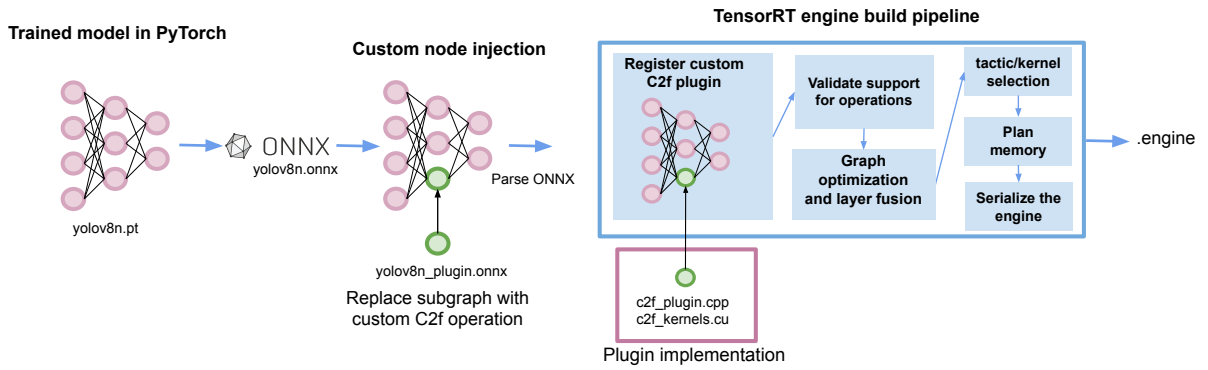


Figure 5.9: TensorRT engine build workflow with custom plugin injection. A trained YOLOv8 model is exported from PyTorch to ONNX, a selected C2f subgraph is replaced by a custom plugin node, and TensorRT builds a target-specific engine by parsing the modified graph, validating supported operators and registered plugins, applying graph optimizations, selecting tactics, planning memory, and serializing the final engine.

### 5.9.1 Plugin Architecture and Weight Management

The plugin replicates the YOLOv8n model.2 C2f forward pass, illustrated in Figure 5.10: `cv1` ( $1 \times 1$ ,  $32 \rightarrow 2C_{\text{half}}=32$ ), channel split into  $x_1$  and  $x_2$  ( $C_{\text{half}}=16$  channels each), one bottleneck stage with `m.0.cv1` ( $3 \times 3$ ,  $16 \rightarrow 16$ ) and `m.0.cv2` ( $3 \times 3$ ,  $16 \rightarrow 16$ ) with a residual shortcut, concatenation of  $[x_1 \mid x_2 \mid \text{m.0.out}]$  into  $3C_{\text{half}}=48$  channels, and `cv2` ( $1 \times 1$ ,  $48 \rightarrow 32$ ). All convolutions use folded batch-normalisation weights: bias vectors absorb the BN parameters offline so that inference requires only a convolution followed by a bias add.

Weights are exported from the trained PyTorch model in a custom binary format and loaded once during `initialize()` into device memory; `terminate()` frees them.

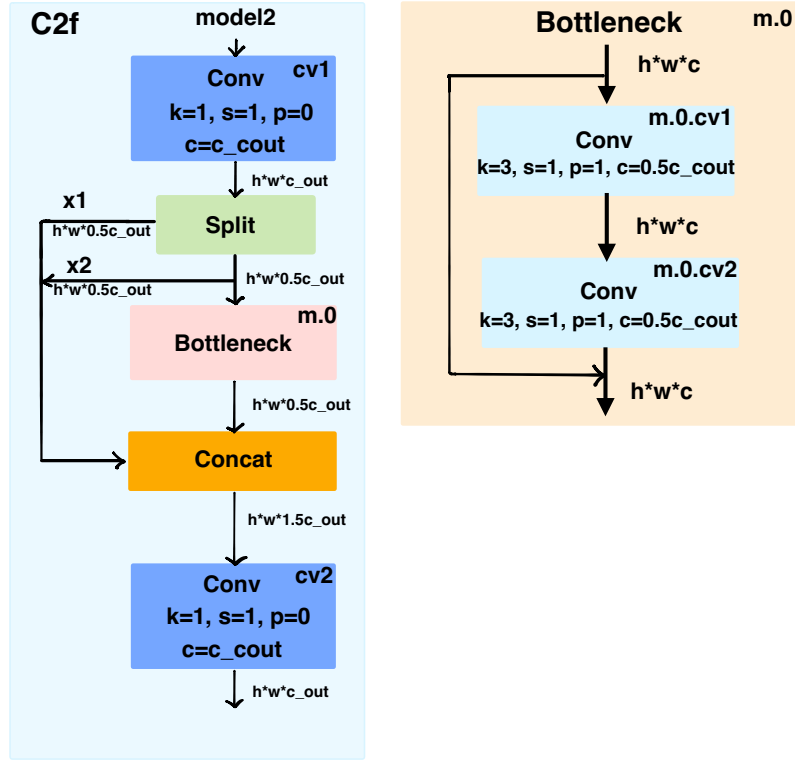


Figure 5.10: Architecture of the C2f (Cross Stage Partial with two convolutions) block, model.2 layer used in YOLOv8, shown for  $n=1$  bottleneck stage. **Left:** the outer block applies a  $1 \times 1$  projection (cv1) to produce  $c_{out}$  channels, splits the result into  $x_1$  and  $x_2$  (each  $0.5 c_{out}$  channels), routes  $x_2$  through one bottleneck, and concatenates  $[x_1, x_2, m.0(x_2)]$  into  $1.5 c_{out}$  channels before the final  $1 \times 1$  projection (cv2). **Right:** the bottleneck (m.0) consists of two convolutions with an optional residual shortcut that adds the block input to its output when the channel count is preserved.

### 5.9.2 Fused Shared-Memory Kernel

The plugin executes all four convolutions, bias additions, and SiLU activations in a single CUDA kernel (`fused\_c2f\_model2\_kernel`), keeping all intermediate feature maps in shared memory without writing them to global memory. An  $8 \times 8$ -pixel output tile is assigned to each thread block, and the four convolution stages execute sequentially from shared memory with the residual shortcut applied in registers. Table 5.32 shows the per-component breakdown.

The plugin successfully eliminates the Split copy nodes and all intermediate global-memory writes. However, the convolution cost increases from 0.581 ms to 4.889 ms ( $8.4 \times$ ). The default engine dispatches `sm80\_xmma\_fprop` kernels (observed in the Nsight Systems kernel trace), whose names encode `tf32f32` accumulation, confirming TF32 Tensor Core execution on Orin Ampere; the plugin kernel uses scalar FP32 arithmetic and cannot

Table 5.32: Model.2 C2f execution cost: default TensorRT engine versus plugin engine. Times are median values from the TensorRT layer profiler.

| Component               | Default engine               | Plugin engine              |
|-------------------------|------------------------------|----------------------------|
| 4 convolutions          | 0.581 ms (TF32 Tensor Cores) | 4.889 ms (FP32 CUDA cores) |
| SiLU activations        | 0.380 ms (4 PWN kernels)     | fused into kernel          |
| Split / slice copies    | 0.248 ms                     | eliminated                 |
| Boundary reformat nodes | —                            | 0.249 ms                   |
| <b>model.2 total</b>    | <b>1.209 ms</b>              | <b>5.138 ms</b>            |

utilise the Tensor Core units. The savings from eliminating copies (0.248 ms) and fusing SiLU (0.380 ms) are outweighed by this compute regression, yielding a net increase of 3.929 ms for model.2 that propagates directly to the 4.00 ms end-to-end gap.

### 5.9.3 Summary

The experiment confirms that the identified overhead sources, Split copies and unfused activations, can be structurally removed by a plugin. The performance gap is entirely attributable to the absence of Tensor Core utilisation in the custom kernel: TensorRT’s advantage lies not only in graph-level fusion but in access to hardware-specific Tensor Core kernels that general-purpose FP32 CUDA code cannot match. It might be possible to close this gap is CUTLASS-based fused convolutions, which provide Tensor Core execution with arbitrary activations. However, the development of such convolutions is not trivial and still requires a lot of manual testing and debugging.

## 5.10 Evaluation of Activation Function Simplification: SiLU vs. ReLU

The kernel-level analysis in Section 5.8 identified unfused SiLU activations as the second-largest source of overhead in the default TensorRT engine, consuming 19.3% of total engine time. This section investigates whether substituting SiLU with ReLU eliminates this overhead and quantifies the resulting latency reduction, along with the accuracy implications of the change.

### 5.10.1 Why SiLU Cannot Be Fused by TensorRT

YOLOv8n uses SiLU (Sigmoid Linear Unit) [55] as its activation function throughout, defined as  $\text{SiLU}(x) = x \cdot \sigma(x)$ , where  $\sigma(x) = (1 + e^{-x})^{-1}$  is the logistic sigmoid. This is a

composite expression: unlike ReLU ( $\max(0, x)$ ), SiLU requires two distinct element-wise operations, a sigmoid and a multiplication, that cannot be expressed as a single algebraic operation to a convolution.

TensorRT’s cuDNN-based convolution kernels support a limited set of fused activation epilogues: ReLU, clipped ReLU, sigmoid, tanh, and their composites where the intermediate result is not reused. SiLU requires the input  $x$  to be available both as the argument to sigmoid and as the multiplicand, which means the activation cannot be fused into the convolution’s output stage. TensorRT therefore emits the activation as two separate pointwise kernels: `PWN(Sigmoid)` followed by `PWN(Mul)`, sometimes merged by the graph optimiser into a single `PWN(PWN(Sigmoid), PWN(Mul))` node, but always requiring a global-memory round-trip between the convolution and the activation.

ReLU, by contrast, is a supported cuDNN convolution epilogue. TensorRT fuses it directly into the convolution kernel, so the activation is applied in registers as the output tile is being written to global memory, no second kernel launch and no DRAM re-read of the output tensor.

### 5.10.2 Activation Substitution Methodology

To isolate the structural cost of unfused SiLU from accuracy effects, the experiment replaces all 57 SiLU instances in the pre-trained YOLOv8n model with ReLU via direct weight substitution in PyTorch without any subsequent fine-tuning or retraining. The substitution traverses the module tree and replaces every `nn.SiLU` with `nn.ReLU(inplace=True)`, preserving all convolutional, batch-normalisation (already folded), and structural weights unchanged.

The resulting model is exported to ONNX (opset 17, `simplify=True`) and compiled into a TensorRT FP32 engine using the same build flags as the default engine. This procedure yields an engine that is structurally identical to the SiLU baseline in all respects except the activation function. Profiling is performed with the same Nsight Systems setup and inference harness as the default engine, making the two profiles directly comparable.

### 5.10.3 TensorRT Graph-Level Impact

Figure 5.11 illustrates the per-layer profiling difference between the two engines using `model.2/cv1` as a representative example. In the SiLU engine, TRT reports two separate profiling entries for this layer: the convolution (`Conv`, 0.126 ms) and the activation (`PWN(PWN(Sigmoid), PWN(Mul))`, 0.112 ms), for a combined cost of **0.238 ms**. In the ReLU engine, TRT reports a single fused entry (`Conv+Relu`, 0.124 ms), a reduction of  $\approx 1.9\times$ .

The convolution time itself is almost identical in both cases; the difference comes entirely from the additional activation kernel in the SiLU path.

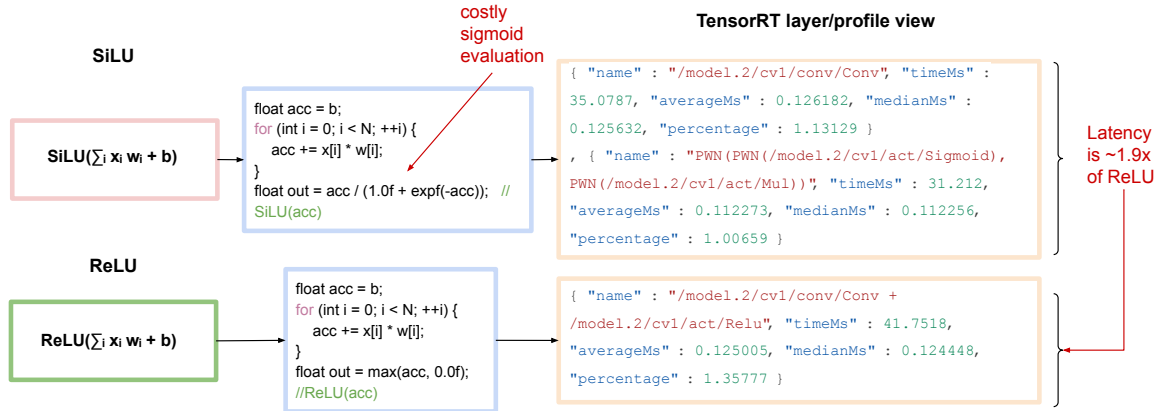


Figure 5.11: TensorRT per-layer profiling for model.2/cv1: SiLU (top) versus ReLU (bottom). In the ReLU case TensorRT reports a single fused Conv + Relu entry (0.124 ms), whereas the SiLU case produces a separate PWN activation node (0.112 ms) on top of the convolution (0.126 ms), increasing the combined layer cost by approximately  $1.9\times$ .

Across the full network, the SiLU engine has 59 standalone activation nodes (after TRT’s own pointwise fusion), with a total median GPU time of 2.28, ms. None of these nodes appear in the ReLU engine: all 61 convolutional layers in the ReLU engine appear as fused Conv + Relu entries. Table 5.33 shows the overall comparison from the TensorRT layer profiler.

Table 5.33: Comparison between the SiLU and ReLU TensorRT engines (FP32, Orin Nano, batch = 1). Layer profiler values are sums of per-layer median GPU times. End-to-end latency and FPS are from `bench_rigorous.py` (5 repetitions  $\times$  200 runs, 30 warm-up runs; GPU clocks locked during this measurement).

| Metric                        | SiLU engine                    | ReLU engine    |
|-------------------------------|--------------------------------|----------------|
| Convolution layers (sum)      | 6.27 ms                        | 6.25 ms        |
| Activation nodes (sum)        | 2.28 ms                        | 0 ms (fused)   |
| Copy / reformat / other (sum) | 2.59 ms                        | 2.95 ms        |
| <b>Layer profiler total</b>   | <b>11.14 ms</b>                | <b>9.20 ms</b> |
| End-to-end mean latency       | 11.45 ms                       | 9.37 ms        |
| End-to-end std                | $\pm 0.01$ ms                  | $\pm 0.02$ ms  |
| Throughput                    | 87.3 FPS                       | 106.7 FPS      |
| <b>Speedup</b>                | <b>1.22<math>\times</math></b> |                |
| Average power                 | 28 128 mW                      | 26 369 mW      |
| Power reduction               | −6.2 %                         |                |

The 2.28 ms reduction in activation time from the layer profiler (11.14  $\rightarrow$  9.20 ms, −17.4 %)

is slightly lower than the 19.3% share cited in Section 5.8, because the TRT graph optimiser fuses a small fraction of SiLU nodes with preceding or following operations (e.g. `PWN(SiLU+Add)` in residual shortcuts), reducing their standalone cost. The end-to-end benchmark confirms the structural improvement: mean latency drops from 11.45 ms to 9.37 ms (−18.2%), with throughput increasing from 87.3 to 106.7 FPS (+22.2%). Both engines were benchmarked under identical thermal and clock conditions.

#### 5.10.4 Accuracy–Latency Trade-off

The direct weight-substitution approach used here does not involve retraining and therefore does not recover the accuracy the original SiLU activations provided. The SiLU function was introduced in YOLOv8 specifically because it provides a smoother gradient landscape than ReLU [68]. ReLU outputs zero for all negative inputs, which is computationally efficient but can cause neurons to produce zero gradients permanently once their pre-activation falls below zero. SiLU avoids this by preserving a small, smooth gradient for negative values through its  $\sigma(x)$  factor, which in practice leads to improved convergence in deep architectures. A direct weight swap yields a network whose filters were optimised under SiLU gradients but must execute with a different activation shape, degrading detection accuracy.

Validation of the weight-substituted ReLU model on the COCO validation set was not performed. The experiment is therefore restricted to a structural latency measurement. Deploying a ReLU-based YOLOv8n would require either fine-tuning the substituted weights or retraining from scratch with ReLU as the default activation. Prior work on activation function substitution in YOLO architectures suggests that fine-tuning from pretrained weights can partially recover the accuracy gap, though full retraining from scratch is the standard approach used in comparative studies [65, 26].

#### 5.10.5 Summary

Replacing SiLU with ReLU removes 59 standalone activation kernel launches, reduces the layer profiler total by 17.4% (11.14 → 9.20 ms), and delivers a 1.22× end-to-end speedup (11.45 → 9.37 ms, 87.3 → 106.7 FPS) with a 6.2% reduction in average power draw. The savings are consistent with the 19.3% overhead estimate from the kernel analysis: SiLU is the single largest latency target in the FP32 engine that can be addressed without custom CUDA kernels, requiring only a model-level change.

The trade-off is that this change comes at the cost of the activation function that the model was trained with, and recovering the original accuracy requires additional training effort. For latency-critical deployments where a 1–2% mAP reduction is acceptable, the

ReLU swap delivers a meaningful speedup with no hardware-specific engineering. For deployments requiring the full accuracy of the pre-trained model, the alternative is FP16 quantisation, which eliminates the SiLU overhead through a different mechanism (halving the data volume processed by all kernels including activations) and represents a natural next step for precision-aware optimisation.

## 6 Discussion

This chapter interprets the experimental results in the broader context of the research questions, examines causes of unexpected or negative findings, and connects the individual experiments into a coherent picture of which optimization strategies are most effective for YOLOv8n on edge hardware.

### 6.1 Framework effectiveness is platform-dependent

The most consistent finding across all experiments is that no single inference framework dominates across both platforms and all precision settings. On the Jetson Orin Nano GPU, TensorRT was the clear winner in every configuration: FP32 ( $2.21\times$  over PyTorch eager), FP16 ( $4.34\times$ ), and INT8 implicit ( $4.69\times$ ). On the Raspberry Pi 5 CPU, the rankings changed substantially. LiteRT achieved both the lowest FP32 latency (154.1 ms,  $1.88\times$  speedup) and the most efficient INT8 execution (31.05 ms,  $9.4\times$  speedup over PyTorch eager). ONNX Runtime and TVM were competitive with each other in FP32 on the CPU, while TorchCompile consistently underperformed relative to ahead-of-time compiled alternatives.

This platform dependency is not accidental. The runtime advantage of TensorRT on NVIDIA hardware is rooted in low-level GPU-specific kernel selection, Tensor Core dispatch, and aggressive graph-level fusion that general-purpose runtimes cannot replicate. LiteRT’s advantage on Raspberry Pi 5 reflects strong optimization for Arm AArch64 NEON and XNNPACK micro-kernel paths. A deployment decision based on only one platform would therefore produce misleading conclusions about the other.

### 6.2 TVM is competitive but does not exceed TensorRT

Apache TVM with MetaSchedule demonstrated a qualitatively different optimization trajectory depending on precision. In FP32, TVM essentially matched ONNX Runtime on the GPU (23.38 ms vs. 23.18 ms), suggesting that for this model size and architecture, mature CUDA libraries used by both runtimes already close most of the gap that compiler-level scheduling can address. In FP16, however, TVM moved from merely matching to clearly outperforming ONNX Runtime (12.45 ms vs. 16.95 ms), because FP16 enabled ten-

sorted WMMA execution and shifted MetaSchedule into a qualitatively different search space.

Even in the FP16 setting, TensorRT remained substantially faster (6.31 ms vs. 12.45 ms for TVM). The remaining gap is attributable to optimizations that MetaSchedule did not discover within the tuning budget: backend-specific kernel variants, multi-layer fusion across C2f subgraphs, and empirically selected Tensor Core tile configurations that TensorRT accumulates over years of hardware-specific tuning. This shows that compiler-based tuning can unlock hardware potential incrementally, but does not automatically match the depth of a mature vendor runtime for a well-supported architecture.

The long tuning time required by MetaSchedule (approximately 18 hours for FP16) is also a practical limitation for deployment in production contexts where rapid iteration is expected.

### 6.3 Quantization is a deployment trade-off, not a universal gain

The quantization experiments produced the most heterogeneous results of the entire study, and they illustrate that INT8 quantization cannot be treated as a drop-in optimization.

On the Jetson Orin Nano, TensorRT implicit INT8 achieved the best latency result in the entire experiment (5.34 ms,  $4.69\times$  speedup), but at a measurable accuracy cost of approximately 10% in both mAP50 and mAP50–95. TensorRT QDQ INT8 was slower than the implicit path and caused severe accuracy degradation ( $-25.4\%$  mAP50), suggesting that explicit quantization node insertion creates graph boundaries that the TensorRT compiler cannot fully eliminate. ONNX Runtime QDQ INT8 was actually slower than the PyTorch FP32 baseline despite nominally using INT8 values, a result explained by the profiling data showing that the model spent a large fraction of time on QuantizeLinear/DequantizeLinear conversions rather than on useful convolution work.

On the Raspberry Pi 5, the contrast between TFLite INT8 (31.05 ms,  $9.4\times$  speedup) and ONNX Runtime INT8 (133.81 ms,  $2.2\times$  speedup) shows the same principle: latency gains from INT8 depend almost entirely on whether the runtime has an efficient integer execution path for the target hardware. TFLite achieves this through XNNPACK IGEMM kernels with direct INT8 arithmetic; ONNX Runtime is burdened by QDQ conversion overhead.

These results suggest that for YOLOv8n, INT8 quantization is practical only when the deployment path can sustain clean low-precision execution through fused kernels:

TensorRT implicit calibration on GPU or TFLite on Arm CPU. In other configurations, the overhead introduced by quantization can outweigh its benefits.

## 6.4 Structured Pruning is ineffective for this model and hardware

The pruning experiments produced a consistent negative result across both hardware targets and a wide range of sparsity levels. The per-layer sensitivity sweep on the Jetson Orin Nano showed median speedups of  $1.00 \pm 0.02$  across 418 configurations. The full-model pruning and fine-tuning on the RTX 3070 achieved at most 2.3 % relative speedup against its paired baseline, accompanied by a 4.0 % mAP50-95 reduction.

The root cause is the mismatch between channel-level sparsity and the granularity at which cuDNN selects convolution kernels. cuDNN dispatches GEMM and Winograd kernels with fixed warp-level tiling that does not adapt continuously to channel count, so modest channel reductions simply leave warp lanes idle without triggering a faster kernel path. YOLOv8n is also already a compact model, and compact models provide less redundancy for pruning to exploit.

This finding is consistent with the broader quantization observation: the most effective optimization levers are those that change the execution regime of the entire kernel (precision format, Tensor Core utilisation, dedicated micro-kernels), rather than those that reduce the work within an existing kernel regime.

## 6.5 Kernel-Level analysis reveals structural overhead in the optimized engine

The Nsight Systems profiling of the default TensorRT FP32 engine showed that only 59.2 % of engine time is spent on convolutions (useful computation). The remaining 40.8 % is structural overhead: unfused SiLU activations (19.3 %), format copies and reformats (15.1 %), and other non-compute operations (6.4 %). This overhead profile motivates the two targeted experiments that followed: the C2f plugin (addressing format copies and the Split copy nodes) and the SiLU→ReLU substitution (addressing unfused activations).

The fact that these two overhead sources together account for roughly 36 % of total engine time shows that even well-optimized TensorRT engines can contain significant avoidable cost at FP32. Identifying these sources required kernel-level profiling beyond what framework-level benchmarks reveal.

## 6.6 The C2f Plugin confirms both the source and the limit of the overhead

The custom TensorRT plugin for the C2f block successfully removed the C2f Split copy nodes (6.7% of engine time) and the associated format conversion overhead. However, the plugin achieved approximately the same end-to-end latency as the default TensorRT engine, because the removal of copy overhead was offset by the loss of Tensor Core utilisation.

The default TensorRT engine dispatches the `sm80_xmma` kernel family, which uses TF32 Tensor Core arithmetic (4× throughput increase on Ampere) and NHWC-native layout, eliminating the NCHW↔NHWC conversions that the custom CUDA path incurs. The hand-written fused kernel processes output pixels sequentially per thread and cannot exploit the warp-cooperative 16×8×8 accumulation required by `wmma` operations.

The result is instructive: the plugin demonstrates that a hand-written CUDA kernel can replicate the structural graph transformation (fusion of four convolutions) while preserving correctness, but it cannot reproduce the hardware-specific Tensor Core performance without adopting a CUTLASS-style block-tiling strategy. This is a general lesson about the limits of simple custom CUDA kernels when competing with vendor-optimised math libraries on modern Ampere-class hardware.

## 6.7 SiLU replacement delivers a clear structural speedup at an accuracy cost

The SiLU→ReLU experiment produced the clearest positive latency result among the targeted optimizations: a 1.22× end-to-end speedup (11.45 ms→9.37 ms), 87.3→106.7 FPS, and a 6.2% reduction in average power draw. Unlike the plugin, this improvement has no counterbalancing hardware-level regression, because the ReLU epilogue is fused into the convolution kernel without any additional launch.

The trade-off is accuracy. The substitution was performed without retraining, and the network’s weights were optimised under SiLU gradients. COCO validation accuracy was not evaluated for the weight-substituted model, and recovering it would require either fine-tuning or full retraining. This places the SiLU replacement in the same category as INT8 quantization: a deployment trade-off suitable when a moderate accuracy reduction is acceptable, not a free performance improvement.

For deployments where the original accuracy must be preserved, FP16 inference is

the more appropriate alternative. TensorRT FP16 achieves 6.31 ms ( $4.34\times$  over PyTorch FP32 eager) with no measurable accuracy degradation ( $\Delta\text{mAP}_{50} = +0.03\%$  on the GPU side), and it eliminates the SiLU overhead through a different mechanism: half-precision reduces the data volume processed by all kernels, including the activation and convolution stages.

## 6.8 Unexplored Deployment Configurations

Several deployment configurations were identified but not experimentally evaluated in this work. All experiments used batch size 1, which is representative of low-latency real-time deployment but does not reflect throughput-optimised scenarios where larger batches amortise kernel launch overhead. On the hardware side, the study was limited to a CUDA GPU and an Arm CPU. Alternative targets, dedicated NPUs (Hailo, Qualcomm AI Engine), the Vulkan compute API on the Raspberry Pi 5 VideoCore GPU, FPGAs, and mobile SoCs, represent structurally different execution models where the overhead patterns identified here may manifest differently or not at all. These configurations are discussed as concrete future directions in Chapter 8.

## 7 Conclusion

This thesis investigated how compiler-aware optimization affects the inference performance of YOLOv8n on two edge hardware platforms, the NVIDIA Jetson Orin Nano GPU and the Raspberry Pi 5 CPU, across a range of deployment strategies: framework selection, numerical precision reduction, INT8 quantization, structured pruning, low-level TensorRT plugin development, and activation function substitution.

The central finding is that the choice of inference framework and precision format is the dominant factor in edge deployment performance, and the most effective optimizations are those that change the hardware execution regime rather than those that reduce work within an existing regime.

**Framework and Precision Results.** On the Jetson Orin Nano GPU, TensorRT consistently achieved the best result at every precision level:  $2.21\times$  over PyTorch FP32 at full precision,  $4.34\times$  with FP16, and  $4.69\times$  with implicit INT8 calibration. On the Raspberry Pi 5 CPU, LiteRT delivered the best FP32 result ( $1.88\times$  over PyTorch) and the most effective INT8 path with TFLite INT8 ( $9.4\times$ , 31.05 ms). All frameworks, including FP16 variants, preserved YOLOv8n detection accuracy within 0.1% mAP50 relative to the PyTorch FP32 baseline, confirming that reduced-precision floating-point deployment is accuracy-neutral for this model.

**Quantization.** INT8 quantization produces the largest absolute latency reductions (up to  $9.4\times$  on CPU) but introduces measurable accuracy degradation in all tested configurations ( $-6$  to  $-10\%$  mAP50 on GPU,  $-6\%$  on CPU). The magnitude of both the speedup and the accuracy loss depends critically on the deployment path: TensorRT implicit calibration and TFLite INT8 achieved substantial gains, while ONNX Runtime QDQ INT8 provided no practical benefit over the FP32 baseline on either platform.

**Structured Pruning.** Structured channel pruning of YOLOv8n did not yield meaningful latency reductions on either hardware target. The per-layer sensitivity sweep on the Orin Nano showed speedups of  $1.00 \pm 0.02$  across 418 configurations, and the full-model pruning experiment on the RTX 3070 achieved at most 2.3% relative speedup at a 4.0% mAP50-95 cost.

**Kernel-Level Analysis.** GPU kernel profiling revealed that the default TensorRT FP32 engine spends only 59.2% of its time on useful convolution work. The remain-

ing 40.8 % is structural overhead: 19.3 % from unfused SiLU activations and 15.1 % from format copies and reformats. This analysis identified two concrete targets for further optimization and motivated the plugin and activation substitution experiments.

**Custom TensorRT Plugin.** The custom C2f plugin successfully removed the Split copy overhead (6.7 % of engine time) and confirmed that graph-level fusion of the C2f subgraph is feasible. However, the plugin achieved the same overall latency as the default engine because the removal of copy overhead was offset by the loss of Tensor Core utilisation: the default engine’s TF32 Tensor Core kernels are more efficient than a hand-written FP32 CUDA kernel that processes output pixels sequentially per thread. The plugin demonstrated the structural feasibility of the approach while exposing the performance gap between custom CUDA code and vendor-optimized Tensor Core libraries.

**Activation Function Substitution.** Replacing SiLU with ReLU throughout the network, without any retraining, eliminated all 59 standalone PWN activation kernels and delivered a  $1.22\times$  end-to-end speedup (11.45 ms $\rightarrow$ 9.37 ms, 87.3 $\rightarrow$ 106.7 FPS) with a 6.2 % power reduction. This is the largest single-technique improvement achieved without changing the precision format. The trade-off is that the weight-substituted model has not been validated for accuracy loss, and recovering the original detection quality requires retraining.

**Overall.** Across all experiments, the most practically useful optimizations for latency-critical deployments without accuracy compromise are FP16 TensorRT on GPU and LiteRT FP32 on CPU. For deployments where some accuracy loss is acceptable, TFLite INT8 on CPU and TensorRT implicit INT8 on GPU offer substantially lower latency at the cost of measurable accuracy degradation. Framework selection and precision format account for a larger fraction of the achievable performance range than any structural modification to the model itself.

## 8 Future Work

The experiments in this thesis open several directions for further investigation.

- **CUTLASS-based Tensor Core plugin.** The C2f plugin demonstrated that a custom CUDA kernel can structurally fuse the C2f subgraph, but cannot match TensorRT’s `sm80\_xmma` Tensor Core throughput. Re-implementing the fused convolutions with CUTLASS would provide the `wmma`-aligned block tiling required for Tensor Core execution with arbitrary activations, potentially combining the structural overhead elimination shown here with the hardware arithmetic efficiency of the default engine.
- **Activation function retraining.** The `SiLU`→`ReLU` weight substitution was not evaluated for accuracy. Retraining YOLOv8n from scratch with `ReLU` would quantify the true accuracy cost, and a shorter fine-tuning experiment would determine how much of the mAP gap can be recovered without full retraining.
- **FP16 plugin.** The C2f plugin operated in FP32. An FP16 version paired with the TensorRT FP16 engine would combine structural fusion with reduced-precision Tensor Core arithmetic, which is the configuration most likely to produce a net improvement over the default engine.
- **TensorRT-aware pruning.** All pruning experiments used PyTorch FP32. TensorRT’s engine-builder operates on the pruned ONNX graph directly and could select a different kernel configuration at a sufficiently reduced channel count. Evaluating structured pruning combined with TRT compilation would determine whether the cuDNN kernel-granularity limitation observed in PyTorch applies equally to TRT-compiled engines.
- **Quantization-aware training (QAT).** Post-training quantization introduced  $-6$  to  $-10\%$  mAP degradation across INT8 paths. QAT inserts simulated quantization noise during training, typically recovering most of the accuracy loss, and would allow the large latency gains of TFLite INT8 and TensorRT implicit INT8 to be used without the current accuracy penalty.
- **Advanced pruning and quantization methods.** This thesis applied standard structured channel pruning and post-training quantization. Recent research has produced more effective alternatives: magnitude-based and gradient-based sensitivity scoring for pruning, mixed-precision quantization that assigns different bit-widths

per layer based on sensitivity, and joint pruning-quantization methods that co-optimize both simultaneously. Applying these approaches to YOLOv8n could yield better accuracy–efficiency trade-offs than the methods explored here.

- **Broader deployment targets.** The hardware comparison was limited to an NVIDIA GPU and an Arm CPU. Natural extensions include: inference via the Vulkan compute API on the Raspberry Pi 5 GPU, which exposes the VideoCore VII for general-purpose compute; dedicated NPU accelerators such as Hailo-8 and Qualcomm AI Engine, which use dataflow-based execution models that differ structurally from CUDA; FPGA-based deployment, where the compute graph can be mapped to custom logic with precise control over precision, latency, and power; and mobile SoCs (e.g. Qualcomm Snapdragon, Apple Neural Engine) that represent the largest class of real-world edge devices.
- **Large language model inference at the edge.** Edge optimization of convolutional vision models, as studied in this thesis, represents one approach to the broader challenge of efficient on-device inference. A growing and substantially different research direction is the deployment of large language models (LLMs) on resource-constrained hardware. LLMs involve fundamentally different bottlenecks - dominated by memory bandwidth for weight loading rather than arithmetic throughput - and require different optimization strategies: aggressive weight quantization (4-bit, 2-bit), speculative decoding, and KV-cache compression. Investigating how compiler-aware techniques such as those evaluated here translate to the LLM inference setting, and whether edge-class hardware can support useful LLM deployment with acceptable latency and power, is a high-impact direction for future work.

## 9 Description of Use of Artificial Intelligence

In working on this Master's Thesis, I have used artificial intelligence (AI) tools in the following limited ways. I used ChatGPT (OpenAI) and Claude (Anthropic) to assist with idea development, clarifying concepts, and improving the structure and language of the text (for example, by suggesting alternative formulations, checking grammar, and helping to simplify explanations). AI was also used to get help with technical formatting issues in  $\text{\LaTeX}$  and to debug small code examples by explaining error messages and proposing corrections.

All scientific content, including the problem formulation, choice of methods, implementation, experiments, analysis, and conclusions, has been developed, reviewed, and academically assessed by me. AI tools were not used to generate entire sections of the report, to perform literature searches independently, or to answer the assignment in its entirety. I have not uploaded sensitive or confidential information to the AI tools. I am solely responsible for the academic content of this assignment.

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# A Demonstration (Appendix A)

Table A.1: Per-layer performance statistics of YOLOv8n on Jetson Orin Nano (GPU, FP32). Regime is derived from the roofline knee  $AI_{knee} \approx 14.56$  FLOPs/byte (using measured BW=55.68 GB/s and peak=810.68 GFLOP/s): Memory-bound if  $AI < 13.10$ , Knee/transition if  $13.10 \leq AI \leq 16.02$ , Compute-bound if  $AI > 16.02$ .

| Layer                 | Time (ms) | AI (FLOPs/byte) | Perf (GFLOP/s) | Regime            |
|-----------------------|-----------|-----------------|----------------|-------------------|
| model.0.conv          | 18.39     | 7.713           | 4.8            | Memory-bound      |
| model.1.conv          | 12.74     | 23.955          | 18.5           | Compute-bound     |
| model.2.cv1.conv      | 8.84      | 7.995           | 5.9            | Memory-bound      |
| model.2.m.0.cv1.conv  | 4.81      | 35.899          | 24.5           | Compute-bound     |
| model.2.m.0.cv2.conv  | 4.72      | 35.899          | 25.0           | Compute-bound     |
| model.2.cv2.conv      | 9.84      | 9.593           | 8.0            | Memory-bound      |
| model.3.conv          | 6.93      | 47.291          | 34.1           | Compute-bound     |
| model.4.cv1.conv      | 3.94      | 15.920          | 13.3           | Knee / transition |
| model.4.m.0.cv1.conv  | 2.93      | 70.416          | 40.2           | Compute-bound     |
| model.4.m.0.cv2.conv  | 2.74      | 70.416          | 43.1           | Compute-bound     |
| model.4.m.1.cv1.conv  | 2.77      | 70.416          | 42.6           | Compute-bound     |
| model.4.m.1.cv2.conv  | 2.69      | 70.416          | 43.8           | Compute-bound     |
| model.4.cv2.conv      | 5.36      | 21.192          | 19.6           | Compute-bound     |
| model.5.conv          | 5.37      | 85.714          | 43.9           | Compute-bound     |
| model.6.cv1.conv      | 2.13      | 30.769          | 24.6           | Compute-bound     |
| model.6.m.0.cv1.conv  | 2.15      | 122.034         | 54.9           | Compute-bound     |
| model.6.m.0.cv2.conv  | 2.00      | 122.034         | 58.9           | Compute-bound     |
| model.6.m.1.cv1.conv  | 2.01      | 122.034         | 58.6           | Compute-bound     |
| model.6.m.1.cv2.conv  | 1.93      | 122.034         | 61.1           | Compute-bound     |
| model.6.cv2.conv      | 3.48      | 40.506          | 30.1           | Compute-bound     |
| model.7.conv          | 4.60      | 97.959          | 51.3           | Compute-bound     |
| model.8.cv1.conv      | 1.46      | 48.485          | 35.9           | Compute-bound     |
| model.8.m.0.cv1.conv  | 2.38      | 118.033         | 49.5           | Compute-bound     |
| model.8.m.0.cv2.conv  | 2.32      | 118.033         | 50.7           | Compute-bound     |
| model.8.cv2.conv      | 1.92      | 55.491          | 41.1           | Compute-bound     |
| model.9.cv1.conv      | 1.07      | 35.165          | 24.5           | Compute-bound     |
| model.9.cv2.conv      | 2.66      | 59.813          | 39.4           | Compute-bound     |
| model.12.cv1.conv     | 4.62      | 45.283          | 34.1           | Compute-bound     |
| model.12.m.0.cv1.conv | 2.13      | 122.034         | 55.5           | Compute-bound     |
| model.12.m.0.cv2.conv | 2.00      | 122.034         | 59.1           | Compute-bound     |
| model.12.cv2.conv     | 3.01      | 36.641          | 26.1           | Compute-bound     |
| model.15.cv1.conv     | 6.76      | 23.821          | 23.3           | Compute-bound     |
| model.15.m.0.cv1.conv | 2.93      | 70.416          | 40.3           | Compute-bound     |

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# APPENDIX A. DEMONSTRATION (APPENDIX A)

| Layer                 | Time (ms) | AI (FLOPs/byte) | Perf (GFLOP/s) | Regime                     |
|-----------------------|-----------|-----------------|----------------|----------------------------|
| model.15.m.0.cv2.conv | 2.69      | 70.416          | 43.8           | Compute-bound              |
| model.15.cv2.conv     | 4.67      | 19.085          | 16.8           | Compute-bound              |
| model.16.conv         | 2.87      | 53.731          | 41.1           | Compute-bound              |
| model.18.cv1.conv     | 3.01      | 36.641          | 26.1           | Compute-bound              |
| model.18.m.0.cv1.conv | 2.18      | 122.034         | 54.2           | Compute-bound              |
| model.18.m.0.cv2.conv | 2.02      | 122.034         | 58.5           | Compute-bound              |
| model.18.cv2.conv     | 2.98      | 36.641          | 26.4           | Compute-bound              |
| model.19.conv         | 2.75      | 73.096          | 42.9           | Compute-bound              |
| model.21.cv1.conv     | 1.95      | 55.491          | 40.4           | Compute-bound              |
| model.21.m.0.cv1.conv | 2.42      | 118.033         | 48.8           | Compute-bound              |
| model.21.m.0.cv2.conv | 2.27      | 118.033         | 51.9           | Compute-bound              |
| model.21.cv2.conv     | 1.94      | 55.491          | 40.5           | Compute-bound              |
| model.22.cv2.0.0.conv | 8.39      | 137.799         | 56.2           | Compute-bound              |
| model.22.cv2.0.1.conv | 8.27      | 137.799         | 57.1           | Compute-bound              |
| model.22.cv2.0.2      | 3.95      | 15.919          | 13.3           | Knee / transition          |
| model.22.cv2.1.0.conv | 3.94      | 154.839         | 59.9           | Compute-bound              |
| model.22.cv2.1.1.conv | 2.06      | 122.034         | 57.4           | Compute-bound              |
| model.22.cv2.1.2      | 0.84      | 15.681          | 15.6           | Compute-bound              |
| model.22.cv2.2.0.conv | 2.30      | 107.063         | 51.2           | Compute-bound              |
| model.22.cv2.2.1.conv | 0.92      | 83.721          | 32.2           | Compute-bound              |
| model.22.cv2.2.2      | 0.51      | 14.798          | 6.4            | Compute-bound              |
| model.22.cv3.0.0.conv | 11.51     | 152.381         | 51.3           | Compute-bound              |
| model.22.cv3.0.1.conv | 13.53     | 170.414         | 54.5           | Compute-bound              |
| model.22.cv3.0.2      | 4.97      | 19.874          | 16.5           | Compute-bound              |
| model.22.cv3.1.0.conv | 5.35      | 173.494         | 55.1           | Compute-bound              |
| model.22.cv3.1.1.conv | 3.30      | 146.939         | 55.8           | Compute-bound              |
| model.22.cv3.1.2      | 1.09      | 19.506          | 18.9           | Compute-bound              |
| model.22.cv3.2.0.conv | 2.83      | 115.663         | 52.0           | Compute-bound              |
| model.22.cv3.2.1.conv | 1.25      | 94.737          | 37.0           | Compute-bound              |
| model.22.cv3.2.2      | 0.56      | 18.161          | 9.1            | Compute-bound              |
| model.22.dfl.conv     | 1.89      | 0.471           | 0.6            | Strongly memory-bound      |
| Full Model (THOP)     | 327.00    | 157.765         | 13.54          | Mixed / Overhead-dominated |

Table A.2: Per-layer performance statistics of YOLOv8n on Jetson Orin Nano (GPU, FP32). Regime is derived from the roofline knee  $AI_{knee} \approx 14.56$  FLOPs/byte (using measured BW=55.68 GB/s and peak=810.68 GFLOP/s): Memory-bound if  $AI < 13.10$ , Knee/transition if  $13.10 \leq AI \leq 16.02$ , Compute-bound if  $AI > 16.02$ .

| Layer                 | Time (ms) | AI (FLOPs/byte) | Perf (GFLOP/s) | Regime            |
|-----------------------|-----------|-----------------|----------------|-------------------|
| model.0.conv          | 0.78      | 7.713           | 114.1          | Memory-bound      |
| model.1.conv          | 0.81      | 23.955          | 292.6          | Compute-bound     |
| model.2.cv1.conv      | 0.26      | 7.995           | 204.2          | Memory-bound      |
| model.2.m.0.cv1.conv  | 0.45      | 35.899          | 259.8          | Compute-bound     |
| model.2.m.0.cv2.conv  | 0.44      | 35.899          | 267.2          | Compute-bound     |
| model.2.cv2.conv      | 0.30      | 9.593           | 263.5          | Memory-bound      |
| model.3.conv          | 0.46      | 47.291          | 508.0          | Compute-bound     |
| model.4.cv1.conv      | 0.20      | 15.920          | 262.0          | Knee / transition |
| model.4.m.0.cv1.conv  | 0.57      | 70.416          | 208.7          | Compute-bound     |
| model.4.m.0.cv2.conv  | 0.56      | 70.416          | 211.8          | Compute-bound     |
| model.4.m.1.cv1.conv  | 0.56      | 70.416          | 211.6          | Compute-bound     |
| model.4.m.1.cv2.conv  | 0.55      | 70.416          | 213.3          | Compute-bound     |
| model.4.cv2.conv      | 0.23      | 21.192          | 446.4          | Compute-bound     |
| model.5.conv          | 0.40      | 85.714          | 593.7          | Compute-bound     |
| model.6.cv1.conv      | 0.18      | 30.769          | 296.6          | Compute-bound     |
| model.6.m.0.cv1.conv  | 0.24      | 122.034         | 494.4          | Compute-bound     |
| model.6.m.0.cv2.conv  | 0.23      | 122.034         | 502.4          | Compute-bound     |
| model.6.m.1.cv1.conv  | 0.24      | 122.034         | 501.2          | Compute-bound     |
| model.6.m.1.cv2.conv  | 0.23      | 122.034         | 502.4          | Compute-bound     |
| model.6.cv2.conv      | 0.23      | 40.506          | 462.6          | Compute-bound     |
| model.7.conv          | 0.52      | 97.959          | 455.0          | Compute-bound     |
| model.8.cv1.conv      | 0.18      | 48.485          | 284.6          | Compute-bound     |
| model.8.m.0.cv1.conv  | 0.34      | 118.033         | 351.1          | Compute-bound     |
| model.8.m.0.cv2.conv  | 0.33      | 118.033         | 358.1          | Compute-bound     |
| model.8.cv2.conv      | 0.19      | 55.491          | 424.5          | Compute-bound     |
| model.9.cv1.conv      | 0.19      | 35.165          | 136.6          | Compute-bound     |
| model.9.cv2.conv      | 0.18      | 59.813          | 577.3          | Compute-bound     |
| model.12.cv1.conv     | 0.24      | 45.283          | 647.4          | Compute-bound     |
| model.12.m.0.cv1.conv | 0.25      | 122.034         | 480.6          | Compute-bound     |
| model.12.m.0.cv2.conv | 0.24      | 122.034         | 499.3          | Compute-bound     |
| model.12.cv2.conv     | 0.20      | 36.641          | 396.8          | Compute-bound     |
| model.15.cv1.conv     | 0.26      | 23.821          | 601.7          | Compute-bound     |
| model.15.m.0.cv1.conv | 0.56      | 70.416          | 209.2          | Compute-bound     |
| model.15.m.0.cv2.conv | 0.55      | 70.416          | 213.2          | Compute-bound     |
| model.15.cv2.conv     | 0.21      | 19.085          | 372.1          | Compute-bound     |
| model.16.conv         | 0.28      | 53.731          | 417.4          | Compute-bound     |
| model.18.cv1.conv     | 0.20      | 36.641          | 402.2          | Compute-bound     |
| model.18.m.0.cv1.conv | 0.24      | 122.034         | 488.6          | Compute-bound     |
| model.18.m.0.cv2.conv | 0.24      | 122.034         | 498.5          | Compute-bound     |

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| Layer                 | Time (ms) | AI (FLOPs/byte) | Perf (GFLOP/s) | Regime                     |
|-----------------------|-----------|-----------------|----------------|----------------------------|
| model.18.cv2.conv     | 0.20      | 36.641          | 402.5          | Compute-bound              |
| model.19.conv         | 0.36      | 73.096          | 327.2          | Compute-bound              |
| model.21.cv1.conv     | 0.18      | 55.491          | 426.3          | Compute-bound              |
| model.21.m.0.cv1.conv | 0.34      | 118.033         | 351.7          | Compute-bound              |
| model.21.m.0.cv2.conv | 0.33      | 118.033         | 361.8          | Compute-bound              |
| model.21.cv2.conv     | 0.18      | 55.491          | 429.7          | Compute-bound              |
| model.22.cv2.0.0.conv | 0.57      | 137.799         | 830.1          | Compute-bound              |
| model.22.cv2.0.1.conv | 0.56      | 137.799         | 842.6          | Compute-bound              |
| model.22.cv2.0.2      | 0.26      | 15.919          | 205.0          | Knee / transition          |
| model.22.cv2.1.0.conv | 0.45      | 154.839         | 524.0          | Compute-bound              |
| model.22.cv2.1.1.conv | 0.24      | 122.034         | 493.1          | Compute-bound              |
| model.22.cv2.1.2      | 0.22      | 15.681          | 58.7           | Knee / transition          |
| model.22.cv2.2.0.conv | 0.41      | 107.063         | 288.2          | Compute-bound              |
| model.22.cv2.2.1.conv | 0.21      | 83.721          | 139.2          | Compute-bound              |
| model.22.cv2.2.2      | 0.22      | 14.798          | 14.9           | Knee / transition          |
| model.22.cv3.0.0.conv | 0.66      | 152.381         | 891.4          | Compute-bound              |
| model.22.cv3.0.1.conv | 0.83      | 170.414         | 892.5          | Compute-bound              |
| model.22.cv3.0.2      | 0.29      | 19.874          | 284.8          | Compute-bound              |
| model.22.cv3.1.0.conv | 0.42      | 173.494         | 707.2          | Compute-bound              |
| model.22.cv3.1.1.conv | 0.39      | 146.939         | 471.0          | Compute-bound              |
| model.22.cv3.1.2      | 0.22      | 19.506          | 91.9           | Compute-bound              |
| model.22.cv3.2.0.conv | 0.38      | 115.663         | 385.9          | Compute-bound              |
| model.22.cv3.2.1.conv | 0.25      | 94.737          | 184.1          | Compute-bound              |
| model.22.cv3.2.2      | 0.22      | 18.161          | 22.8           | Compute-bound              |
| model.22.dfl.conv     | 0.27      | 0.471           | 3.9            | Strongly memory-bound      |
| Full Model (THOP)     | 26.02     | 157.765         | 170.22         | Mixed / Overhead-dominated |